

Terminology, Positioning, and Imaging Principles

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PART ONE ■ TERMINOLOGY AND POSITIONING

GENERAL, SYSTEMIC, AND SKELETAL ANATOMY AND ARTHROLOGY

General Anatomy

Anatomy is the study, classification, and description of the structure and organs of the human body, whereas **physiology** deals with the processes and functions of the body, or how the body parts work. In the living subject, it is almost impossible to study anatomy without also studying some physiology. However, radiographic study of the human body is primarily a study of the anatomy of the various systems, with less emphasis on the physiology. Consequently, anatomy of the human system is emphasized in this radiographic anatomy and positioning textbook.

NOTE: Phonetic respelling¹ of anatomic and positioning terms is included throughout this text to facilitate correct pronunciation of the terms commonly used in medical radiography.

STRUCTURAL ORGANIZATION

Several levels of structural organization make up the human body. The lowest level of organization is the **chemical level**. All chemicals necessary for maintaining life are composed of **atoms**, which are joined in various ways to form **molecules**. Various chemicals in the form of molecules are organized to form **cells**.

Cells

The cell is the basic structural and functional unit of all living tissue. Every single part of the body, whether muscle, bone, cartilage, fat, nerve, skin, or blood, is composed of cells.

Tissues

Tissues are cohesive groups of similar cells that, together with their intercellular material, perform a specific function. The four basic types of tissue are as follows:

1. **Epithelial** (*ep'-i-the'-le-al*): Tissues that cover internal and external surfaces of the body, including the lining of vessels and organs, such as the stomach and the intestines
2. **Connective**: Supportive tissues that bind together and support various structures
3. **Muscular**: Tissues that make up the substance of a muscle
4. **Nervous**: Tissues that make up the substance of nerves and nerve centers

Organs

When complex assemblies of tissues are joined to perform a specific function, the result is an organ. Organs usually have a specific shape. Examples of organs of the human body are the kidneys, heart, liver, lungs, stomach, and brain.

System

A system consists of a group or an association of organs that have a similar or common function. The urinary system, consisting of the kidneys, ureters, bladder, and urethra, is an example of a body system. The total body comprises **10 individual body systems**.

Organism

The 10 systems of the body when functioning together make up the total organism—one living being (Fig. 1.1).

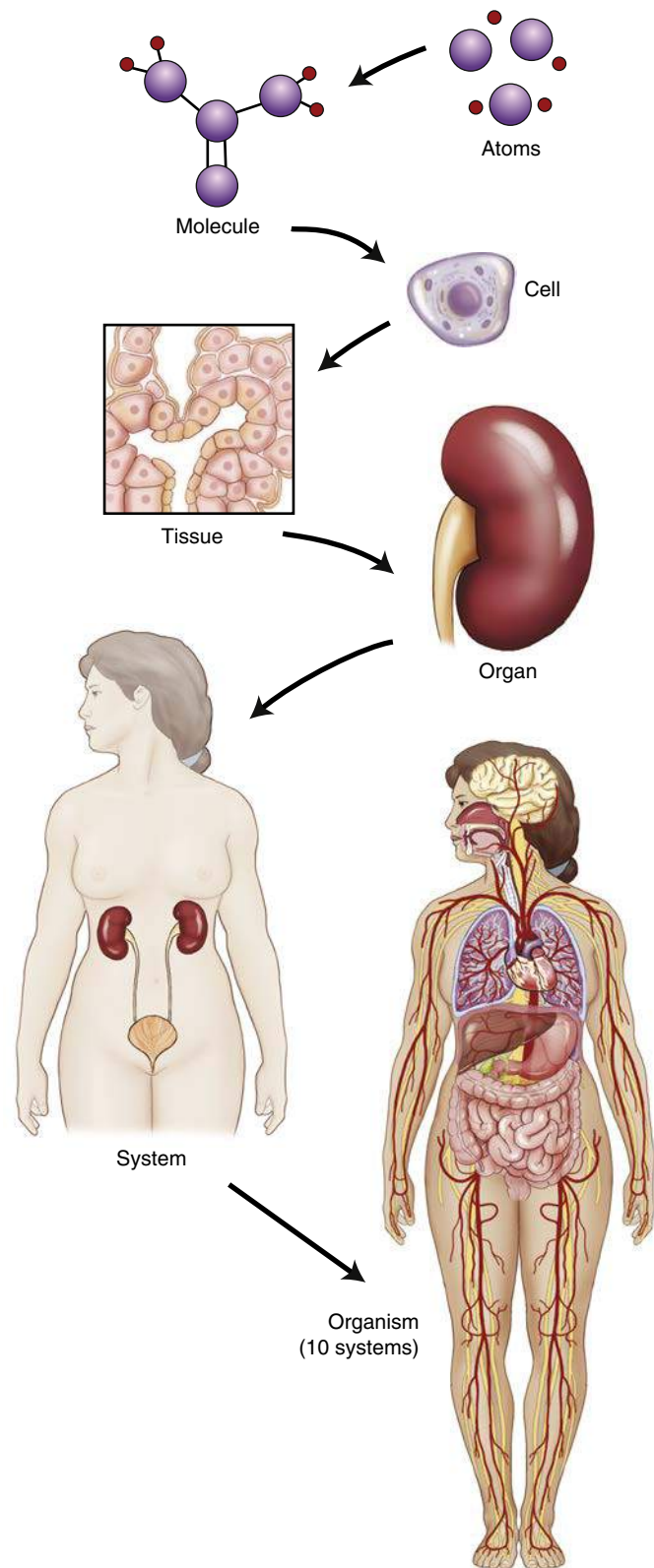


Fig. 1.1 Levels of human structural organization.

Systemic Anatomy**BODY SYSTEMS**

The human body is a structural and functional unit made up of 10 lesser units called *systems*. These 10 systems are the (1) skeletal, (2) circulatory, (3) digestive, (4) respiratory, (5) urinary, (6) reproductive, (7) nervous, (8) muscular, (9) endocrine, and (10) integumentary (*in-teg'-u-men'-tar-e*) systems.

Skeletal System

The skeletal system (Fig. 1.2) is important for the technologist to learn. The skeletal system includes the **206 separate bones** of the body and their associated cartilages and joints. The study of bones is termed **osteology**, whereas the study of joints is called **arthrology**.

The four functions of the skeletal system are as follows:

1. Support and protect many soft tissues of the body
2. Allow movement through interaction with the muscles to form a system of levers
3. Produce blood cells
4. Store calcium

Circulatory System

The circulatory system (Fig. 1.3) is composed of the following:

- The **cardiovascular organs**—heart, blood, and blood vessels
- The **lymphatic system**—lymph nodes, lymph vessels, lymph glands, and spleen

The six functions of the circulatory system are as follows:

1. Distribute oxygen and nutrients to the cells of the body
2. Transport cell waste and carbon dioxide from the cells
3. Transport water, electrolytes, hormones, and enzymes
4. Protect against disease
5. Prevent hemorrhage by forming blood clots
6. Assist in regulating body temperature

Digestive System

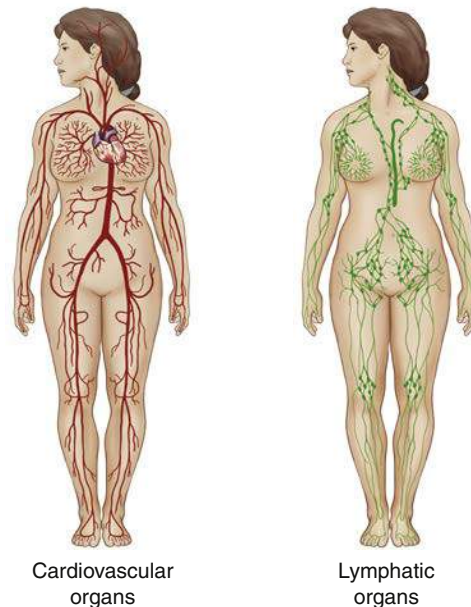
The digestive system includes the alimentary canal and certain accessory organs (Fig. 1.4). The alimentary canal is made up of the mouth, pharynx, esophagus, stomach, small intestine, large intestine, and anus. Accessory organs of digestion include the salivary glands, liver, gallbladder, and pancreas.

The twofold function of the digestive system is as follows:

1. Prepare food for absorption by the cells through numerous physical and chemical breakdown processes
2. Eliminate solid wastes from the body



Fig. 1.2 Skeletal system.



Cardiovascular
organs

Lymphatic
organs

Fig. 1.3 Circulatory system.



Fig. 1.4 Digestive system.

Respiratory System

The respiratory system is composed of two lungs and a series of passages that connect the lungs to the outside atmosphere (Fig. 1.5). The structures that make up the passageway from the exterior to the alveoli of the lung interior include the nose, mouth, pharynx, larynx, trachea, and bronchial tree.

The three primary functions of the respiratory system are as follows:

1. Supply oxygen to the blood and eventually to the cells
2. Eliminate carbon dioxide from the blood
3. Assist in regulating the acid-base balance of the blood

Urinary System

The urinary system includes the organs that produce, collect, and eliminate urine. The organs of the urinary system consist of the kidneys, ureters, bladder, and urethra (Fig. 1.6).

The four functions of the urinary system are as follows:

1. Regulate the chemical composition of the blood
2. Eliminate many waste products
3. Regulate fluid and electrolyte balance and volume
4. Maintain the acid-base balance of the body

Reproductive System

The reproductive system is made up of organs that produce, transport, and store the germ cells (Fig. 1.7). The testes in the male and the ovaries in the female produce mature germ cells. Transport and storage organs of the male include the vas deferens, prostate gland, and penis. The organs of reproduction in the female are the ovaries, uterine (fallopian) tubes, uterus, and vagina (Fig. 1.7).

The function of the reproductive system is to reproduce the organism.



Fig. 1.5 Respiratory system.

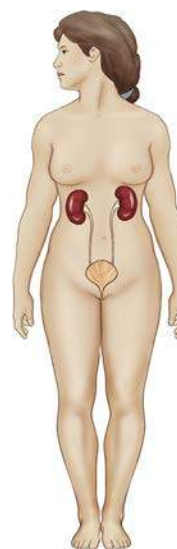
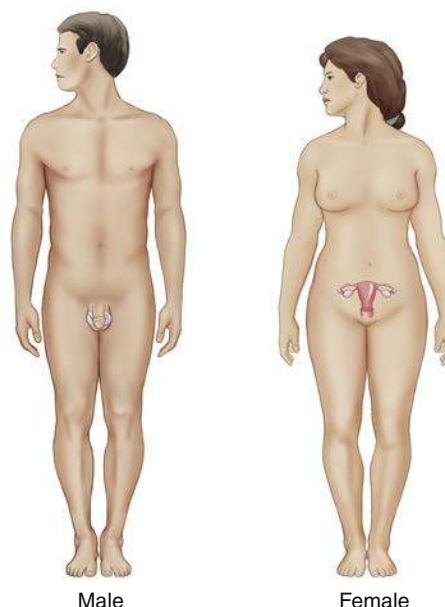


Fig. 1.6 Urinary system.



Male

Female

Fig. 1.7 Reproductive system.

Nervous System

The nervous system is composed of the brain, spinal cord, nerves, ganglia, and special sense organs such as the eyes and ears (Fig. 1.8).

The function of the nervous system is to coordinate voluntary and involuntary body activities and transmit electrical impulses to various parts of the body and the brain.

Muscular System

The muscular system (Fig. 1.9), which includes all muscle tissues of the body, is subdivided into three types of muscles: (1) **skeletal**, (2) **smooth**, and (3) **cardiac**.

Most of the muscle mass of the body is skeletal muscle, which is striated and under voluntary control. The voluntary muscles act in conjunction with the skeleton to allow body movement. About 43% of the weight of the human body is accounted for by voluntary or striated skeletal muscle.

Smooth muscle, which is involuntary, is located in the walls of hollow internal organs such as blood vessels, the stomach, and intestines. These muscles are called *involuntary* because their contraction usually is not under voluntary or conscious control.

Cardiac muscle is found only in the walls of the heart and is involuntary but striated.

The three functions of muscle tissue are as follows:

1. Allow movement, such as locomotion of the body or movement of substances through the alimentary canal
2. Maintain posture
3. Produce body heat

Endocrine System

The endocrine system includes **all the ductless glands** of the body (Fig. 1.10). These glands include the testes, ovaries, pancreas, adrenals, thymus, thyroid, parathyroid, pineal, and pituitary. The placenta acts as a temporary endocrine gland.

Hormones, which are the secretions of the endocrine glands, are released directly into the bloodstream.

The function of the endocrine system is to regulate bodily activities through the various hormones carried by the cardiovascular system.

Integumentary System

The tenth and final body system is the **integumentary** (*in-teg'-u-men'-tar-e*) system, which is composed of the **skin** and **all structures derived from the skin** (Fig. 1.11). These derived structures include hair, nails, and sweat and oil glands.

The skin is an organ that is essential to life. The skin is the **largest organ of the body**, covering a surface area of approximately 3000 in² (7620 cm²) and constituting 8% of total body mass in the average adult.

The five functions of the integumentary system are as follows:

1. Regulate body temperature
2. Protect the body, within limits, against microbial invasion and mechanical, chemical, and ultraviolet (UV) radiation damage
3. Eliminate waste products through perspiration
4. Receive certain stimuli such as temperature, pressure, and pain
5. Synthesize certain vitamins and biochemicals such as vitamin D



Fig. 1.8 Nervous system.



Fig. 1.9 Muscular system.

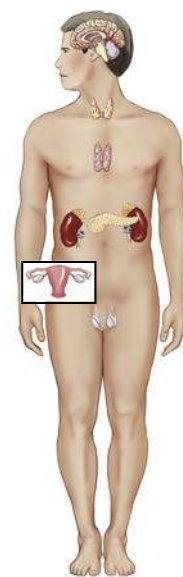


Fig. 1.10 Endocrine system.



Fig. 1.11 Integumentary system.

Skeletal Anatomy

Because a large part of general diagnostic radiography involves examination of the bones and joints, **osteology** (*os'-te-ol'-o-je*) (the study of bones) and **arthrology** (*ar-throl'-o-je*) (the study of joints) are important subjects for the technologist.

OSTEOLOGY

The adult skeletal system is composed of **206 separate bones**, which form the framework of the entire body. Certain cartilages, such as those at the ends of long bones, are included in the skeletal system. These bones and cartilages are united by ligaments and provide surfaces to which the muscles attach. Because muscles and bones must combine to allow body movement, these two systems sometimes are collectively referred to as the *locomotor system*.

The adult human skeleton is divided into the **axial skeleton** and the **appendicular skeleton**.

Axial Skeleton

The **axial** (*ak'-se-al*) skeleton includes all bones that lie on or near the central axis of the body (Table 1.1). The adult axial skeleton consists of **80 bones** and includes the skull, vertebral column, ribs, and sternum (the dark-shaded regions of the body skeleton in Fig. 1.12).

TABLE 1.1 ADULT AXIAL SKELETON

Skull	Cranium	8
	Facial bones	14
Hyoid		1
Auditory ossicles (3 small bones in each ear)		6
Vertebral column	Cervical	7
	Thoracic	12
	Lumbar	5
	Sacral	1
	Coccyx	1
Thorax	Sternum	1
	Ribs	24
Total bones in adult axial skeleton		80

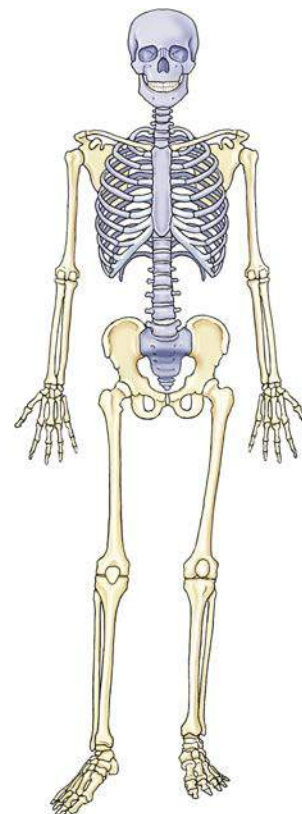


Fig. 1.12 Axial skeleton—80 bones.

Appendicular Skeleton

The second division of the skeleton is the **appendicular** (*ap"-en-dik'-u-lar*) portion. This division consists of all bones of the upper and lower limbs (extremities) and the shoulder and pelvic girdles (the dark-shaded regions in Fig. 1.13). The appendicular skeleton attaches to the axial skeleton. The adult appendicular skeleton comprises **126 separate bones** (Table 1.2).

TABLE 1.2 ADULT APPENDICULAR SKELETON

Shoulder girdles	Clavicles	2
	Scapula (scapulae)	2
Upper limbs	Humerus (humeri)	2
	Ulna (ulnae)	2
	Radius (radii)	2
	Carpals	16
	Metacarpals	10
	Phalanges	28
Pelvic girdle	Hip bones (innominate bones)	2
Lower limbs	Femur (femora)	2
	Tibia	2
	Fibula (fibulae)	2
	Patella (patellae)	2
	Tarsals	14
	Metatarsals	10
	Phalanges	28
Total bones in adult appendicular skeleton		126
Entire number of separate bones in adult skeleton ^a		206

^aThis includes the two sesamoid bones anterior to the knees: the right and left patellae.

Sesamoid Bones

A sesamoid bone is a special type of small, oval-shaped bone that is embedded in certain tendons (most often near joints). Although sesamoid bones are present even in a developing fetus, they are not counted as part of the normal axial or appendicular skeleton except for the two patellae, the largest sesamoid bones. The other most common sesamoid bones are located in the posterior foot at the base of the first toe (Figs. 1.14 and 1.15).

In the upper limb, sesamoid bones are found most commonly in tendons near the anterior (palmar) surface of the hand at the base of the thumb. Others may be found in tendons of other upper or lower limb joints.

Sesamoid bone may be fractured by trauma; sesamoid bones can be demonstrated radiographically or by computed tomography (CT).

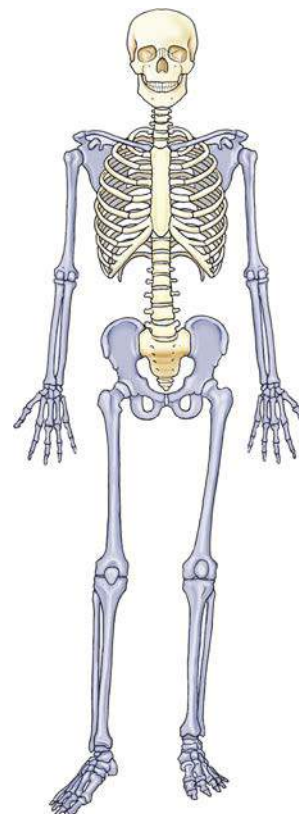
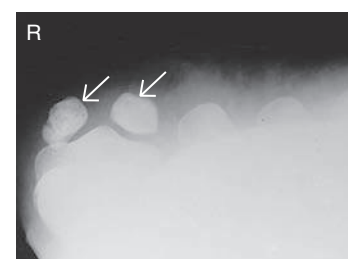
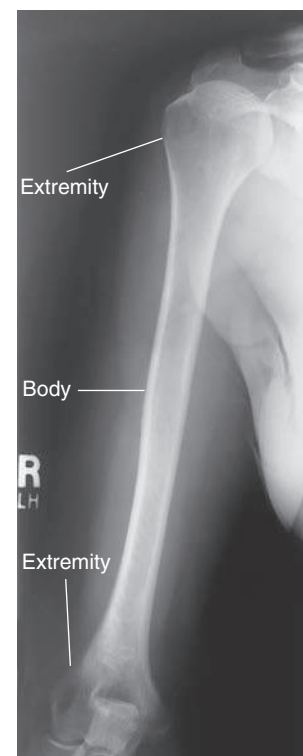
CLASSIFICATION OF BONES

Each of the 206 bones of the body can be classified according to shape as follows:

- Long bones
- Short bones
- Flat bones
- Irregular bones

Long Bones

Long bones consist of a **body** and **two ends** or **extremities**. Long bones are found only in the appendicular skeleton. (Fig. 1.16 is a radiograph of a humerus, a typical long bone of the upper arm.)

**Fig. 1.13** Appendicular skeleton—126 bones.**Fig. 1.14** Sesamoid bones on the posterior base of the first toe.**Fig. 1.15** Sesamoid bones. Tangential projection (base of first toe).**Fig. 1.16** Long bone (humerus).

Composition The outer shell of most bones is composed of hard or dense bone tissue known as **compact bone**, or **cortex**, meaning an external layer. Compact bone has few intercellular empty spaces and serves to protect and support the entire bone.

The **body** (older term is **shaft**) contains a thicker layer of compact bone than is found at the ends, to help resist the stress of the weight placed on them.

Inside the shell of compact bone and especially at both ends of each long bone is found **spongy**, or **cancellous**, bone. Cancellous bone is highly porous and usually contains red bone marrow, which is responsible for the production of red blood cells.

The body of a long bone is hollow. This hollow portion is known as the **medullary** (*med'-u-lar'-e*) **cavity**. In adults, the medullary cavity usually contains fatty yellow marrow. A dense fibrous membrane, the **periosteum** (*per'-e-os'-te-am*), covers bone except at the articulating surfaces. The articulating surfaces are covered by a layer of **hyaline cartilage** (Fig. 1.17).

Hyaline (*hi'-ah-lin*), meaning glassy or clear, is a common type of cartilage or connecting tissue. Its name comes from the fact that it is not visible with ordinary staining techniques, and it appears "clear" or glassy in laboratory studies. It is present in many places, including within the covering over ends of bones, where it is called **articular cartilage**.

The **periosteum** is essential for bone growth, repair, and nutrition. Bones are richly supplied with blood vessels that pass into them from the periosteum. Near the center of the body of long bones, a **nutrient artery** passes obliquely through the compact bone via a **nutrient foramen** into the medullary cavity.

Short Bones

Short bones are roughly cuboidal and are found only in the wrists and ankles. Short bones consist mainly of cancellous tissue with a thin outer covering of compact bone. The eight **carpal bones** of each wrist (Fig. 1.18) and the seven **tarsal bones** of each foot are short bones.

Flat Bones

Flat bones consist of two plates of compact bone with cancellous bone and bone marrow between them. Examples of flat bones are the bones that make up the **calvaria** (skull cap) (Fig. 1.19), **sternum**, **ribs**, and **scapulae**.

The narrow space between the two layers of compact bone of flat bones within the cranium is known as the **diploe** (*dip'-lo-e*). Flat bones provide protection for interior contents and broad surfaces for muscle attachment.

Irregular Bones

Bones that have peculiar shapes are lumped into one final category—irregular bones. **Vertebrae** (Fig. 1.20), **facial bones**, **bones of the base of the cranium**, and **bones of the pelvis** are examples of irregular bones.

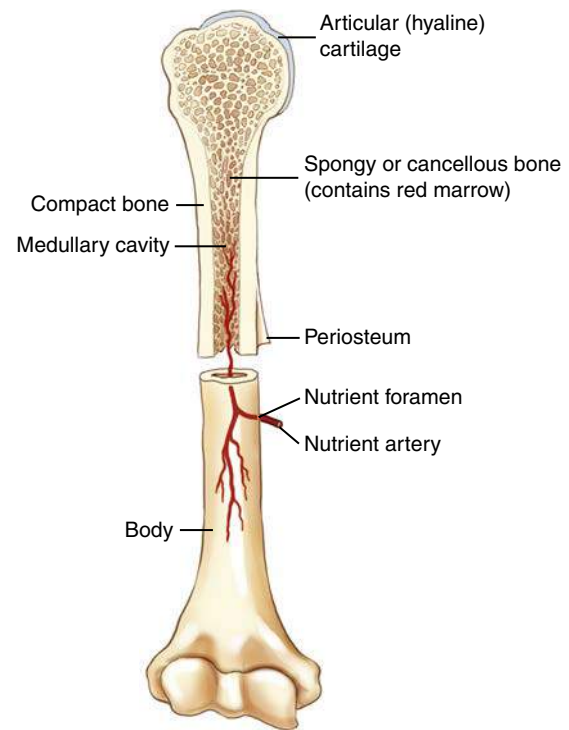


Fig. 1.17 Long bone.



Fig. 1.18 Short bones (carpals).



Fig. 1.19 Flat bones (calvaria).



Fig. 1.20 Irregular bone (vertebra).

DEVELOPMENT OF BONES

The process by which bones form within the body is known as **ossification** (*os-i-fi-ka-shun*). The embryonic skeleton is composed of fibrous membranes and hyaline cartilage. Ossification begins at about the sixth embryonic week and continues until adulthood.

Blood Cell Production

In adults, **red blood cells (RBCs)** are produced by the red bone marrow of certain flat and irregular bones such as the **sternum**, **ribs**, **vertebrae**, and **pelvis**, as well as the ends of the long bones.

Bone Formation

Two types of bone formation are known. When bone replaces membranes, the ossification is called **intramembranous** (*in-tra-mem'-brah-nus*). When bone replaces cartilage, the result is **endochondral** (*en-do-kon'-dral*) (intracartilaginous) ossification.

Intramembranous Ossification Intramembranous ossification occurs rapidly and takes place in bones that are needed for protection, such as sutures of the flat bones of the calvaria (skullcap), which are centers of growth in early bone development.

Endochondral Ossification Endochondral ossification, which is much slower than intramembranous ossification, occurs in most parts of the skeleton, especially in the long bones.

Primary and Secondary Centers of Endochondral Ossification (Fig. 1.21)

The first center of ossification, which is called the **primary center**, occurs in the midbody area. This primary center of ossification in growing bones is called the **diaphysis** (*di-af-i-sis*). This becomes the **body** in a fully developed bone.

Secondary centers of ossification appear near the ends of the limbs of long bones. Most secondary centers appear after birth, whereas most primary centers appear before birth. Each secondary center of ossification is called an **epiphysis** (*e-pif-i-sis*). Epiphyses of the distal femur and the proximal tibia are the first to appear and may be present at birth in a term newborn. Cartilaginous plates, called **epiphyseal plates**, are found between the metaphysis and each epiphysis until skeletal growth is complete. The **metaphysis** is the wider portion of a long bone adjacent to the epiphyseal plate. The metaphysis is the area where bone growth in length occurs. Growth in the length of bones results from a longitudinal increase in these epiphyseal cartilaginous plates. This is followed by progressive ossification through endochondral bone development until all the cartilage has been replaced by bone, at which time growth of the skeleton is complete. This process of epiphyseal fusion of the long bones occurs progressively from the age of puberty to **full maturity**, which is **between the ages of 20 and 25 years**.¹ However, the time for each bone to complete growth varies for different regions of the body. On average, the female skeleton matures more quickly than the male skeleton. Also, geography, socioeconomic, genetic factors, and disease affect epiphyseal fusion.¹

Fig. 1.22 shows a radiograph of the knee region of a 6-year-old child. Primary and secondary centers of endochondral ossification or bone growth are well demonstrated and labeled.

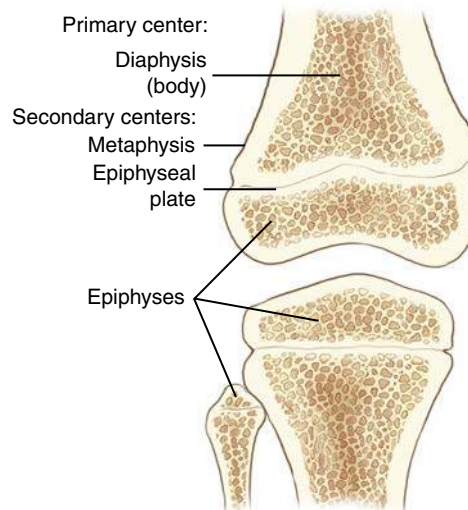


Fig. 1.21 Endochondral ossification.

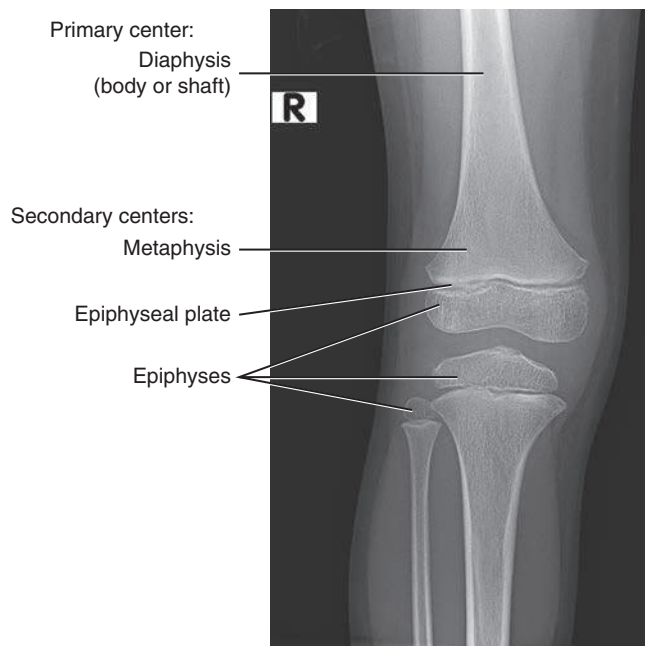


Fig. 1.22 Knee region (6-year-old child).

Arthrology (Joints)

The study of joints or articulations is called **arthrology**. It is important to understand that movement does not occur in all joints. The first two types of joints to be described are immovable joints and only slightly movable joints, which are held together by several fibrous layers, or cartilage. These joints are adapted for growth rather than for movement.

CLASSIFICATION OF JOINTS

Functional

Joints may be classified according to their function in relation to their mobility or lack of mobility as follows:

- **Synarthrosis** (*sin'-ar-thro'-sis*)—immovable joint
- **Amphiarthrosis** (*am'-fe-ar-thro'-sis*)—joint with limited movement
- **Diarthrosis** (*di'-ar-thro'-sis*)—freely movable joint

Structural

The primary classification system of joints, described in *Gray's Anatomy*² and used in this textbook, is a **structural classification** based on the **three types of tissue that separate the ends of bones** in the different joints. These three classifications by tissue type, along with their subclasses, are as follows:

1. Fibrous (*fi'-brus*) joints
 - Syndesmosis (*sin'-des-mo'-sis*)
 - Suture (*su'-tur*)
 - Gomphosis (*gom-fo'-sis*)
2. Cartilaginous (*kar'-ti-laj'-i-nus*) joints
 - Symphysis (*sim'-fi-sis*)
 - Synchondrosis (*sin'-kon-dro'-sis*)
3. Synovial (*si-no'-ve-al*) joints

Fibrous Joints

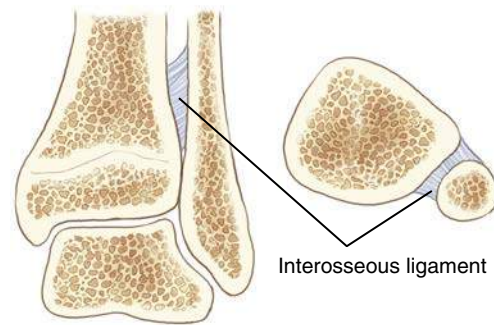
Fibrous joints lack a joint cavity. The adjoining bones, which are nearly in direct contact with each other, are held together by fibrous connective tissue. Three types of fibrous joints are syndesmoses, which are slightly movable; sutures, which are immovable; and gomphoses, a unique type of joint with only very limited movement (Fig. 1.23).

Syndesmoses¹ Syndesmoses are fibrous types of articulations that are held together by interosseous ligaments and slender fibrous cords that allow slight movement at these joints. Some earlier references restricted the fibrous syndesmosis classification to the inferior tibiofibular joint. However, fibrous-type connections also may occur in other joints, such as the sacroiliac junction with its massive interosseous ligaments that in later life become almost totally fibrous articulations. The carpal and tarsal joints of the wrist and foot also include interosseous membranes that can be classified as syndesmosis-type joints that are only slightly movable, or amphiarthrodial.

Sutures Sutures are found only between bones in the skull. These bones make contact with one another along interlocking or serrated edges and are held together by layers of fibrous tissue, or sutural ligaments. Movement is very limited at these articulations; in adults, these are considered **immovable**, or **synarthrodial**, joints.

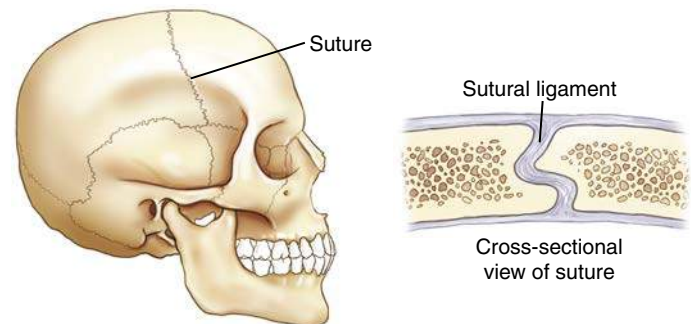
Limited expansion- or compression-type movement at these sutures can occur in the infant skull (e.g., during the birth process). However, by adulthood, active bone deposition partially or completely obliterates these suture lines.

Gomphoses A **gomphosis** joint is the third unique type of fibrous joint, in which a conical process is inserted into a socket-like portion of bone. This joint or fibrous union—which, strictly speaking, does not occur between bones but between the roots of the teeth and the alveolar sockets of the mandible and the maxillae—is a specialized type of articulation that allows only very limited movement.



Distal tibiofibular joint

1. Syndesmosis—Amphiarthrodial (slightly movable)



Skull suture

2. Suture—Synarthrodial (immovable)



Roots of teeth

3. Gomphosis—Amphiarthrodial (only limited movement)

Fig. 1.23 Fibrous joints—three types.

Cartilaginous Joints

Cartilaginous joints also lack a joint cavity, and the articulating bones are held together tightly by cartilage. Similar to fibrous joints, cartilaginous joints allow little or no movement. These joints are synarthrodial or amphiarthrodial and are held together by two types of cartilage: symphyses and synchondroses (Fig. 1.24).

Symphyses The essential feature of a symphysis is the **presence of a broad, flattened disk of fibrocartilage** between two contiguous bony surfaces. These fibrocartilage disks form relatively thick pads that are capable of being compressed or displaced, allowing some movement of these bones, which makes these joints **amphiarthrodial** (slightly movable).

Examples of such symphyses are the intervertebral disks (between bodies of the vertebrae), between the manubrium (upper portion) and body of the sternum, and the symphysis pubis (between the two pubic bones of the pelvis).

Synchondroses A typical synchondrosis is a **temporary form of joint** wherein the connecting **hyaline cartilage** (which on long bones is called an *epiphyseal plate*) is converted into bone at adulthood. These temporary types of growth joints are considered **synarthrodial** or immovable.

Examples of such joints are the epiphyseal plates between the epiphyses and the metaphysis of long bones and at the three-part union of the pelvis, which forms a cup-shaped acetabulum for the hip joint.

Synovial Joints

Synovial joints are freely movable joints, most often found in the upper and lower limbs, which are characterized by a **fibrous capsule that contains synovial fluid** (Fig. 1.25). The ends of the bones that make up a synovial joint may make contact but are completely separate and contain a joint space or cavity, which allows for a wide range of movement at these joints. Synovial joints are generally **diarthrodial**, or freely movable. (Exceptions include the sacroiliac joints of the pelvis, which are amphiarthrodial, or slightly movable.)

The exposed ends of these bones contain thin protective coverings of **articular cartilage**. The **joint cavity**, which contains a viscous lubricating **synovial fluid**, is enclosed and surrounded by a **fibrous capsule** that is reinforced by strengthening **accessory ligaments**. These ligaments limit motion in undesirable directions. The inner surface of this fibrous capsule is thought to secrete the lubricating synovial fluid.

Movement Types of Synovial Joints There are a considerable number and variety of synovial joints, and they are grouped according to the **seven types of movement** that they permit. These are listed in order from the least to the greatest permitted movement.

NOTE: The preferred name is listed first, followed by a synonym in parentheses. (This practice is followed throughout this textbook.)

Plane (gliding) joints This type of synovial joint permits the least movement, which, as the name implies, is a **sliding or gliding motion between the articulating surfaces**.

Examples of plane joints are the **intermetacarpal**, **carpometacarpal**, and **intercarpal** joints of the hand and wrist (Fig. 1.26). The right and left lateral **atlantoaxial joints** between C1 and C2 vertebrae are also classified as plane, or gliding, joints; they permit some rotational movement between these vertebrae, as is described in Chapter 8.

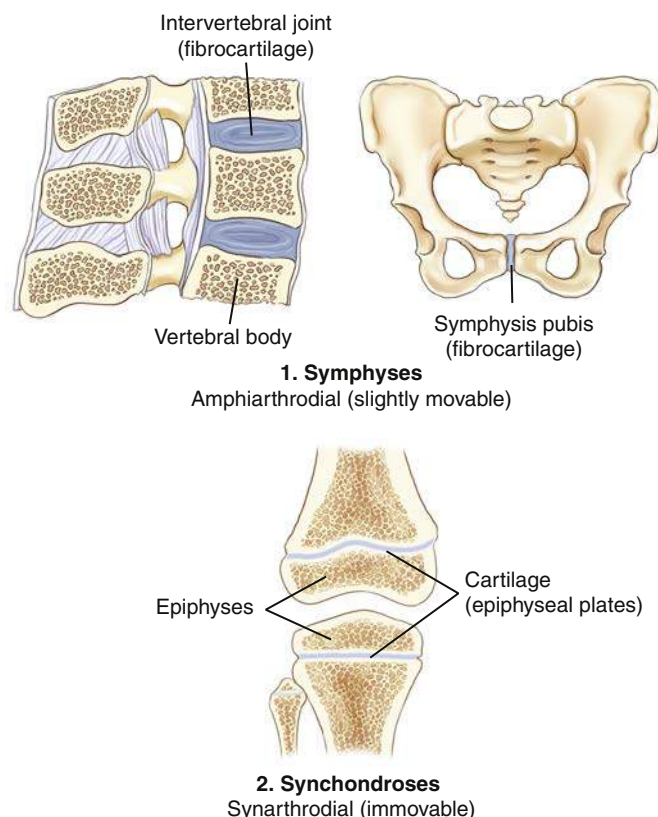


Fig. 1.24 Cartilaginous joints—two types.

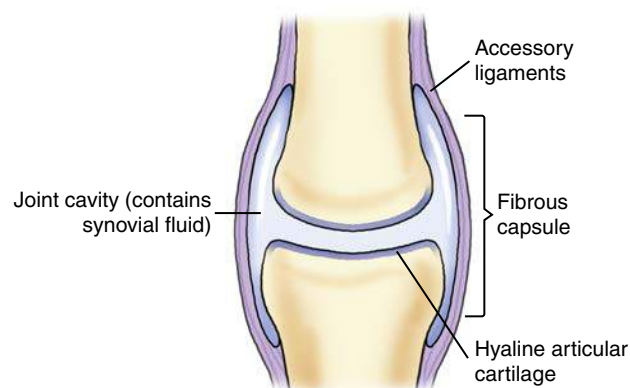


Fig. 1.25 Synovial joints—diarthrodial (freely movable).

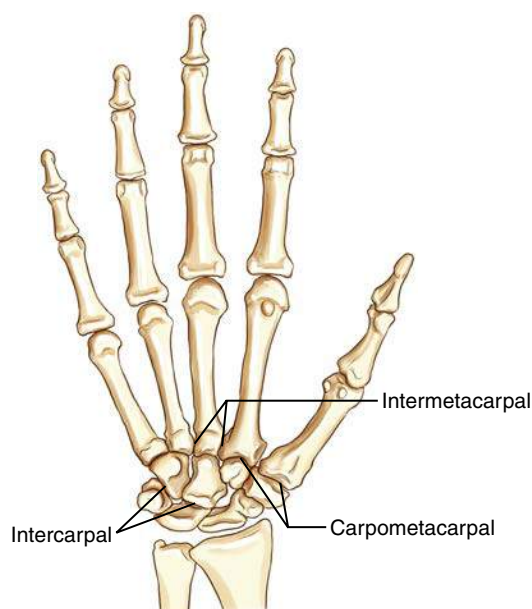


Fig. 1.26 Plane (gliding) joints.

Ginglymus (hinge) joints The articular surfaces of ginglymi, or ginglymus (*jīn'-gli-mus*) joints, are molded to each other in such a way that they permit **flexion and extension movements** only. The articular fibrous capsule on this type of joint is thin on surfaces where bending takes place, but strong collateral ligaments firmly secure the bones at the lateral margins of the fibrous capsule.

Examples of ginglymi include the **interphalangeal joints** of fingers and toes and the **elbow joint** (Fig. 1.27).

Pivot (trochoid) joints The pivot or trochoid (*tro'-koid*) joint is formed by a bony, pivot-like process that is surrounded by a ring of ligaments or a bony structure or both. This type of joint allows **rotational movement** around a single axis.

Examples of pivot joints are the **proximal and distal radioulnar joints** of the forearm, which demonstrate this pivot movement during rotation of the hand and wrist.

Another example is the joint **between the first and second cervical vertebrae**. The odontoid process (dens) of the axis (C2) forms the pivot, and the anterior arch of the atlas (C1), combined with ligaments, forms the ring (Fig. 1.28).

Ellipsoid (condylar) joints In the ellipsoid (*e-lip'-soid*) joint, movement occurs primarily in one plane and is combined with a slight degree of rotation at an axis at right angles to the primary plane of movement. The rotational movement is limited by associated ligaments and tendons.

This type of joint allows primarily four directional movements: **flexion and extension** and **abduction and adduction**. **Circumduction** movement also occurs; this results from conelike sequential movements of flexion, abduction, extension, and adduction.

Examples of ellipsoidal joints include the metacarpophalangeal joints of the fingers, the radiocarpal (wrist joint), and the metatarsophalangeal joints of the toes (Fig. 1.29).

Saddle (sellar) joints The term saddle, or *sellar* (*sel'-ar*), describes this joint structure well in that the ends of the bones are shaped concave-convex and are positioned opposite each other (Fig. 1.30). (Two saddle-like structures fit into each other.)

Movements of this biaxial type of saddle joint are the same as for ellipsoidal joints—**flexion, extension, adduction, abduction, and circumduction**.

The best example of a true *saddle* joint is the **first carpometacarpal joint** of the thumb. Other sellar joints include the ankle and the calcaneocuboid joints. Although the ankle joint was classified as a ginglymus in earlier references, current references classify it as a saddle joint.³

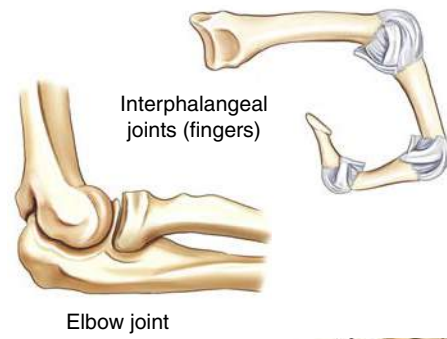


Fig. 1.27 Ginglymus (hinge) joints.

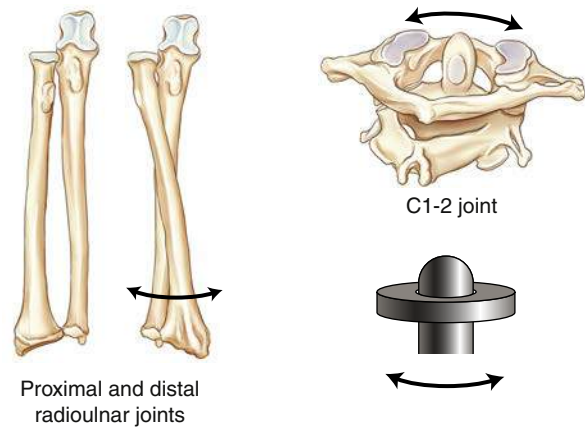


Fig. 1.28 Pivot (trochoid) joints.

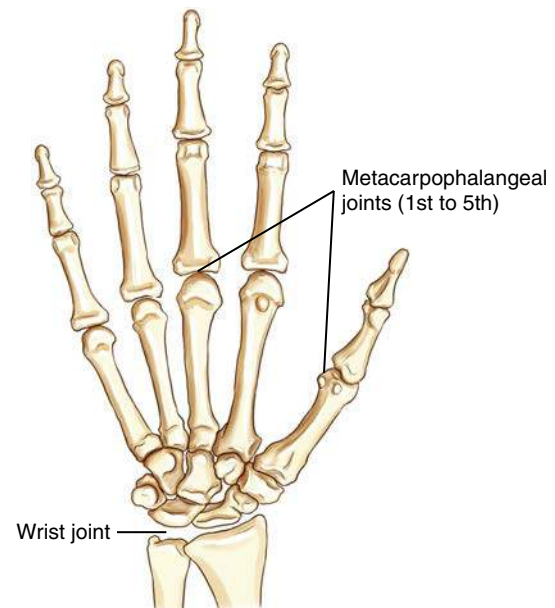


Fig. 1.29 Ellipsoid (condylar) joints.

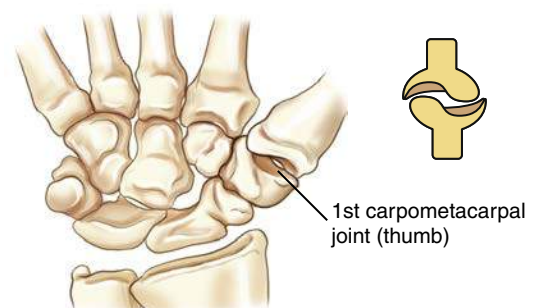


Fig. 1.30 Saddle (sellar) joints.

Ball-and-socket (spheroidal) joints The ball-and-socket or spheroidal (*sfe'-roid-el*) joint allows for the greatest freedom of motion. The distal bone (humerus) that makes up the joint is capable of motion around an almost indefinite number of axes, with one common center.

The greater the depth of the socket, the more limited is the movement. However, the deeper joint is stronger and more stable. For example, the hip joint is a much stronger and more stable joint than the shoulder joint, but the range of movement is more limited in the hip.

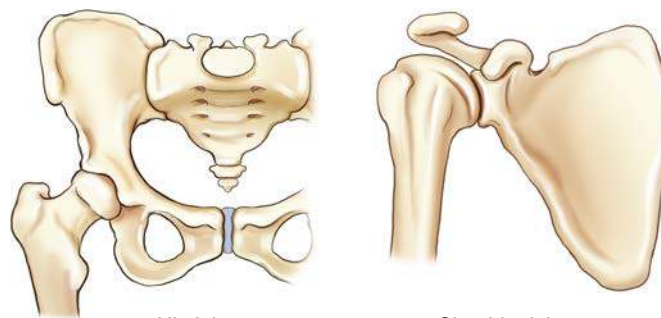
Movements of ball-and-socket joints include flexion, extension, abduction, adduction, circumduction, and medial and lateral rotation.

Two examples of ball-and-socket joints are the **hip joint** and the **shoulder joint** (Fig. 1.31).

Bicondylar joints³ Bicondylar joints usually provide movement in a single axis, such as flexion and extension. They can permit limited rotation. Bicondylar joints are formed by two convex condyles, which may be encased by a fibrous capsule.

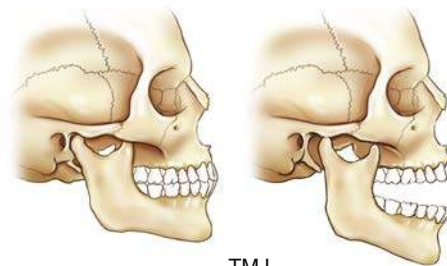
Two examples of bicondylar joints are the knee (formerly classified as ginglymus) and the temporomandibular joint (TMJ) (Fig. 1.32).

See Table 1.3 for a summary of joint classification.



Hip joint

Shoulder joint

Fig. 1.31 Ball-and-socket (spheroidal) joints.

TMJ



Knee

Fig. 1.32 Bicondylar joints.**TABLE 1.3 SUMMARY OF JOINT CLASSIFICATION**

JOINT CLASSIFICATION	MOBILITY CLASSIFICATION	MOVEMENT TYPES	MOVEMENT DESCRIPTION	EXAMPLES
Fibrous Joints				
Syndesmoses	Amphiarthrodial (slightly movable)	—	—	Distal tibiofibular, sacroiliac, carpal, and tarsal joints
Sutures	Synarthrodial (immovable)	—	—	Skull sutures
Gomphoses	Very limited movement	—	—	Areas around roots of teeth
Cartilaginous Joints				
Symphyses	Amphiarthrodial (slightly movable)	—	—	Intervertebral disks Symphysis pubis
Synchondroses	Synarthrodial (immovable)	—	—	Epiphyseal plates of long bones and between the three parts of the pelvis
Synovial joints	Diarthrodial (freely movable) except for the sacroiliac joints (synovial joints with only very limited motion [amphiarthrodial])	Plane (gliding)	Sliding or gliding	Intermetacarpal, intercarpal, and carpometacarpal joints, C1 on C2 vertebrae
		Ginglymi (hinge)	Flexion and extension	Interphalangeal joints of fingers, toes, and elbow joints
		Pivot (trochoid)	Rotational	Proximal and distal radioulnar and between C1 and C2 vertebrae (atlantoaxial joint)
		Ellipsoid (condylar)	Flexion and extension Abduction and adduction Circumduction	Metacarpophalangeal and wrist joints
		Saddle (sellar)	Flexion and extension Abduction and adduction Circumduction	First carpometacarpal joint (thumb), ankle, and calcaneocuboid joints
		Ball and socket (spheroidal)	Flexion and extension Abduction and adduction Circumduction Medial and lateral rotation	Hip and shoulder joints
		Bicondylar	Movement primarily along one axis with some limited rotation	Knee and temporomandibular joints

NOTE: Arthrology is the study of joints. The nomenclature for joints described in this chapter will be used in subsequent chapters throughout the text.

Body Habitus

Body habitus is generally defined as the build, physique, and general shape of the human body. The size, dimensions, and shape of the patient's body affect the positioning of specific regions of the body such as the respiratory, gastrointestinal, and biliary systems.

Body habitus is classified into four general body styles:

1. **Sthenic:** Approximately 50% of the population falls into this category. For the purpose of radiographic positioning, sthenic body styles are considered average in shape and internal organ location (Fig. 1.33).
2. **Hyposthenic:** A body style which is more slender than the sthenic body habitus. Approximately 35% of the population is classified as hyposthenic (Fig. 1.34).
3. **Hypersthenic:** A body style which has a broad frame as compared to the sthenic body habitus. Approximately 5% of the population is classified as hypersthenic (Fig. 1.35).
4. **Asthenic:** Approximately 10% of the population is very thin or slender with a long and narrow body build. More slight in stature than the hyposthenic patient.



Fig. 1.33 Sthenic body habitus.



Fig. 1.34 Hyposthenic body habitus.



Fig. 1.35 Hypersthenic body habitus.

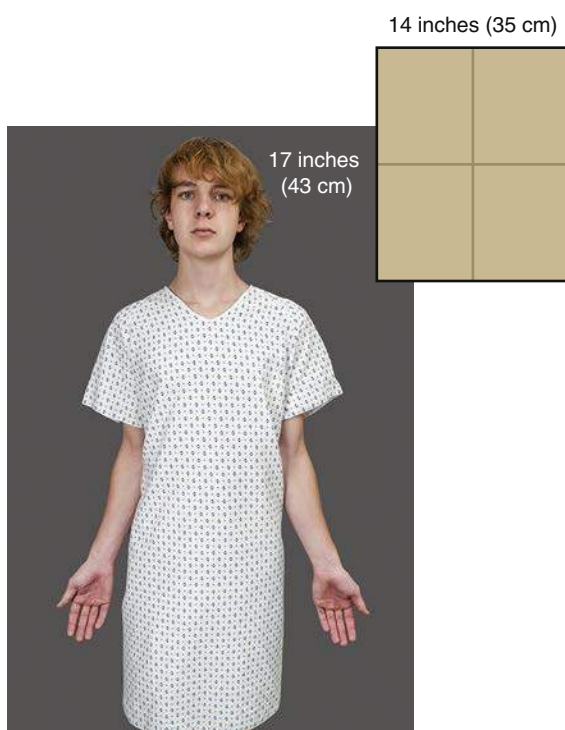


Fig. 1.36 Portrait alignment of image receptor.



Fig. 1.37 Landscape alignment of image receptor Ditto.

IMPACT OF BODY HABITUS ON RADIOGRAPHIC POSITIONING

The technologist must consider the patient's body habitus and alter centering and image receptor (IR) placement accordingly. This is especially true during adult chest radiography (described in Chapter 2). For the hyposthenic and asthenic patient, the image receptor is typically placed in **portrait** (lengthwise) alignment because the lungs are generally longer than those of the hypersthenic patient (Fig. 1.36). For the hypersthenic patient, the image receptor is typically placed in **landscape** (crosswise) alignment because the lungs are generally shorter in length and more broad in width than those of the hyposthenic or asthenic patient (Fig. 1.37). IR placement for the sthenic adult patient may be portrait or landscape, depending on the person's age, height, and even pathology. Other anatomic regions also are affected by body habitus. This will be discussed further in Chapter 12, Biliary Tract and Upper Gastrointestinal System.

POSITIONING TERMINOLOGY

1 Radiographic positioning refers to the study of patient positioning performed for **radiographic demonstration or visualization of specific body parts on image receptors**. The radiologic technologist must clearly understand the correct use of positioning terminology. This section lists, describes, and illustrates the commonly used terms consistent with the positioning and projection terminology as approved and published by the American Registry of Radiologic Technologists (ARRT).⁴

Throughout this text, named positions (i.e., with the proper name of the person who first described a specific position or procedure) are referred to as **methods**, such as the Towne, Waters, and Caldwell methods. The ARRT concurs regarding the use of the named method in parentheses after the projection or position term. The description of radiographic positions by the proper name method is becoming less common.

General Terms

Radiograph (*ra'-de-o-graf*): (1) An image of a patient's anatomic part(s), as produced by the action of x-rays on an image receptor (Fig. 1.38). If the radiograph is produced with the use of traditional film-screen (analog) technology, the image is captured and displayed on film; if the radiograph is produced via digital technology, the image is viewed and stored on display monitors. (2) The term *radiograph* refers to the recording medium *and* the image.

Radiography (*ra'-de-og'-rah-fe*): The process and procedures of producing a radiograph.

Image receptor (IR): The device that responds to the ionizing radiation to create the radiographic image after it exits the patient; refers to both analog (film-based) cassettes and digital acquisition devices.

Central ray (CR): Refers to the centermost portion of the x-ray beam emitted from the x-ray tube—the portion of the x-ray beam that has the least divergence.

Radiographic Examination or Procedure A radiologic technologist is shown positioning a patient for a routine chest examination or procedure (Fig. 1.39). A radiographic examination involves five general functions:

1. Positioning of body part and alignment with the IR and CR
2. Application of radiation protection measures and devices
3. Selection of exposure factors (radiographic technique)
4. Instructions to the patient related to respiration (breathing) and initiation of the x-ray exposure
5. Processing of the IR (analog) [chemical processing] or digital processing systems

Anatomic Position The *anatomic* (*an'-ah-tom'-ik*) position is a reference position that defines specific surfaces and planes of the body. It also defines anatomic directional terms such as anterior, posterior, medial, lateral, superior, and inferior regions of the body. The anatomic position is an upright position with arms abducted slightly (down), hands by sides with palms forward, and head and feet together and directed straight ahead (Fig. 1.40).

Viewing Radiographs A general rule in viewing radiographs is to display them so that the **patient is facing the viewer**, with the patient in the **anatomic position**.



Fig. 1.38 Chest radiograph.



Fig. 1.39 Radiographic examination.



Fig. 1.40 Anatomic position.

Body Planes, Sections, and Lines (Fig. 1.41)

Positioning terms that describe CR angles or relationships between body parts often are related to **imaginary planes** that pass through the body in the **anatomic position**. The study of CT, MRI (magnetic resonance imaging), and sonography (diagnostic medical ultrasound) emphasizes sectional anatomy, which also involves the primary body planes and sections as described subsequently.

PLANE: STRAIGHT LINE SURFACE CONNECTING TWO POINTS

Four common planes used in medical imaging are the sagittal plane, coronal plane, horizontal (axial) plane, and oblique plane.

Sagittal Plane

A sagittal (*saj'-i-tal*) plane is any **longitudinal** plane that divides the body into **right and left parts**.

The **midsagittal plane**, sometimes called the **median plane**, is a midline sagittal plane that divides the body into **equal right and left parts**. It passes approximately through the sagittal suture of the skull. Any plane parallel to the midsagittal or median plane is called a **sagittal plane**.

Coronal Plane

A coronal (*ko-ro'-nal*) plane is any **longitudinal** plane that divides the body into **anterior and posterior parts**.

The **midcoronal plane** divides the body into approximately **equal anterior and posterior parts**. It is called a coronal plane because it passes approximately through the coronal suture of the skull. Any plane parallel to the midcoronal or frontal plane is called a **coronal plane**.

Horizontal (Axial) Plane

A horizontal (axial) plane is any **transverse** plane that passes through the body at **right angles to a longitudinal plane**, dividing the body into superior and inferior portions.

Oblique Plane

An oblique plane is a **longitudinal** or **transverse** plane that is at an angle or slant and is **not parallel** to the sagittal, coronal, or horizontal plane.

SECTIONAL IMAGE OF BODY PART

Longitudinal Sections—Sagittal, Coronal, and Oblique

These sections or images run **lengthwise** in the direction of the long axis of the body or any of its parts, regardless of the position of the body (erect or recumbent). Longitudinal sections or images may be taken in the **sagittal, coronal, or oblique plane**.

Transverse or Axial Sections (Cross-Sections)

Sectional images are at right angles along any point of the longitudinal axis of the body or its parts (Fig. 1.42)

Sagittal, Coronal, and Axial Images

CT, magnetic resonance imaging (MRI), and sonography images are obtained in these three common orientations or views. These common orientations are sagittal, coronal, and transverse (axial). (MRI sectional images are shown in Figs. 1.43 through 1.45.)

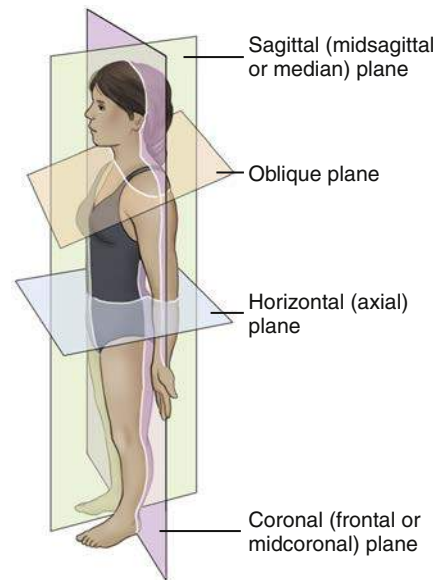
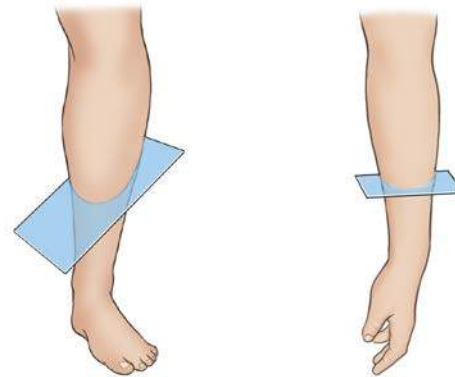


Fig. 1.41 Sagittal, coronal, oblique, and horizontal body planes.



Oblique transverse plane or section of leg

Transverse (axial or cross-sectional) plane or section of arm

Fig. 1.42 Transverse and oblique sections of body parts.



Fig. 1.43 Sagittal image.

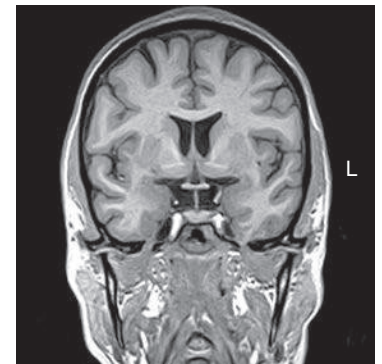


Fig. 1.44 Coronal image.

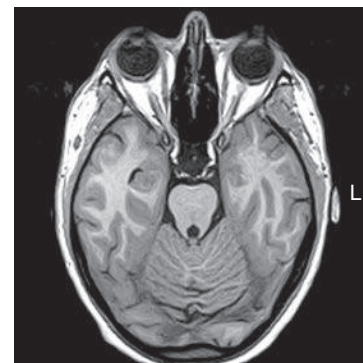


Fig. 1.45 Transverse (axial) image.

PLANES OF THE SKULL (FIG. 1.46)

Base Plane of Skull

This precise transverse plane is formed by connecting the lines from the infraorbital margins (inferior edge of bony orbits) to the superior margin of the external auditory meatus (EAM), the external opening of the ear. This sometimes is called the **Frankfort horizontal plane**,¹ as used in orthodontics and cranial topography to measure and locate specific cranial points or structures.

Occlusal Plane

This horizontal plane is formed by the biting surfaces of the upper and lower teeth with jaws closed (used as a reference plane of the head for cervical spine and skull radiography).

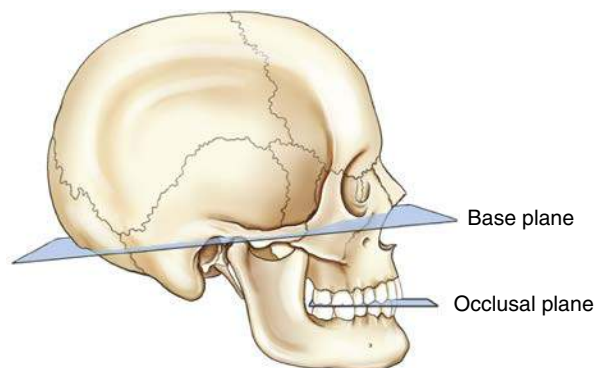


Fig. 1.46 Planes of skull.

Body Surfaces and Parts**TERMS FOR THE BACK AND FRONT PORTIONS OF THE BODY****Posterior or Dorsal (FIG. 1.47)**

Posterior (*pos-te'-re-or*) or dorsal (*dor'-sal*) refers to the **back half** of the patient, or the part of the body seen when the person is viewed from the back; includes the bottoms of the feet and the backs of the hands as demonstrated in the anatomic position.

Anterior or Ventral

Anterior (*an-te'-re-or*) or ventral (*ven'-tral*) refers to the **front half** of the patient, or the part seen when viewed from the front; includes the tops of the feet and the fronts or palms of the hands in the anatomic position.

TERMS FOR SURFACES OF THE HANDS AND FEET

Three terms are used in radiography to describe specific surfaces of the upper and lower limbs.

Plantar

Plantar (*plan'-tar*) refers to the **sole** or **posterior** surface of the foot.

Dorsal

Foot Dorsal (*dor'-sal*) refers to the **top** or **anterior** surface of the foot (dorsum pedis).

Hand Dorsal also refers to the **back** or **posterior** aspect of the hand (dorsum manus) (Fig. 1.48).

NOTE: The term **dorsum** (or **dorsal**) in general refers to the vertebral or posterior part of the body. However, when used in relationship with the foot, *dorsum* (dorsum pedis) specifically refers to the **upper surface**, or **anterior aspect**, of the foot opposite the sole, whereas for the hand (dorsum manus), it refers to the back or posterior surface opposite the palm.¹

Palmar

Palmar (*pal'-mar*) refers to the **palm of the hand**; in the anatomic position, the same as the **anterior or ventral** surface of the hand.¹

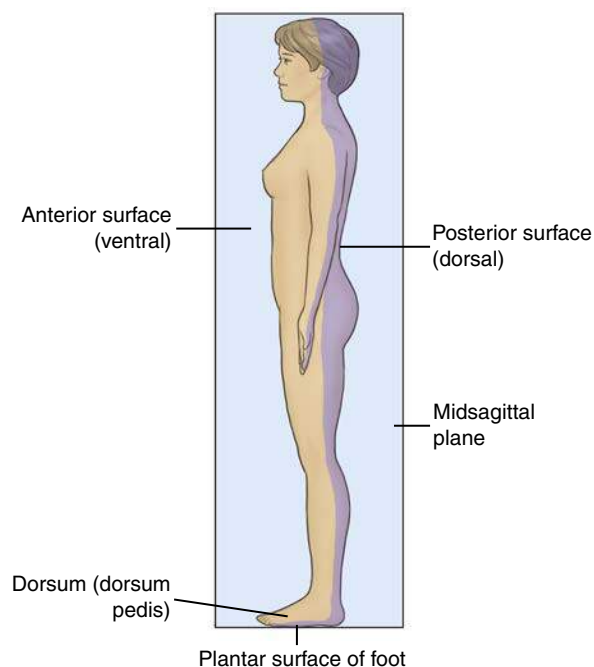


Fig. 1.47 Posterior vs. anterior.

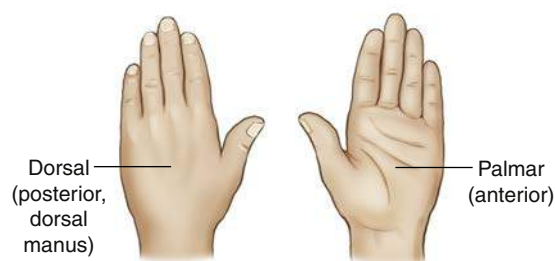


Fig. 1.48 Dorsal and palmar surfaces of hand.

Radiographic Projections

Projection is a positioning term that describes the **direction or path of the CR of the x-ray beam** as it passes through the patient, projecting an image onto the IR. Although the term **position** is used in the clinical setting, the term **projection** is considered to be the most accurate term for describing how the procedure is performed. Therefore, the term **projection** is used most frequently throughout this text.

COMMON PROJECTION TERMS

Posteroanterior (PA) Projection

Posteroanterior (*pos'-ter-o-an-te'-re-or*) (PA) projection refers to a projection of the CR from **posterior to anterior**.

The term *posteroanterior* combines the two terms *posterior* and *anterior* into one word, abbreviated as PA. The CR enters at the posterior surface and exits at the anterior surface (**PA projection**) (Fig. 1.49).

This projection assumes a **true PA** without intentional rotation, which requires the CR to be perpendicular to the coronal body plane and parallel to the sagittal plane, unless some qualifying oblique or rotational term is used to indicate otherwise.

Anteroposterior (AP) Projection

Anteroposterior (*an'-ter-o-pos-te'-re-or*) (AP) projection refers to a projection of the CR from **anterior to posterior**, the opposite of PA. *Anteroposterior* combines the two terms *anterior* and *posterior* into one word.

Anteroposterior describes the direction of travel of the CR, which enters at an anterior surface and exits at a posterior surface (**AP projection**) (Fig. 1.50).

The term assumes a **true AP** without rotation unless a qualifier term also is used, indicating it to be an oblique projection.

AP Oblique Projection

An AP projection of the upper or lower limb that is rotated is called *oblique*. This is not a true AP projection and **must also include a qualifying term** that indicates which way it is rotated, such as medial or lateral rotation (Fig. 1.51). (For an oblique projection of the whole body, see **oblique position** descriptions later in this chapter.) With an AP oblique projection, the CR enters the anterior surface and exits the posterior surface of the body or body part.

PA Oblique Projection

A PA projection of the upper limb with lateral rotation (from PA) is shown in Fig. 1.52. (This is applicable to both upper and lower limbs.) This projection is described as a PA oblique. It **must also include a qualifying term** that indicates which way it is rotated. With a PA oblique projection, the CR enters the posterior surface and exits the anterior surface of the body or body part.

Mediolateral and Lateromedial Projections

A **lateral** projection is described by the **path of the CR**. Two examples are the **mediolateral** projection of the ankle (Fig. 1.53) and the **lateromedial** projection of the wrist (Fig. 1.54). The medial and lateral sides are determined with the patient in the anatomic position. In the case of the mediolateral ankle projection, the CR enters the medial aspect and exits the lateral aspect of the ankle.



Fig. 1.49 PA projection.



Fig. 1.50 AP projection.



Fig. 1.51 AP oblique projection—medial rotation (from AP).



Fig. 1.52 PA oblique projection—lateral rotation (from PA).



Fig. 1.53 Mediolateral projection (ankle).



Fig. 1.54 Lateromedial projection (wrist).

Body Positions

In radiography, the term *position* is used in two ways, first as **general body positions**, as described next, and second as **specific body positions**, which are described in the pages that follow.

GENERAL BODY POSITIONS

The eight most commonly used general body positions in medical imaging are as follows:

1. **Supine** (*soo'-pine*): **Lying on back**, facing upward (Fig. 1.55)
2. **Prone** (*prohn*): **Lying on abdomen**, facing downward (head may be turned to one side) (Fig. 1.56)
3. **Erect** (*e-rectt'*) (upright): An **upright position**, to stand or sit erect
4. **Recumbent** (*re-kum'-bent*) (reclining): **Lying down in any position** (prone, supine, or on side)
 - **Dorsal recumbent**: Lying on back (supine)
 - **Ventral recumbent**: Lying face down (prone)
 - **Lateral recumbent**: Lying on side (right or left lateral)
5. **Trendelenburg**⁵ (*tren-del-en-berg*) position: A recumbent position with the body tilted with the **head lower than the feet** (Fig. 1.57)
6. **Fowler**⁶ (*fow'-ler*) position: A recumbent position with the body tilted with the **head higher than the feet** (Fig. 1.58)
7. **Sims position** (semiprone position): A recumbent oblique position with the patient lying on the **left anterior side**, with the right knee and thigh flexed and the left arm extended down behind the back. A **modified Sims** position as used for insertion of the rectal tube for a barium enema is shown in Fig. 1.59 (demonstrated in Chapter 13).
8. **Lithotomy** (*li-thot'-o-me*) position: A **recumbent** (supine) position with knees and hip flexed and thighs abducted and rotated externally, supported by ankle supports (Fig. 1.60). This position is seen frequently in the surgical suite for certain urinary studies.



Fig. 1.57 Trendelenburg position—head lower than feet.



Fig. 1.58 Fowler position—feet lower than head.

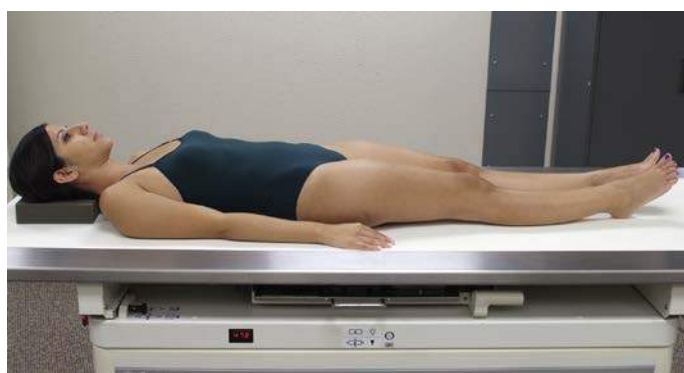


Fig. 1.55 Supine position.

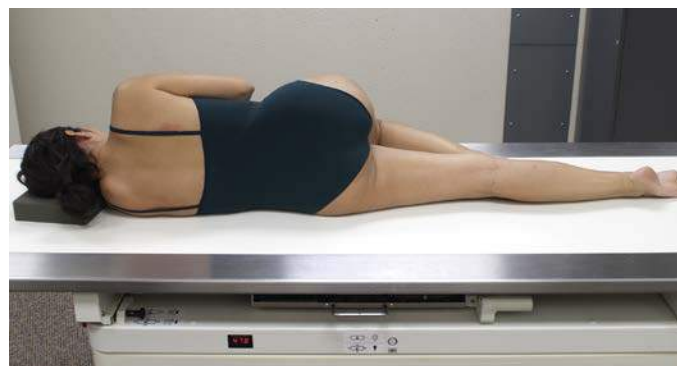


Fig. 1.59 Modified Sims position.



Fig. 1.56 Prone position.



Fig. 1.60 Lithotomy position. (From Chitlik A: Safe positioning for robotic-assisted laparoscopic prostatectomy, *AORN J* 90[1]:39, 2011.)

SPECIFIC BODY POSITIONS

In addition to identifying general body positions, the term *position* is used in radiography to refer to a specific body position described by the body part closest to the IR (oblique and lateral) or by the surface on which the patient is lying (decubitus).

Lateral Position

Lateral (*lat'-er-al*) position refers to the side of, or a side view.

Specific lateral positions are described by the **side of the body closest to the IR** or the **body part from which the CR exits**. A **right lateral** position is shown with the right side of the body closest to the IR in the erect position (Fig. 1.61). Fig. 1.62 demonstrates a recumbent **left lateral** position.

A true lateral position is always 90°, or perpendicular, or at a right angle, to a true AP or PA projection. If it is not a true lateral, it is an oblique position.

Oblique Position⁵

Oblique (*ob-lek'*, or *ob-lik'*)⁷ (*oh bleek'*, or *oh blike'*) position refers to an angled position in which neither the sagittal nor the coronal body plane is perpendicular or at a right angle to the IR.

Oblique body positions of the thorax, abdomen, or pelvis are described by the **side of the body closest to the IR** or the **body part from which the CR exits**.

Left and Right Posterior Oblique (LPO and RPO) Positions

LPO and RPO describe the specific oblique positions in which the **left or right posterior** aspect of the body is closest to the IR. A left posterior oblique (LPO) is demonstrated in both the erect (Fig. 1.63) and recumbent (Fig. 1.64) positions.

The CR exits from the left or right posterior aspect of the body.

NOTE: These also can be referred to as **AP oblique projections** because the CR enters an anterior surface and exits posteriorly. However, this is **not a complete description** and requires a specific position clarifier such as **LPO or RPO position**. Therefore, throughout this text, these body obliques are referred to as **positions** and not **projections**.

Obliques of upper and lower limbs are described correctly as AP and PA oblique but require the use of either **medial** or **lateral rotation** as a qualifier (see Figs. 1.51 and 1.52).

Right and Left Anterior Oblique (RAO and LAO) Positions

RAO and LAO refer to oblique positions in which the **right or left anterior** aspect of the body is closest to the IR and can be erect or recumbent general body positions. (A right anterior oblique [RAO] is shown in both examples (Figs. 1.65 and 1.66).

NOTE: These also can be described as **PA oblique projections** if a position clarifier is added, such as an RAO or LAO **position**.

It is *not* correct to use these oblique terms or the abbreviations LPO, RPO, RAO, or LAO as projections because they do not describe the direction or path of the CR; rather, these are **positions**.



Fig. 1.63 Erect LPO position.



Fig. 1.64 Recumbent LPO position.



Fig. 1.65 Erect RAO position.



Fig. 1.61 Erect R lateral position.

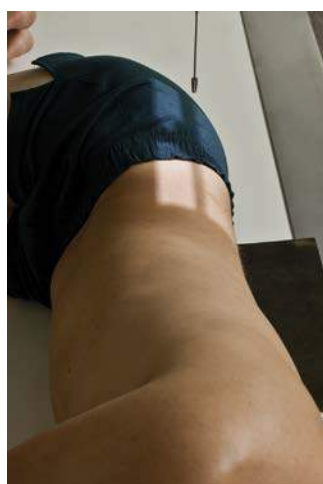


Fig. 1.62 Recumbent L lateral position.



Fig. 1.66 Recumbent RAO position.

Decubitus (Decub) Position

The word **decubitus** (*de-ku'bi-tus*) literally means to lie down, or the position assumed in lying down.

This body position, meaning to **lie on a horizontal surface**, is designated according to the **surface on which the body is resting**. This term describes a patient who is lying on one of the following body surfaces: **back** (dorsal), **front** (ventral), or **side** (right or left lateral).

In radiographic positioning, decubitus is *always* performed with the CR **horizontal**.

Decubitus positions are essential for detecting air-fluid levels or free air in a body cavity such as the chest or abdomen, where the air rises to the uppermost part of the body cavity. Decubitus positions are often performed if the patient cannot assume erect position.

Right or Left Lateral Decubitus Position—AP or PA Projection

In this position, the patient lies on the side, and the x-ray beam is directed horizontally from anterior to posterior (AP) (Fig. 1.67) or from posterior to anterior (PA) (Fig. 1.68).

The AP or PA projection is important as a qualifying term with decubitus positions to denote the direction of the CR.

This position is either a **left lateral decubitus** (see Fig. 1.67) or a **right lateral decubitus** (see Fig. 1.68).

NOTE: The decubitus position is identified according to the dependent side (side down) and the AP or PA projection indication. Example: Left lateral decubitus (PA projection) is with the patient lying on left side facing the image receptor. The CR enters the posterior side and exits the anterior side.

Dorsal Decubitus Position—Left or Right Lateral

In this position, the patient is **lying on the dorsal** (posterior) surface with the **x-ray beam directed horizontally**, exiting from the side closest to the IR (Fig. 1.69).

The position is named according to the surface on which the patient is lying (dorsal or ventral) and by the side closest to the IR (right or left).

Ventral Decubitus Position—Right or Left Lateral

In this position, the patient is lying on the ventral (anterior) surface with the x-ray beam directed horizontally, exiting from the side closest to the IR (Fig. 1.70).

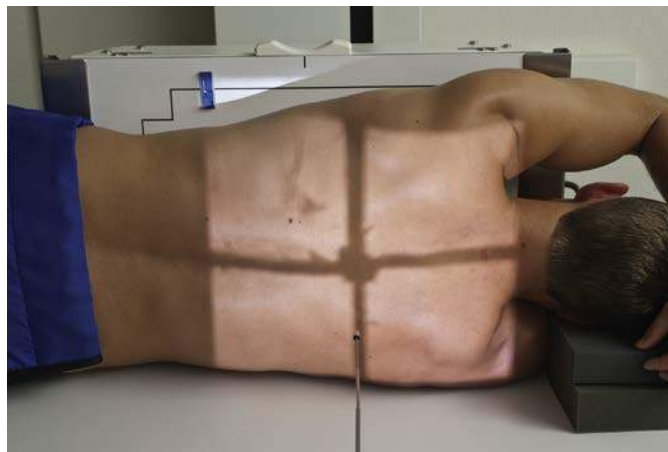


Fig. 1.68 Right lateral decubitus position (PA projection).



Fig. 1.69 Dorsal decubitus position (L lateral).



Fig. 1.70 Ventral decubitus position (R lateral).



Fig. 1.67 Left lateral decubitus position (AP projection).

Additional Special-Use Projection Terms

Following are some additional terms commonly used to describe projections. These terms, as shown by their definitions, also refer to the path or projection of the CR and are projections rather than positions.

Axial Projection

Axial (*ak'-se-al*) refers to the **long axis** of a structure or part (around which a rotating body turns or is arranged).

Special application—AP or PA axial: In radiographic positioning, the term *axial* is used to describe **any angle of the CR of 10° or more along the long axis of the body or body part.**⁷ However, in a true sense, an axial projection would be directed along, or parallel to, the long axis of the body or part. The term *semiaxial*, or “partly” axial, more accurately describes any angle along the axis that is not truly perpendicular or parallel to the long axis. However, for the sake of consistency with other references, the term ***axial projection*** is used throughout this text to describe both axial and semiaxial projections, as defined earlier and as illustrated in Figs. 1.71 through 1.73.

Inferosuperior and Superoinferior Axial Projections Inferosuperior axial projections are frequently performed for the shoulder and hip, where the CR enters below or inferiorly and exits above or superiorly (see Fig. 1.73).

The opposite of this is the **superoinferior** axial projection, such as a special nasal bone projection (see Fig. 1.71).

Tangential Projection

Tangential (*ta'-jen'-shal*) means touching a curve or surface at only one point.

This is a special use of the term *projection* to describe the CR that skims a body part to project the anatomy into profile and free of superimposition of surrounding body structures.

Examples Following are two examples or applications of the term *tangential projection*:

- Tangential projection of zygomatic arch (Fig. 1.74)
- Tangential projection of patella (Fig. 1.75)

AP Axial Projection—Lordotic Position

This is a **specific AP axial chest projection** for demonstrating the apices of the lungs. It also is called the **AP lordotic position**. In this case, the long axis of the body rather than the CR is angled.

The term **lordotic** comes from **lordosis**, a term that denotes curvature of the cervical and lumbar spine (see Chapters 8 and 9). As the patient assumes this position (Fig. 1.76), the lumbar lordotic curvature is exaggerated, making this a descriptive term for this special chest projection.



Fig. 1.72 AP axial (semiaxial) projection (CR 37° caudal).

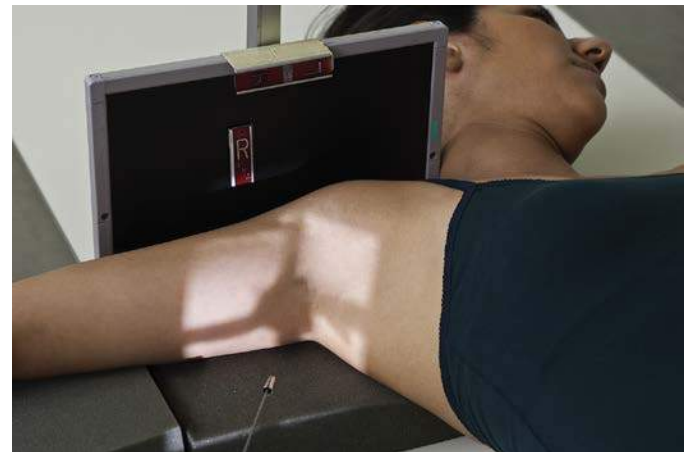


Fig. 1.73 Inferosuperior axial projection.

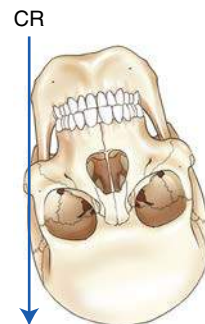


Fig. 1.74 Tangential projection (zygomatic arch).

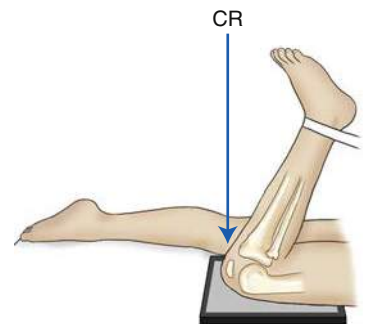


Fig. 1.75 Tangential projection (patella).

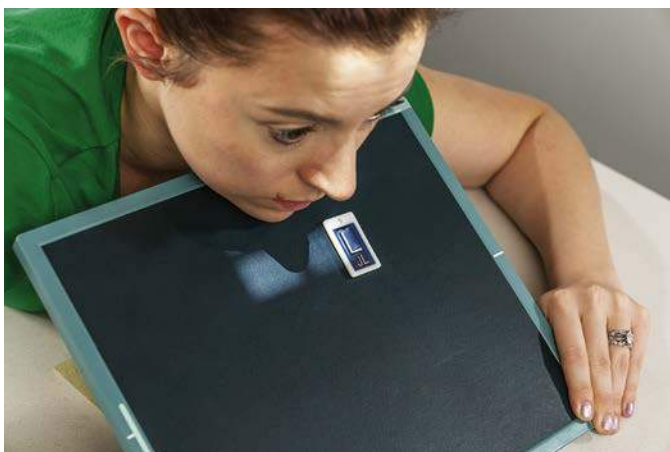


Fig. 1.71 Superoinferior (axial) projection.



Fig. 1.76 AP lordotic chest position.

Transthoracic Lateral Projection (Right Lateral Position)

This is a lateral projection through the thorax. It requires a qualifying positioning term (right or left lateral position) to indicate which shoulder is closest to the IR and is being examined (Fig. 1.77).

NOTE: This is a special adaptation of the projection term, indicating that the CR passes through the thorax even though it does not include an entrance or exit site. In practice, this is a common lateral shoulder projection and is referred to as a **right** or **left transthoracic lateral shoulder**.

Dorsoplantar and Plantodorsal Projections

These are secondary terms for AP or PA projections of the foot.

Dorsoplantar (DP) describes the path of the CR from the **dorsal** (anterior) surface to the **plantar** (posterior) surface of the foot (Fig. 1.78).

A special plantodorsal projection of the heel bone (calcaneus) is called an **axial plantodorsal projection (PD)** because the angled CR enters the plantar surface of the foot and exits the dorsal surface (Fig. 1.79).

NOTE: The term **dorsum** for the **foot** refers to the anterior surface, **dorsum pedis** (see Fig. 1.47).

Parietoacanthial and Acanthioparietal Projections

The CR enters at the cranial **parietal** bone and exits at the **acanthion** (junction of nose and upper lip) for the **parietoacanthial projection** (Fig. 1.80).

The opposite CR direction would describe the **acanthioparietal projection** (Fig. 1.81).

These are also known as **PA Waters** and **AP reverse Waters** methods and are used to visualize the facial bones.

Submentovertical (SMV) and Verticosubmental (VSM) Projections

These projections are used for the **skull** and **mandible**.

The CR enters below the chin, or **mentum**, and exits at the vertex or top of the skull for the **submentovertical (SMV) projection** (Fig. 1.82).

The less common, opposite projection of this would be the **verticosubmental (VSM) projection**, entering at the top of the skull and exiting below the mandible (not shown).

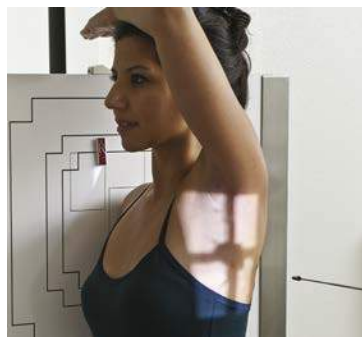


Fig. 1.77 Transthoracic lateral shoulder projection (R lateral shoulder position).



Fig. 1.78 AP or dorsoplantar (DP) projection of foot.



Fig. 1.79 Axial plantodorsal (PD) projection of calcaneus.



Fig. 1.80 Parietoacanthial projection (PA Waters position).



Fig. 1.81 Acanthioparietal projection.

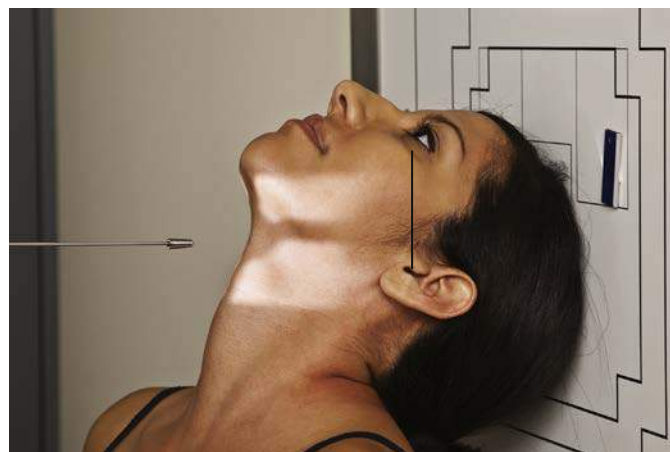


Fig. 1.82 Submentovertical (SMV) projection.

Relationship Terms

Following are paired positioning or anatomic terms that are used to describe relationships to parts of the body with opposite meanings.

Medial Versus Lateral

Medial (*me-de-al*) versus lateral refers to **toward** versus **away from** the center, or median plane.

In the anatomic position, the medial aspect of any body part is the “inside” part closest to the median plane, and the lateral part is away from the center, or away from the median plane or midline of the body (Fig. 1.83).

Examples In the anatomic position, the thumb is on the lateral aspect of the hand. The lateral part of the abdomen and thorax is the part away from the median plane.

Proximal Versus Distal

Proximal (*prok-si-mal*) is **near the source** or beginning, and distal (*dis-tal*) is **away from**. In regard to the upper and lower limbs, proximal and distal would be the part closest to or away from the trunk, the source or beginning of that limb (see Fig. 1.83).

Examples The elbow is proximal to the wrist. The finger joint closest to the palm of the hand is called the *proximal interphalangeal (PIP) joint*, and the joint near the distal end of the finger is the *distal interphalangeal (DIP) joint* (see Chapter 4).

Cephalad Versus Caudad

Cephalad (*sef-ah-lad*) means **toward** the head end of the body; caudad (*kaw-dad*) means **away from** the head end of the body.

A **cephalad angle** is any angle toward the head end of the body (Fig. 1.84; also see Fig. 1.86). (*Cephalad*, or *cephalic*, literally means “head” or “toward the head.”)

A **caudad angle** is any angle toward the feet or away from the head end (Fig. 1.85). (*Caudad* or *caudal* comes from *cauda*, literally meaning “tail.”)

In human anatomy, cephalad and caudad also can be described as **superior** (toward the head) or **inferior** (toward the feet).

NOTE: As is shown in Figs. 1.84, 1.85, and 1.86, these terms are correctly used to describe the direction of the CR angle for axial projections along the entire length of the body, not just projections of the head.

Interior (Internal, Inside) Versus Exterior (External, Outer)

Interior is inside of something, nearer to the center, and exterior is situated on or near the outside.

The prefix **intra-** means **within** or **inside** (e.g., intravenous: inside a vein).

The prefix **inter-** means situated **between things** (e.g., intercostal: located between the ribs).

The prefix **exo-** means **outside** or **outward** (e.g., exocardial: something that develops or is situated outside the heart).

Superficial Versus Deep

Superficial is nearer the skin surface; deep is farther away.

Example The cross-sectional drawing in Fig. 1.87 shows that the humerus is deep compared with the skin of the arm, which is superficial.

Another example would be a superficial tumor or lesion, which is located near the surface, compared with a deep tumor or lesion, which is located deeper within the body or part.

Ipsilateral Versus Contralateral

Ipsilateral (*ip-si-lat-er-al*) is on the same side of the body or part; contralateral (*kon-tra-lat-er-al*) is on the opposite side.

Example The right thumb and the right great toe are ipsilateral; the right knee and the left hand are contralateral.

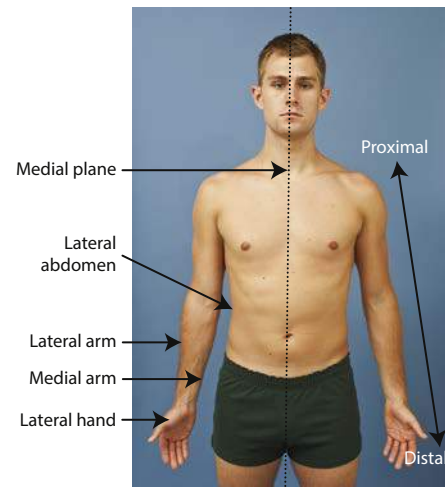


Fig. 1.83 Medial vs. lateral, proximal vs. distal.



Fig. 1.84 Cephalad CR angle (toward head).



Fig. 1.85 Caudad CR angle (away from head).



Fig. 1.86 Cephalic angle (AP axial projection of sacrum).

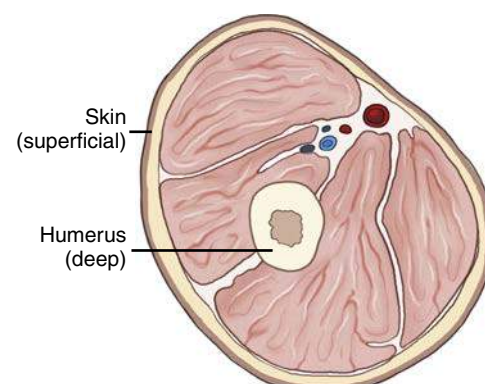


Fig. 1.87 Cross-section of arm.

Terms Related to Movement

The final group of positioning and related terms that every technologist should know relates to various movements. Most of these are listed as paired terms that describe movements in opposite directions.

Flexion Versus Extension

When a joint is flexed or extended, the **angle** between parts is **decreased** or **increased**.

Flexion decreases the angle of the joint (see examples of knee, elbow, and wrist flexions in Fig. 1.88).

Extension increases the angle as the body part moves from a flexed to a straightened position. This is true for the knee, elbow, and wrist joints, as is shown in Fig. 1.88.

Hyperextension

Hyperextension is extending a joint beyond the straight or neutral position.

Abnormal Hyperextension

A hyperextended elbow or knee results when the joint is extended beyond the straightened or neutral position. This is not a natural movement for these two joints and results in injury or trauma.

Normal Flexion and Hyperextension of the Spine

Flexion is bending forward, and extension is returning to the straight or neutral position. A backward bending **beyond the neutral position is hyperextension**. In practice, however, the terms *flexion* and *extension* are commonly used for these two extreme flexion and hyperextension projections of the spine (Fig. 1.89).

Normal Hyperextension of the Wrist

A second example of a special use of the term *hyperextension* concerns the wrist, where the carpal canal (tangential, inferosuperior) projection of the carpals is visualized by a special **hyperextended wrist movement** in which the wrist is extended beyond the neutral position. This specific wrist movement is also called **dorsiflexion** (backward or posterior flexion) (Fig. 1.90A).

Acute Flexion of the Wrist

An acute or full flexion of the wrist is required for a special tangential projection for a carpal bridge projection of the posterior aspect of the wrist (see Fig. 1.90B).

Ulnar Deviation Versus Radial Deviation of the Wrist

Deviation literally means “to turn aside” or “to turn away from the standard or course.”⁸

Ulnar deviation (Fig. 1.91A) is to turn or bend the hand and wrist from the natural position toward the ulnar side, and **radial deviation** (Fig. 1.91B) is toward the radial side of the wrist.

NOTE: Earlier editions of this textbook and other positioning references have defined these wrist movements as ulnar and radial flexion movements because they describe specific flexion movements toward either the ulna or the radius.⁹ However, because practitioners in the medical community, including orthopedic physicians, commonly use the terms *ulnar* and *radial deviation* for these wrist movements, this text also has changed this terminology to *ulnar* and *radial deviation movements* to prevent confusion and to ensure consistency with other medical references.

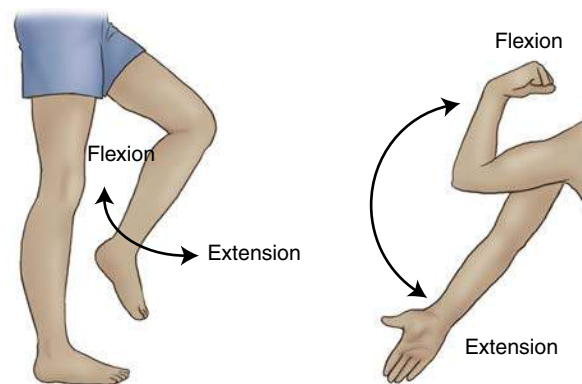


Fig. 1.88 Flexion vs. extension.

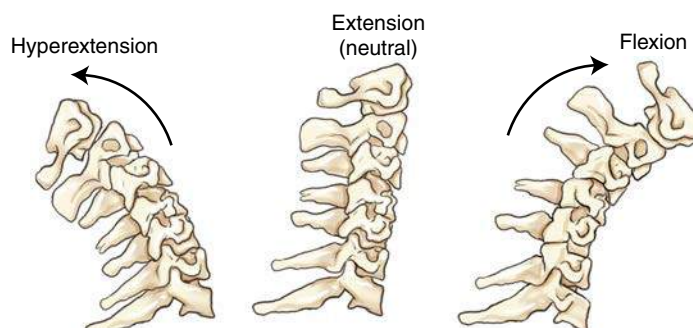


Fig. 1.89 Hyperextension, extension, and flexion of spine.

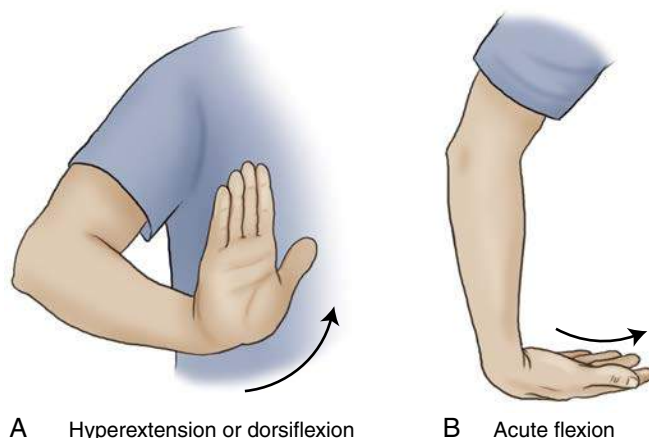


Fig. 1.90 Wrist extension and flexion movements. A, Hyperextension. B, Acute flexion.

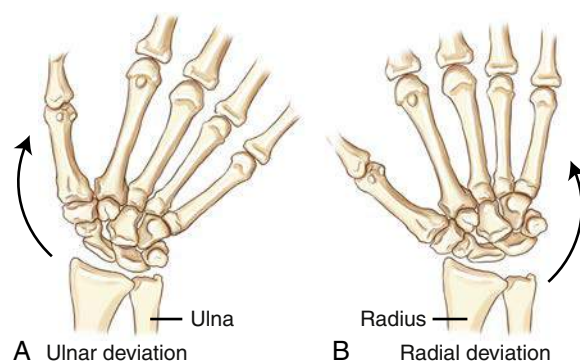


Fig. 1.91 Deviation wrist movements. A, Ulnar deviation. B, Radial deviation.

Dorsiflexion Versus Plantar Flexion of the Foot and Ankle

Dorsiflexion of the Foot This term means to **decrease the angle (flex)** between the dorsum (top of foot) and the lower leg, moving foot and toes upward (Fig. 1.92A).

Plantar Flexion of the Foot Extending the ankle joint, moving foot and toes downward from the normal position; flexing or decreasing the angle toward the plantar (posterior) surface of the foot (see Fig. 1.92B).

NOTE: See preceding page for dorsiflexion of the wrist (see Fig. 1.90A) compared with dorsiflexion of the foot (see Fig. 1.92A).

Eversion Versus Inversion

Eversion (*e-ver'-zhun*) is an **outward stress movement** of the foot at the ankle joint (Fig. 1.93).

Inversion (*in-ver'-zhun*) is **inward stress** movement of the foot as applied to the foot without rotation of the leg (Fig. 1.94).

The plantar surface (sole) of the foot is turned or rotated away from the median plane of the body (the sole faces in a more lateral direction) for eversion and toward the median plane for inversion.

The leg does not rotate, and stress is applied to the medial and lateral aspects of the ankle joint for evaluation of possible widening of the joint space (ankle mortise).

Valgus Versus Varus¹

Valgus (*val'-gus*) describes an abnormal position in which a part or limb is forced outward from the midline of the body. *Valgus* sometimes is used to describe **eversion stress** of the ankle joint.

Varus (*va'-rus*) describes an abnormal position in which a part or limb is forced inward toward the midline of the body. The term *varus stress* sometimes is used to describe **inversion stress** applied at the ankle joint.

NOTE: The terms *valgus* and *varus* are also used to describe the loss of normal alignment of bones due to fracture (see Chapter 15).

Medial (Internal) Rotation Versus Lateral (External) Rotation

Medial rotation is a rotation or turning of a body part with movement of the **anterior** aspect of the part **toward the inside**, or median, plane (Fig. 1.95A).

Lateral rotation is a rotation of an **anterior** body part **toward the outside**, or away from the median plane (Fig. 1.95B).

NOTE: In radiographic positioning, these terms describe movement of the **anterior** aspect of the part that is being rotated. In the forearm movements (see Fig. 1.95A and B), the anterior aspect of the forearm moves medially or internally on medial rotation and laterally or externally on lateral rotation. Another example is the medial and lateral oblique projections of the knee, in which the **anterior** part of the knee is rotated medially and laterally in either the AP or PA projections (see Chapter 6).

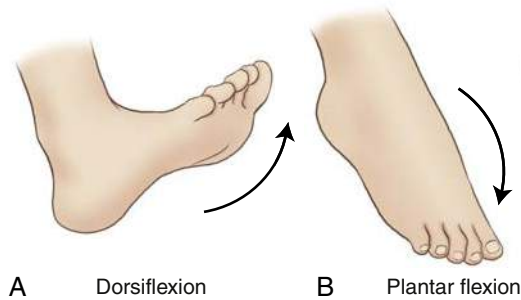


Fig. 1.92 Movements of ankle and foot. A, Dorsiflexion. B, Plantar flexion.



Fig. 1.93 Eversion (valgus stress).



Fig. 1.94 Inversion (varus stress).

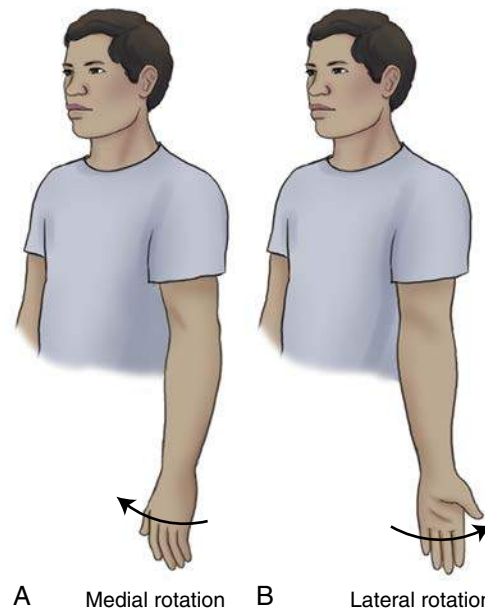


Fig. 1.95 Rotational movements of upper limb. A, Medial (internal) rotation. B, Lateral (external) rotation.

Abduction Versus Adduction

Abduction (*ab-duk'-shun*) is the lateral movement of the arm or leg **away** from the body.

Another application of this term is the abduction of the fingers or toes, which means spreading them apart (Fig. 1.96A).

Adduction (*ah-duk'-shun*) is a movement of the arm or leg **toward** the body, to draw toward a center or medial line (Fig. 1.96B).

Adduction of the fingers or toes means moving them together or toward each other.

Supination Versus Pronation

Supination (*su'-pi-na'-shun*) is a rotational movement of the hand into the anatomic position (palm up in supine position or forward in erect position) (Fig. 1.97A). This movement rotates the radius of the forearm laterally along its long axis.

Pronation (*pro-na'-shun*) is a rotation of the hand into the opposite of the anatomic position (palm down or back) (Fig. 1.97B).

NOTE: To help remember these terms, relate them to the body positions of supine and prone. *Supine* or *supination* means face up or palm up, and *prone* or *pronation* means face down or palm down.

Protraction Versus Retraction

Protraction (*pro-trak'-shun*) is a **movement forward** from a normal position (Fig. 1.98A).

Retraction (*re-trak'-shun*) is a **movement backward** or the condition of being drawn back (Fig. 1.98B).

Example Protraction is moving the jaw forward (sticking the chin out) or drawing the shoulders forward. Retraction is the opposite of this—that is, moving the jaw backward or squaring the shoulders, as in a military stance.

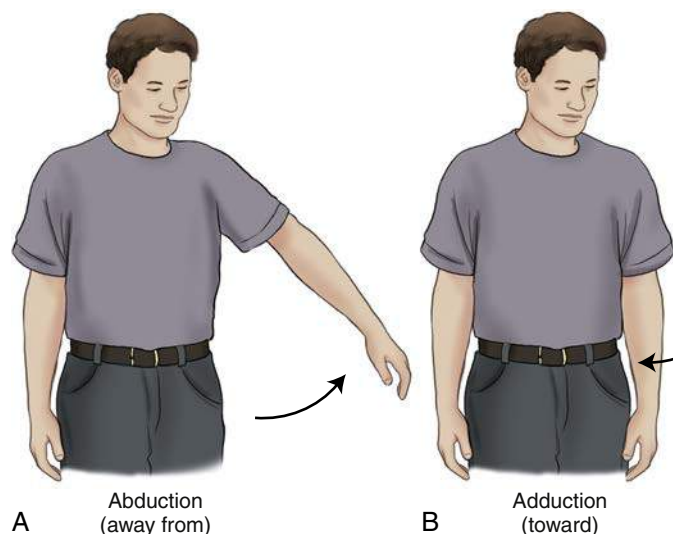


Fig. 1.96 Movements of upper limbs. A, Abduction. B, Adduction.

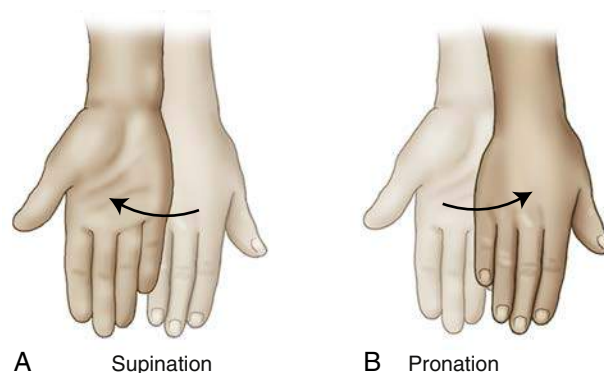


Fig. 1.97 Movements of hand. A, Supination. B, Pronation.

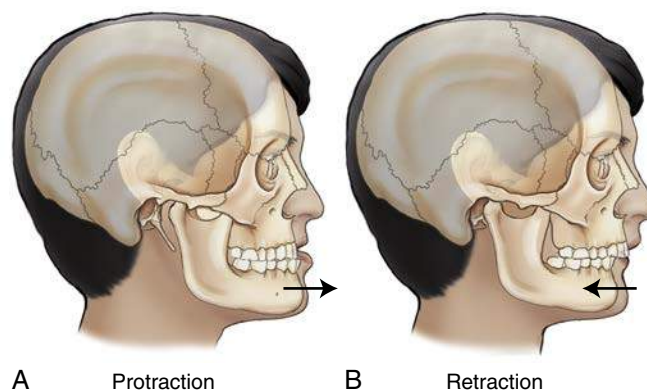


Fig. 1.98 Movements of mandible. A, Protraction. B, Retraction.

Elevation Versus Depression

Elevation is a lifting, raising, or moving of a part superiorly (Fig. 1.99A).

Depression is a letting down, lowering, or moving of a part inferiorly (Fig. 1.99B).

Example Shoulders are elevated when they are raised, as when shrugging the shoulders. Depressing the shoulders is lowering them.

Circumduction

Circumduction (*ser'-kum-duk'-shun*) means **to move around in the form of a circle** (Fig. 1.100). This term describes sequential movements of flexion, abduction, extension, and adduction, resulting in a cone-type movement at any joint where the four movements are possible (e.g., fingers, wrist, arm, leg).

Rotation Versus Tilt

Rotate is to turn or rotate a body part on its axis. In Fig. 1.101, the midsagittal plane of the entire body, including the head, is **rotated**.

Tilt is a slanting or tilting movement with respect to the long axis. Fig. 1.102 demonstrates no rotation of the head but a **tilting** (slanting) of the midsagittal plane of the head, which therefore is not parallel to the tabletop.

Understanding the difference between these two terms is important in cranial and facial bone positioning (see Chapter 11).

See Table 1.4 for a summary of positioning-related terminology.

Summary of Projections and Positions

The three terms **position**, **projection**, and **view** are sometimes confusing and may be used incorrectly in practice. These terms should be understood and used correctly (Table 1.5).

Position

Position is a term that is used to indicate the patient's general physical position, such as supine, prone, recumbent, erect, or Trendelenburg.

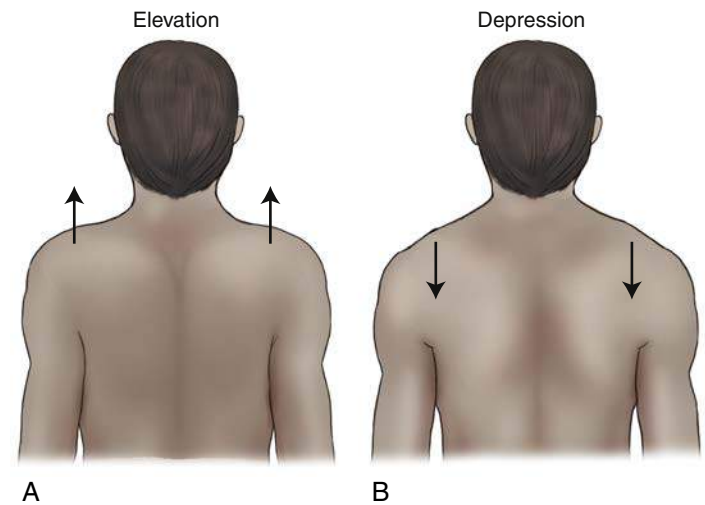


Fig. 1.99 Movements of shoulders. A, Elevation. B, Depression.

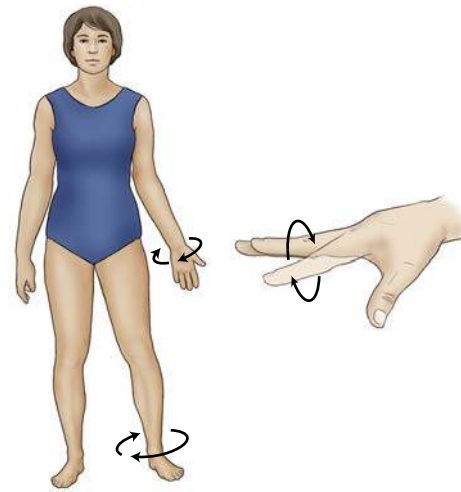


Fig. 1.100 Circumduction movements.

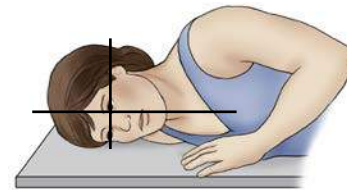


Fig. 1.101 Rotation—midsagittal plane rotated.

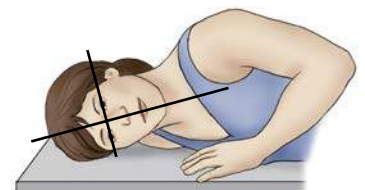


Fig. 1.102 Tilt—midsagittal plane of head tilted.

TABLE 1.4 SUMMARY OF POSITIONING-RELATED TERMS

Body Planes, Sections, and Lines	Relationship Terms
Longitudinal planes or sections	Medial vs. lateral
• Sagittal	Proximal vs. distal
• Coronal	Cephalad vs. caudad
• Oblique	Ipsilateral vs. contralateral
Transverse planes or sections	Internal vs. external
• Horizontal, axial, or cross-section	Superficial vs. deep
• Oblique	Lordosis vs. kyphosis (scoliosis)
Base plane	Movement Terms
Occlusal plane	Flexion vs. extension (acute flexion vs. hyperextension)
Infraorbitomeatal line (IOML)	Ulnar vs. radial deviation
Body Surfaces	Dorsiflexion vs. plantar flexion
Posterior	Eversion vs. inversion
Anterior	Valgus vs. varus
Plantar	Medial vs. lateral rotation
Dorsum	Abduction vs. adduction
Palmar	Supination vs. pronation
	Protraction vs. retraction
	Elevation vs. depression
	Tilt vs. rotation
	Circumduction
	Cephalad vs. caudad

TABLE 1.5 SUMMARY OF PROJECTIONS AND POSITIONS

PROJECTIONS (PATH OF CR)	GENERAL BODY POSITIONS	SPECIFIC BODY POSITIONS
Posteroanterior (PA)	Anatomic	R or L lateral
Anteroposterior (AP)	Supine	Oblique
Mediolateral	Prone	Left posterior oblique (LPO)
Lateromedial	Erect (upright)	Right posterior oblique (RPO)
AP or PA oblique	Recumbent	Left anterior oblique (LAO)
AP or PA axial	Trendelenburg	Right anterior oblique (RAO)
Tangential	Sims	Decubitus
Transthoracic	Fowler	Left lateral decubitus
Dorsoplantar (DP)	Lithotomy	Right lateral decubitus
Plantodorsal (PD)		Ventral decubitus
Inferosuperior axial		Dorsal decubitus
Superoinferior axial		Lordotic
Axiolateral		
Submentovertex (SMV)		
Verticosubmental (VSM)		
Parietoacanthial		
Acanthioparietal		
Craniocaudal		

Position also is used to describe **specific body positions** by the anatomy closest to the IR, such as **lateral** and **oblique positions**.

The term *position* should be “restricted to discussion of the patient’s physical position.”¹¹

Projection

Projection is a correct positioning term that describes or refers to the **path or direction of the central ray**, projecting an image onto an IR.

The term *projection* should be “restricted to discussion of the path of the central ray.”¹⁰

View

View is *not* a correct positioning term in the United States.

View describes the anatomy or body part as seen by the IR or other recording medium, such as a fluoroscopic screen. In the United States, the term *view* should be “**restricted to discussion of a radiograph or image.**”¹⁰

POSITIONING PRINCIPLES

Evaluation Criteria

The goal of every technologist should be to take not just a “passable” radiograph but rather an optimal one that can be evaluated by a **definable standard**, as described under **evaluation criteria** (Fig. 1.103).

An example of a three-part radiographic image evaluation as used in this text for a lateral forearm is shown on the right. The positioning photo and the resulting optimal radiograph (Figs. 1.104 and 1.105) are shown for this lateral forearm, as described in Chapter 4.

EVALUATION CRITERIA FORMAT

The technologist should review and compare radiographs using this standard to determine how close to an optimal image was achieved. A systematic method of learning how to critique radiographs is to break the evaluation down into these **three parts**.

1. **Anatomy demonstrated:** Describes precisely what anatomic parts and structures should be clearly visualized on that image (radiograph).
2. **Position:** Generally evaluates four issues: (1) placement of body part in relationship to the IR, (2) positioning factors that are important for the projection, (3) correct centering of anatomy, and (4) collimation.
3. **Exposure:** Describes how exposure factors or technique (kilovoltage [kVp], milliamperage [mA], and time [milliseconds]) can be evaluated for optimum exposure for that body part. **No motion** is a first priority, and a description of how the presence or absence of motion can be determined is listed. (Motion is included with exposure criteria because exposure time is the primary controlling factor for motion.)

Sample Lateral Forearm Evaluation Criteria

Anatomy Demonstrated: • Lateral projection of entire radius and ulna; proximal row of carpals, elbow, and distal end of humerus; and pertinent soft tissues such as fat pads and stripes of wrist and elbow joints.

Position: • Long axis of forearm aligned with long axis of IR.
• Elbow flexed 90°. • No rotation from true lateral as evidenced by the following: • Head of the ulna should be superimposed over the radius. • Humeral epicondyles should be superimposed. • Radial head should superimpose the coronoid process with radial tuberosity seen in profile.
• Collimation to **area of interest**.

Exposure: • Optimum density (brightness) and contrast with no motion will reveal sharp cortical margins and clear, bony trabecular markings and fat pads and stripes of the wrist and elbow joints.



Fig. 1.103 Technologist viewing digital images on monitor.



Fig. 1.104 Accurate positioning for lateral forearm.



Fig. 1.105 Lateral forearm.

Image Markers and Patient Identification

A **minimum** of two types of markers should be imprinted on every radiographic image. These are (1) **patient identification and date** and (2) **anatomic side markers**.

PATIENT IDENTIFICATION AND DATE (FILM-SCREEN CASSETTE [ANALOG] SYSTEMS)

With analog (film) systems, patient information, which includes data such as name, date, case number, and institution, are photofashed on the film in the space provided by a lead block in the film cassette (Fig. 1.106A). Each IR has a marker on the exterior indicating this area where the patient ID, including the date, should be placed. A general rule for most chest studies is to place the patient ID information at the top margin of the IR on chests. The patient ID marker is always placed where it is least likely to cover essential anatomy. The anatomic side markers are and should still be placed in a manner on the IR so that they are legible and esthetically correct (Fig. 1.106B). They must be within the collimation field so that they provide a permanent indicator of the correct side of the body or anatomic part.

Digital Systems

With photostimulable storage phosphor (PSP) cassette-based systems, often a bar code system imprints the patient information before or after exposure (Fig. 1.107). With digital imaging systems, patient identification is entered during registration and prior to exposure.

ANATOMIC SIDE MARKER

A right or left marker must also appear on every radiographic image correctly indicating the patient's right or left side or which limb is being radiographed, the right or the left. This may be provided as the word "Right" or "Left" or just the initials "R" or "L." This side marker preferably should be placed directly on the IR inside the lateral portion of the collimated border of the side being identified, with the placement such that the marker will not be superimposed over essential anatomy.

These radiopaque markers must be placed just within the collimation field so that they will be exposed by the x-ray beam and included on the image.

The two markers, the patient ID, and the anatomic side marker must be placed correctly on *all* radiographic images **including digitally produced images**. Generally, it is an unacceptable practice to write or annotate digitally this information on the image after it is processed because of legal and liability problems caused by potential mis-markings. A **radiograph taken without these two markers often has to be repeated**, which results in unnecessary radiation to the patient, making this a serious error. In the case of digital images, annotating the image to indicate side markers is an unacceptable practice. The exposure should be repeated to ensure the correct anatomy was imaged.

ADDITIONAL MARKERS OR IDENTIFICATION

Certain other markers or identifiers also may be used, such as **technologist initials**, which generally are placed on the R or L marker to identify the specific technologist responsible for the examination. Sometimes the examination room number is also included.

Time indicators are also commonly used; these note the minutes of elapsed time in a series, such as the 20-minute, 30-minute, 1-hour, and 2-hour series of radiographs taken in a small bowel series (SBS) procedure (see Chapter 13).

Another important marker on all decubitus positions is a decubitus marker or some type of indicator such as an **arrow identifying which side is up**. An "upright" or "erect" marker must also

be used to identify erect chest or abdomen positions compared with recumbent, in addition to an arrow indicating which side is up.

Inspiration (INSP) and **expiration (EXP)** markers are used for special comparison PA projections of the chest. **Internal (INT)** and **external (EXT)** markers may be used for rotation projections, such as for the proximal humerus and shoulder. Sample markers are shown in Fig. 1.108.

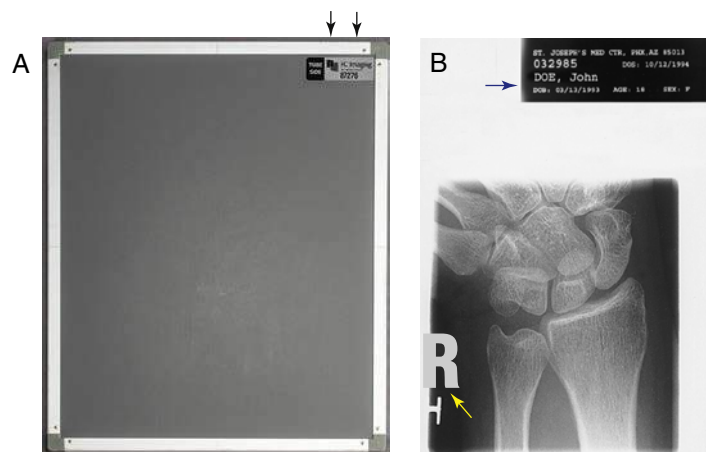


Fig. 1.106 A, Film cassette with patient information in block (arrows). B, Radiograph (blue arrow, Patient identification information; yellow arrow, Anatomic side marker to indicate right wrist).



Fig. 1.107 PSP cassette with bar coding for patient information

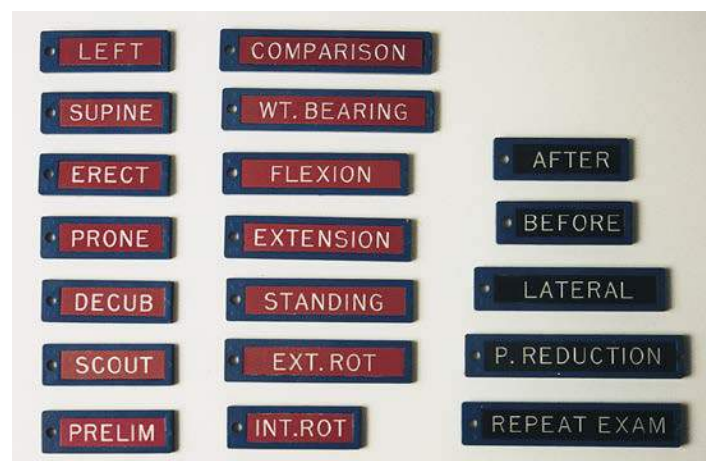


Fig. 1.108 Sample procedure markers.

Professional Ethics and Patient Care

The radiologic technologist is an important member of the health care team who is responsible in general for radiologic examination of patients. This includes being responsible for one's actions under a specific **code of ethics**.

Code of ethics describes the rules of acceptable conduct toward patients and other health care team members as well as personal actions and behaviors as defined within the profession. The ARRT code of ethics¹¹ is provided in [Box 1.1](#). The **announce, communicate, and explain (ACE)** campaign of the American Society of Radiologic Technologists (ASRT) is an initiative to educate patients about the role of the radiologic technologist. ACE is an acronym to help you remember to share and gain important information with and from your patients ([Box 1.2](#)).

Patient Assessment and Clinical History¹²

Once in the radiology imaging suite, the patient will interact directly with the radiologic technologist. Key responsibilities of the technologist are to introduce yourself, ensure you are performing the correct procedure on the correct patient, ask questions and acquire clinical history pertinent to the procedure, and explain the imaging procedure briefly. This protocol must be followed for every patient and for every radiologic procedure. The ASRT recommends the following initial communication with the patient:

- **Introduce yourself:** By name and role as the radiologic technologist performing the procedure.
- **Patient identification by two means:** Ask patient their name and verify by reviewing the patient arm band, patient chart, and/or examination requisition.
- **Verification of procedure(s) ordered:** Make sure it is the correct procedure for the correct patient by examining the patient chart or examination requisition.

- **Acquire clinical history:** Interview patient to determine chief complaints, allergy history, symptoms, duration and frequency of symptoms.
- **Pregnancy status:** Ask females if there is any possibility they might be pregnant. If the response of the patient, is "yes," "maybe," or "I don't know," inform supervisor or physician before proceeding with procedure. Age range for this type of questioning is often determined by the medical facility.
- **Explain procedure:** Explain the radiologic procedure that will be performed. Use terms and language the patient can understand.
- **Provide opportunity for patient to ask questions:** Encourage patient to ask questions about the procedure or other concerns before beginning the procedure.

BOX 1.2 ACE CAMPAIGN

In addition to performing medical imaging procedures, radiologic technologists must also communicate with patients. It is important for patients to understand that radiologic technologists are highly qualified medical imaging professionals who are educated in patient positioning, radiation safety, radiation protection, and equipment protocols. Furthermore, patients should have an understanding of the medical imaging procedure they are undergoing.

To communicate these points to patients, the American Society of Radiologic Technologists (ASRT) recommends that medical imaging professionals use the ACE initiative. The easy-to-remember acronym reminds radiologic technologists to:

- **Announce** your name.
- **Communicate** your credentials.
- **Explain** what you're going to do.

The ACE acronym provides medical imaging professionals with a unique and simple tool to educate patients about the radiologic technologist's role on the health care team.

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BOX 1.1 AMERICAN REGISTRY OF RADIOLOGIC TECHNOLOGISTS CODE OF ETHICS

The Code of Ethics forms the first part of the *Standards of Ethics*. The Code of Ethics shall serve as a guide by which Certificate Holders and Candidates may evaluate their professional conduct as it relates to patients, healthcare consumers, employers, colleagues, and other members of the healthcare team. The Code of Ethics is intended to assist Certificate Holders and Candidates in maintaining a high level of ethical conduct and in providing for the protection, safety, and comfort of patients. The Code of Ethics is aspirational.

1. The radiologic technologist acts in a professional manner, responds to patient needs, and supports colleagues and associates in providing quality patient care.
2. The radiologic technologist acts to advance the principal objective of the profession to provide services to humanity with full respect for the dignity of mankind.
3. The radiologic technologist delivers patient care and service unrestricted by the concerns of personal attributes or the nature of the disease or illness, and without discrimination on the basis of sex, race, creed, religion, or socio-economic status.
4. The radiologic technologist practices technology founded upon theoretical knowledge and concepts, uses equipment and accessories consistent with the purposes for which they were designed, and employs procedures and techniques appropriately.
5. The radiologic technologist assesses situations; exercises care, discretion, and judgment; assumes responsibility for professional decisions; and acts in the best interest of the patient.
6. The radiologic technologist acts as an agent through observation and communication to obtain pertinent information for the physician to aid in the diagnosis and treatment of the patient and recognizes that interpretation and diagnosis are outside the scope of practice for the profession.
7. The radiologic technologist uses equipment and accessories, employs techniques and procedures, performs services in accordance with an accepted standard of practice, and demonstrates expertise in minimizing radiation exposure to the patient, self, and other members of the healthcare team.
8. The radiologic technologist practices ethical conduct appropriate to the profession and protects the patient's right to quality radiologic technology care.
9. The radiologic technologist respects confidences entrusted in the course of professional practice, respects the patient's right to privacy, and reveals confidential information only as required by law or to protect the welfare of the individual or the community.
10. The radiologic technologist continually strives to improve knowledge and skills by participating in continuing education and professional activities, sharing knowledge with colleagues, and investigating new aspects of professional practice.
11. The radiologic technologist refrains from the use of illegal drugs and/or any legally controlled substances which result in impairment of professional judgment and/or ability to practice radiologic technology with reasonable skill and safety to patients.

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Essential Projections

ROUTINE PROJECTIONS

Certain basic projections are listed and described in this text for each radiographic examination or procedure commonly performed throughout the United States and Canada. Routine projections are defined as **projections commonly taken on patients who can cooperate fully**. This varies depending on radiologist and department preference and on geographic differences.

SPECIAL PROJECTIONS

In addition to routine projections, certain special projections are included for each examination or procedure described in this text. These are defined as **projections most commonly taken to demonstrate better specific anatomic parts or certain pathologic conditions or projections that may be necessary for patients who cannot cooperate fully**.

The authors recommend (on the basis of recent survey results) that all students learn and demonstrate proficiency in all essential projections as listed in this text. This includes all routine projections and all special projections as listed and described in each chapter. The following are examples of these routine projection and special projection boxes for Chapter 2. Becoming competent in these projections helps to ensure students are prepared to function as imaging technologists in any part of the United States.

General Principles for Determining Positioning Routines

Two general rules or principles are helpful for remembering and understanding the reasons certain minimum projections are performed for various radiographic examinations.

MINIMUM OF TWO PROJECTIONS (90° FROM EACH OTHER)

The first general rule in diagnostic radiology suggests that a **minimum of two projections** taken as near as possible to 90° from each other are required for most radiographic procedures. Exceptions include an AP mobile (portable) chest, a single AP abdomen (called a KUB—kidneys, ureter, and bladder), and an AP of the pelvis, in which only one projection usually provides adequate information.

Three reasons for this general rule of a minimum of two projections are as follows:

1. Superimposition of anatomic structures

Certain pathologic conditions (e.g., some fractures, small tumors) may not be visualized on one projection only.

2. Localization of lesions or foreign bodies

A minimum of two projections, taken at 90° or as near as possible to right angles of each other, are essential in determining the location of any lesion or foreign body (Fig. 1.109).

Example

Foreign bodies embedded in tissues of the knee. Both AP/PA and lateral projections are necessary to determine the exact location of this “nail.”

3. Determination of alignment of fractures

All fractures require a minimum of two projections, taken at 90° or as near as possible to right angles of each other. This is both to fully visualize the fracture site and to determine alignment of the fractured parts following surgery (Figs. 1.110 and 1.111).

Chest
ROUTINE
• PA, p. XXX • Lateral, p. XXX

Upper Airway
ROUTINE
• Lateral, p. XXX • AP, p. XXX

Chest
SPECIAL
• AP supine or semierect, p. XXX • Lateral decubitus, p. XXX • AP lordotic, p. XXX • Anterior oblique, p. XXX • Posterior oblique, p. XXX

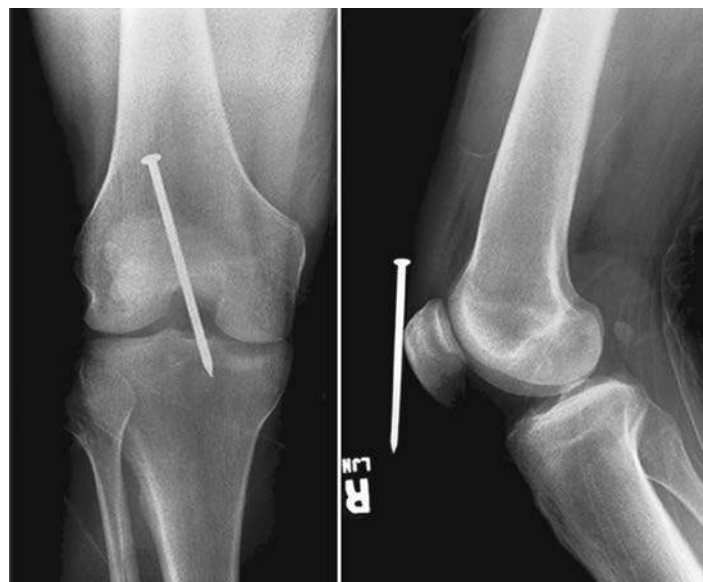


Fig. 1.109 AP and lateral projection for foreign body (nail through anterior knee).



Fig. 1.110 AP humerus projection for postoperative fracture alignment.



Fig. 1.111 Lateral humerus projection for postoperative fracture alignment.

MINIMUM OF THREE PROJECTIONS WHEN JOINTS ARE IN AREA OF INTEREST

This second general rule or principle suggests that all radiographic procedures of the skeletal system involving joints require a minimum of **three** projections rather than only two. These are **AP** or **PA**, **lateral**, and **oblique** projections.

The reason for this rule is that more information is needed than can be provided on only two projections. For example, with multiple surfaces and angles of the bones making up the joint, a small oblique chip fracture or other abnormality within the joint space may not be visualized on either frontal or lateral views but may be well demonstrated in the oblique position.

Following are examples of examinations that generally require three projections as routine (joint is in prime interest area):

- Fingers
- Toes
- Hand
- Wrist (Fig. 1.112)
- Elbow
- Ankle
- Foot
- Knee

Examples of examinations that require two projections as routine include the following:

- Forearm
- Humerus
- Femur
- Hips
- Tibia-fibula (Figs. 1.113 and 1.114)
- Chest

Exceptions to Rules

- Postreduction upper and lower limbs generally require only two projections for checking fracture alignment.
- A pelvis study requires only a single AP projection unless a hip injury is suspected.

Palpation of Topographic Positioning Landmarks

Radiographic positioning requires the location of specific structures or organs within the body, many of which are not visible to the eye from the exterior. Therefore, the technologist must rely on bony landmarks to indicate their location. These bony structures are referred to as **topographic landmarks**. Fig. 1.115 shows examples of topographic landmarks of the pelvis. (see Chapter 2, Chest, p. 70, and Chapter 3, Abdomen, p. 103, for topographic landmarks for chest, abdomen, and pelvis). Topographic landmarks can be located by a process referred to as *palpation*.

PALPATION

Palpation refers to the process of applying light pressure with the fingertips directly on the patient to locate positioning landmarks. **This must be done gently**, because the area being palpated may be painful or sensitive for the patient. Also, **the patient should always be informed of the purpose of this palpation before this process is begun, and patient permission should be obtained.**

NOTE: Palpation of certain landmarks, such as the ischial tuberosity and the symphysis pubis, may be embarrassing for the patient and **may not be permitted by institutional policy**. Technologists should use alternative landmarks as described in later chapters.



Fig. 1.112 Wrist—requires three projections.



Fig. 1.113 AP lower leg projection.



Fig. 1.114 Lateral lower leg projection (same patient as Fig. 1.113).

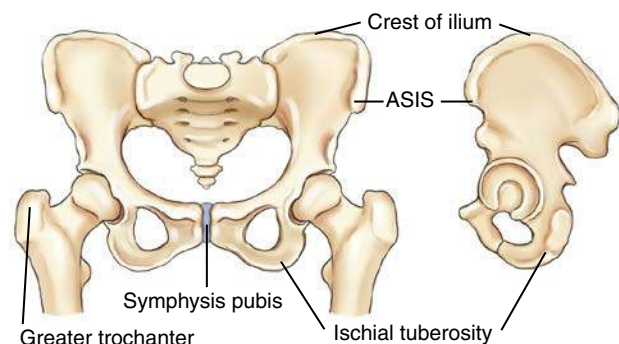


Fig. 1.115 Topographic landmarks of the pelvis.

Image Receptor Alignment

The alignment of the IR to the anatomy is an important consideration. In most cases, the long axis of the anatomic part is aligned to the longest dimension of the IR. This allows for the majority of an anatomic structure to be demonstrated and permits closer collimation of the x-ray field to the anatomy. The example in Fig. 1.116 is an image of a smaller adult chest. The IR is in the **portrait (lengthwise)** alignment in which the long axis of the lungs is aligned to the longest dimension of the IR. In another case, the hypersthenic adult PA chest often requires the IR to be placed in the **landscape (crosswise)** alignment. This permits the broader lateral borders of the lung to be demonstrated (Fig. 1.117). For each position in the text, the terms **landscape** or **portrait** will be listed following the recommended size of IR to indicate how the IR should be aligned to the anatomic part.

Viewing Radiographic Images

The manner in which **PA** and **AP** projection radiographic images are placed for viewing depends on the radiologist's preference and the most common practice in that part of the United States. However, in the United States and Canada, a common and accepted way to place radiographic images for viewing is to display them so that **the patient is facing the viewer**, with the patient in the anatomic position (Fig. 1.118). **This always places the patient's left to the viewer's right.** This is true for **either AP or PA projections** (Figs. 1.119 and 1.120).

Lateral positions are marked R or L by the side of the patient closest to the IR. Placement of lateral radiographic images for viewing varies depending on the radiologist's preference. One common method is to place the image so that the viewer is seeing the image from the same perspective as the x-ray tube. If the left marker is placed anteriorly to the patient, the L would be on the viewer's right (Fig. 1.121). However, some radiologists prefer to view laterals turned 90° and with the anteriorly placed L marker on the viewer's left. Technologists should determine the preferred method for viewing laterals in their department.

PA or AP oblique projections are placed for viewing the same way that a PA or AP projection is placed, with the patient's right to the viewer's left.

Decubitus chest and abdomen projections are generally viewed the way the x-ray tube "sees" them, placed in a portrait alignment with the upside of the patient also on the upper part of the view box (Fig. 1.122).

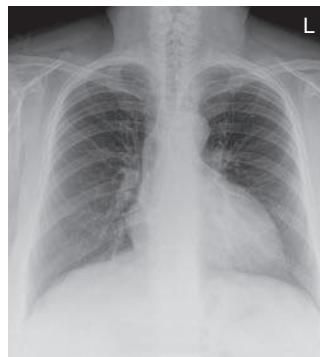


Fig. 1.116 Portrait (lengthwise) IR alignment of PA chest.



Fig. 1.117 Landscape (crosswise) IR alignment of PA chest.



Fig. 1.118 Viewing digital chest radiographs (patient's right always to viewer's left, both PA and AP).



Fig. 1.119 PA chest projection.



Fig. 1.120 AP chest projection.



Fig. 1.121 Left lateral chest.



Fig. 1.122 Left lateral decubitus chest.

Upper and lower limb projections are viewed as projected by the x-ray beam onto the IR; the R or L lead marker appears right-side-up if it has been placed on the IR correctly.

Images that include the digits (hands and feet) generally are placed with the **digits up**. However, other images of the limbs are viewed in the anatomic position with the **limbs hanging down** (Fig. 1.123).

Viewing CT or MRI Images

The generally accepted way of viewing all CT and MRI axial images is similar to that used for conventional radiographs, even though the image represents a thin “slice” or sectional view of anatomic structures. In general, these images are placed **so the patient’s right is to the viewer’s left** (Fig. 1.124).



Fig. 1.123 Viewing digital upper or lower limb images.



Fig. 1.124 Axial (cross-sectional) image (mid thorax level of T5) (patient’s right to viewer’s left).

PART TWO ■ IMAGING PRINCIPLES

Contributor **Andrew Woodward**, MA, RT(R)(CT)(QM)

This period of technologic transition necessitates that students have an understanding of all image acquisition technologies, because they will find themselves working in imaging departments that acquire images by using only digital technology, only film-screen technology, or a combination of both.

This part provides an introduction to radiographic technique and image quality for both film-screen imaging and digital imaging. The study of radiographic technique and image quality includes factors that determine the accuracy with which structures that are being imaged are reproduced in the image. Each of these factors has a specific effect on the final image, and the technologist must strive to maximize these factors to produce the best image possible at the lowest achievable dose.

This part also describes methods of digital image acquisition, discusses the application of digital imaging, and provides an introduction to the important principles of radiation safety.

Technical considerations for both analog and digital imaging technology are discussed in the following sections.

Exposure Factors

For each radiographic image obtained, the radiographer must select *exposure factors* on the control panel of the imaging equipment. The exposure factors required for each examination are determined by numerous variables, including the density/atomic number and thickness of the anatomic part, any pathology present, and image acquisition technology. Fig. 1.125 is an example of a more current radiographic control console.

Exposure factors, sometimes referred to as *technique factors*, include the following:

- **Kilovoltage (kVp)**—controls the energy (penetrating power) of the x-ray beam. This can also be referred to as **kilovoltage peak (kVp)**—the maximum electrical potential used to create the x-ray photons within the x-ray tube.
- **Milliamperage (mA)**—controls the quantity or number of x-rays produced.
- **Exposure time (ms)**—controls the duration of the exposure, usually expressed in milliseconds.

Each of these exposure factors has a specific effect on the signal collected to produce the radiographic image. When performing radiographic procedures, technologists must apply their knowledge of exposure factors and imaging principles to ensure that images obtained are of the **highest quality possible** while exposing patients to the **lowest radiation dose possible**.

Image Quality

- Spatial resolution
- Distortion
- Scatter radiation and grids

SPATIAL RESOLUTION

Spatial resolution is defined as the **recorded sharpness of structures on the image**. Resolution on a radiographic image is demonstrated by the clarity or sharpness of fine structural lines and borders of tissues or structures on the image. Resolution is also known as **detail, recorded detail, image sharpness, or definition**. Resolution of images generally is measured and expressed as line pairs per millimeter (lp/mm), in which a line pair is seen as a single line and an interspace of equal width. The higher the line pair measure, the greater is the resolution; it is typically 5 to 6 lp/mm for general imaging. Lack of visible sharpness or resolution is known as **blur** or **unsharpness**.

Geometric Factors Affecting Spatial Resolution

Geometric factors that control or influence resolution consist of **focal spot size**, **source–image receptor distance (SID)**, and **object–image receptor distance (OID)**. The effect of OID is explained and illustrated in Fig. 1.131.

The use of the **small focal spot** results in **less geometric unsharpness** (Fig. 1.126). To illustrate, a point source is used commonly as the source of x-rays in the x-ray tube; however, the actual source of x-rays is an area on the anode known as the *focal spot*. Most x-ray tubes exhibit dual focus; that is, they have two focal spots: large and small. Use of the small focal spot results in less unsharpness of the image, or an image with a decreased *penumbra*. Penumbra refers to the **unsharp edges of objects in the projected image**. However, even with the use of the small focal spot, some penumbra is present.

Motion The greatest deterrent to image sharpness as related to positioning is *motion*. Two types of motion influence radiographic detail: **voluntary** and **involuntary**.



Fig. 1.125 Example of radiographic console (selecting kVp, mA, and mAs factors).

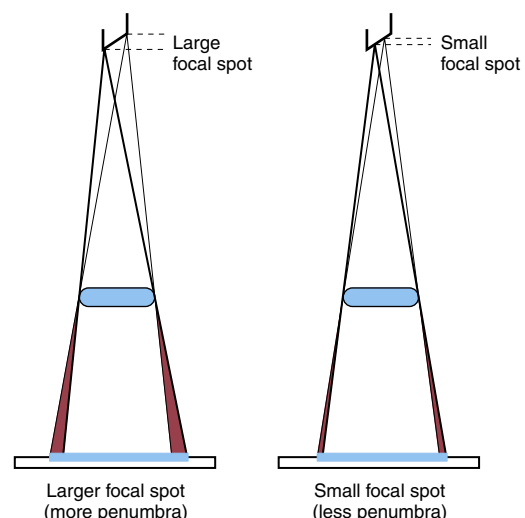


Fig. 1.126 Effect of focal spot size.

Voluntary motion is that which the patient can control. Motion from breathing or movement of body parts during exposure can be prevented or at least minimized by **controlled breathing** and **patient immobilization**. Support blocks, sandbags, or other immobilization devices can be used to reduce motion effectively. These devices are most effective for examination of upper or lower limbs, as will be demonstrated throughout this text.

Involuntary motion cannot be controlled by the patient at will. Therefore, involuntary motion, such as peristaltic action of abdominal organs, tremors, or chills, is more difficult, if not impossible, to control.

If motion unsharpness is apparent on the image, the technologist must determine whether this blurring or unsharpness is due to voluntary or involuntary motion. This determination is important, because these two types of motion can be controlled in various ways.

Difference between voluntary and involuntary motion Voluntary motion is visualized as **generalized blurring of linked structures**, such as blurring of the thoracic bony and soft tissue structures as evident in Fig. 1.127. Voluntary motion can be minimized through the use of high mAs and short exposure times. Increased patient cooperation is another factor that may contribute to decreased voluntary motion; a thorough explanation of the procedure and clear breathing instructions may prove helpful.

Involuntary motion is identified by **localized unsharpness or blurring**. This type of motion is less obvious but can be visualized on abdominal images as localized blurring of the edges of the bowel, with other bowel outlines appearing sharp (gas in the bowel appears as dark areas). Study Fig. 1.128 carefully to see this slight blurring in the left upper abdomen, indicated by arrows. The remaining edges of the bowel throughout the abdomen appear sharp. Fig. 1.127, by comparison, demonstrates overall blurring of the heart, ribs, and diaphragm. A clear explanation of the procedure by the technologist may aid in reducing voluntary motion; however, a decrease in exposure time with an associated increase in mAs is the best and sometimes the only way to minimize motion unsharpness caused by involuntary motion.

Summary of Spatial Resolution Factors

Use of a **small focal spot**, an **increase in SID**, and a **decrease in OID** result in less geometric unsharpness and increased resolution. Patient motion also affects image quality; **short exposure times** and **increased patient cooperation** help to minimize voluntary motion unsharpness. Involuntary motion unsharpness is controlled only by short exposure times.

DISTORTION

The second image quality factor is *distortion*, which is defined as the **misrepresentation of object size or shape** as projected onto radiographic recording media. Two types of distortion have been identified: size distortion (magnification) and shape distortion.

No radiographic image reproduces the exact size of the body part that is being radiographed. This is impossible to do because a degree of magnification or distortion or both always exists as a result of OID and divergence of the x-ray beam. Nevertheless, distortion can be minimized and controlled if some basic principles are used as a guide.

X-Ray Beam Divergence

X-ray beam divergence is a basic but important concept in the study of radiographic positioning. It occurs because x-rays originate from a small source in the x-ray tube (the focal spot) and diverge as they travel to the IR (Fig. 1.129). The field size of the x-ray beam is limited by a collimator that consists of adjustable lead attenuators

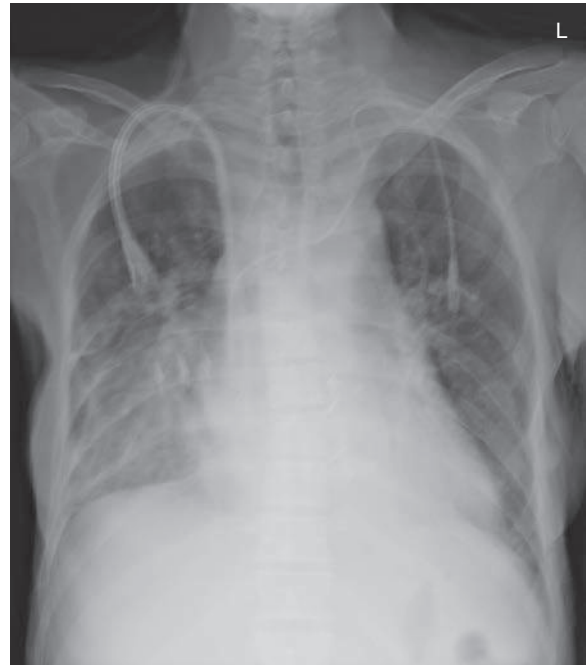


Fig. 1.127 Voluntary motion (breathing and body motion)—blurring of entire chest and overall unsharpness.

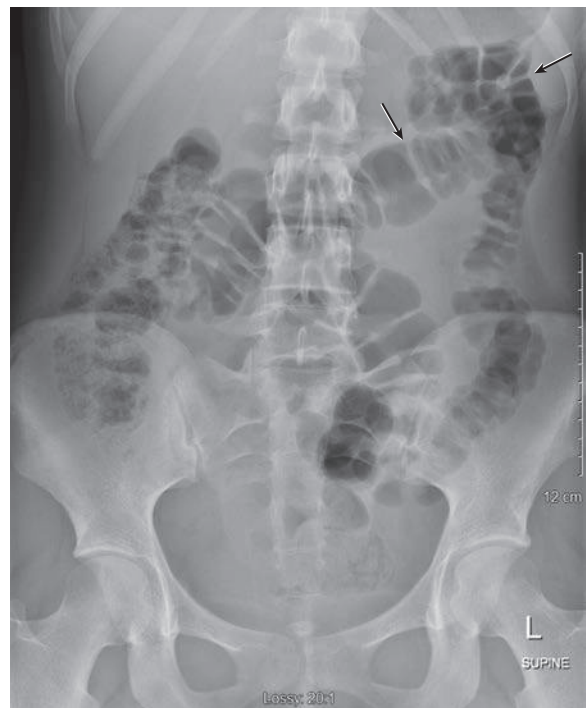


Fig. 1.128 Involuntary motion (from peristaltic action)—localized blurring in upper left abdomen (arrows).

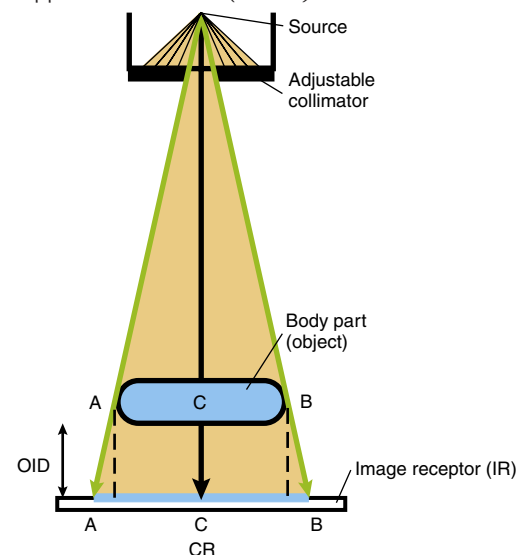


Fig. 1.129 X-ray beam divergence.

or shutters. The collimator and shutters absorb the x-rays on the periphery, controlling the size of the x-ray beam.

The **center point of the x-ray beam**, which is called the *central ray (CR)*, theoretically has no divergence; the **least amount of distortion** is seen at this point on the image. All other aspects of the x-ray beam strike the IR at some angle, with the angle of divergence increasing to the outermost portions of the x-ray beam. The potential for distortion at these outer margins is increased.

Fig. 1.129 demonstrates three points on a body part (marked A, B, and C) as projected onto the IR. Greater magnification is demonstrated at the periphery (A and B) than at the point of the central ray (C). Because of the effect of the divergent x-ray beam, combined with at least some OID, this type of size distortion is inevitable. It is important for technologists to control closely and minimize distortion as much as possible.

Controlling Factors

Following are **four** primary controlling factors of distortion:

1. SID
2. OID
3. Object–image receptor alignment
4. CR alignment/centering

SID The first controlling factor for distortion is SID. The effect of SID on size distortion (magnification) is demonstrated in Fig. 1.130. Note that **less magnification occurs at a greater SID than at a shorter SID**. This is the reason that chest radiographs are obtained at a minimum SID of 72 inches (180 cm) rather than of 40 to 48 inches (100 to 120 cm), which is commonly used for most other examinations. A 72-inch (180-cm) SID results in less magnification of the heart and other structures within the thorax.

Minimum 40-inch (100-cm) SID It has been a long-standing common practice to use 40 inches (100 cm) as the standard SID for most skeletal radiographic examinations. However, in the interest of improving image resolution by decreasing magnification and distortion, it is becoming more common to increase the standard SID to 44 inches or 48 inches (110 cm or 120 cm). Additionally, it has been shown that increasing the SID from 40 to 48 inches reduces the entrance or skin dose even when the requirement for increased mAs is considered. In this textbook, the suggested SID listed on each skeletal positioning page is a **minimum of 40 inches**, with 44 inches or 48 inches recommended if the equipment and departmental protocol allow.

OID The second controlling factor for distortion is OID. The effect of OID on magnification or size distortion is illustrated clearly in Fig. 1.131. The closer the object being radiographed is to the IR, the less are the magnification and shape distortion and the better is the resolution.

Object–Image Receptor Alignment A third important controlling factor of distortion is *object–IR alignment*. This refers to the alignment or plane of the object that is being radiographed in relation to the plane of the image receptor. If the object plane is not parallel to the plane of the IR, distortion occurs. The greater the angle of inclination of the object or the IR, the greater the amount of distortion. For example, if a finger being radiographed is not parallel to the IR, the interphalangeal joint spaces will not be open because of the overlapping of bones, as is demonstrated in Fig. 1.132.

Effect of improper object–IR alignment In Fig. 1.133, the digits (fingers) are supported and aligned **parallel to the image receptor**, resulting in open interphalangeal joints and undistorted phalanges.

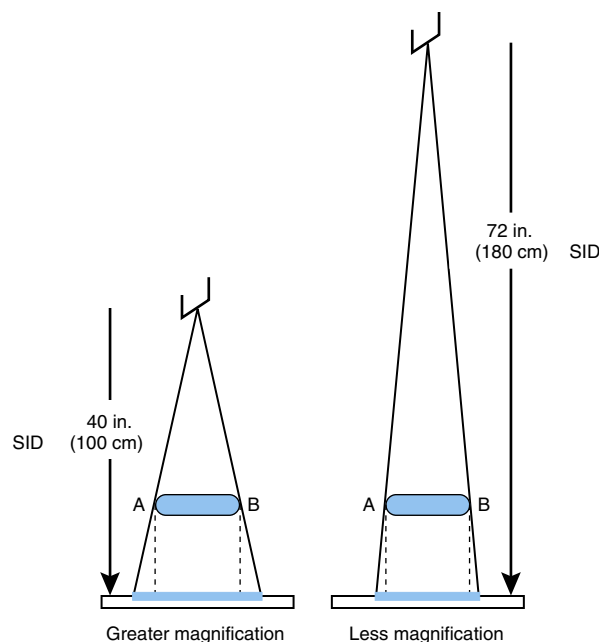


Fig. 1.130 Effect of SID.

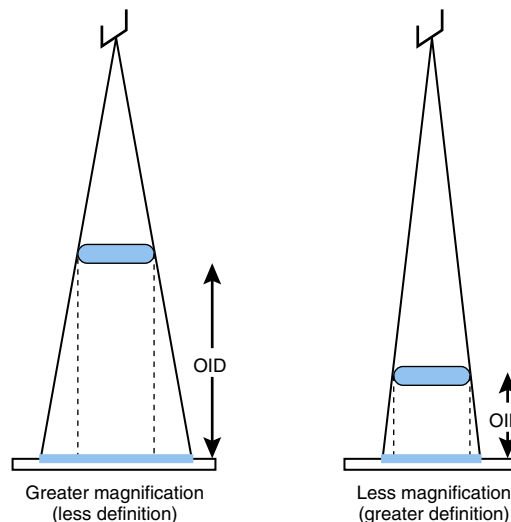


Fig. 1.131 Effect of OID.

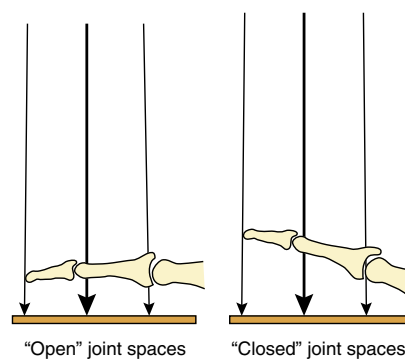


Fig. 1.132 Object alignment and distortion.



Fig. 1.133 Digits parallel to IR—joints open.

In Fig. 1.134, in which the digits are not parallel to the IR, the interphalangeal joints of the digits are not open, and possible pathology within these joint regions may not be visible. Note the open joints of the digits in Fig. 1.135 compared with Fig. 1.136 (see arrows). Additionally, the phalanges will be either foreshortened or elongated.

These examples demonstrate the important principle of correct object-IR alignment. **The plane of the body part that is being imaged must be as near parallel to the plane of the IR as possible** to produce an image of minimal distortion.

CR Alignment The fourth and final controlling factor for distortion is *central ray alignment* (centering), an important principle in positioning. As was previously stated, only the center of the x-ray beam, the CR, has no divergence because it projects that part of the object at 90°, or perpendicular to the plane of the IR (refer to Fig. 1.129). Therefore, the **least possible distortion occurs at the CR**. Distortion increases as the angle of divergence increases from the center of the x-ray beam to the outer edges. For this reason, correct centering or correct central ray alignment and placement is important in minimizing image distortion.

Examples of correct CR placement for an AP knee are shown in Figs. 1.137 and 1.138. The CR passes through the knee joint space with minimal distortion, and the joint space should appear open.

Fig. 1.139 demonstrates correct centering for an AP distal femur, in which the CR is correctly directed perpendicular to the IR and centered to the mid distal femur. However, the knee joint is now exposed to divergent rays (as shown by the arrow), and this causes the knee joint to appear closed (Fig. 1.140).

CR angle For most projections, the CR is aligned **perpendicular**, or 90°, to the plane of the IR. For certain body parts, however, a specific angle of the CR is required, as is indicated by the positioning descriptions in this text as the *CR angle*. This means that the CR is angled from the vertical in a cephalic or caudad direction so as to use distortion intentionally without superimposing anatomic structures.

Summary of Factors That May Affect Distortion

Use of the correct SID while minimizing OID, ensuring the object and IR are aligned, and correctly aligning or centering the CR to the part can minimize distortion on a radiographic image (Table 1.6).

SCATTER RADIATION AND GRIDS

Grids

Because the amount of scatter increases with the thickness of the tissue irradiated, it generally is recommended that a grid should be used for radiography of any body part that is thicker than 10 cm. Depending on the examination, the grid may be portable or may be built into the x-ray equipment. It is positioned between the patient and the IR and absorbs much of the scatter radiation before it hits the IR. Absorption of scatter is a key event that increases image contrast.

Correct Use of Grids An in-depth discussion of grid construction and characteristics is beyond the scope of this text. However,



Fig. 1.134 Digits not parallel to IR—joints not open.

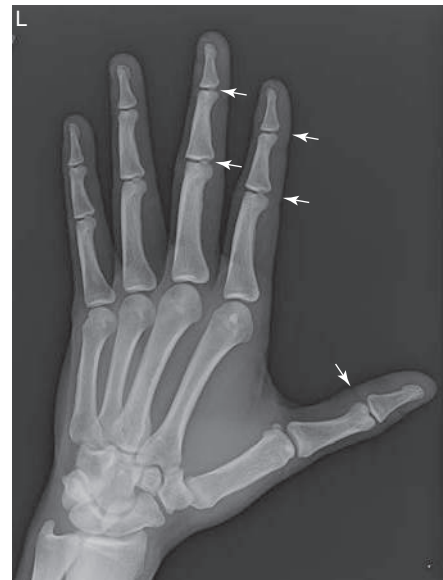


Fig. 1.135 Digits parallel—joints open.



Fig. 1.136 Digits not parallel—joints not open.



Fig. 1.137 Correct CR centering for AP knee.

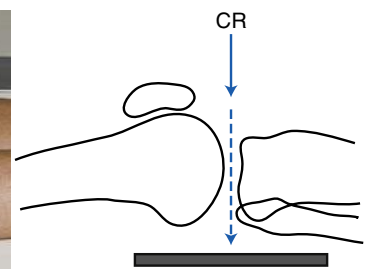


Fig. 1.138 Correct CR centering for knee.

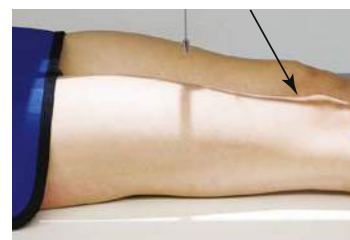


Fig. 1.139 Correct CR centering for AP femur (distortion occurs at knee).

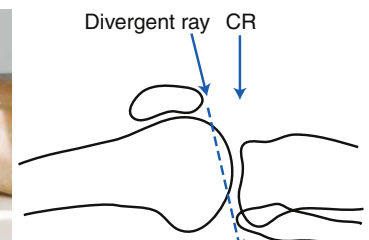


Fig. 1.140 Incorrect CR centering for knee.

TABLE 1.6 SUMMARY OF FACTORS CONTRIBUTING TO DISTORTION

QUALITY FACTOR	PRIMARY CONTRIBUTING FACTORS
Distortion	SID (Source-Image receptor Distance) OID (Object-Image receptor Distance) Object IR alignment CR alignment or centering

several rules must be followed to ensure optimal image quality when grids are used. Incorrect use of grids results in loss of optical density across all or part of the radiographic image; this feature is called *grid cutoff*. Grid cutoff occurs in various degrees and has several causes. Causes of grid cutoff include the following:

1. Off-center grid
2. Off-level grid
3. Off-focus grid
4. Upside-down grid

Off-center grid The CR must be centered along the center axis of the grid. If it is not, lateral decentering is said to occur. The more the CR is off center from the centerline of the grid, the greater is the cutoff that results (Fig. 1.141).

In certain clinical situations in which it is difficult to position the area of interest in the center of the grid, the grid may have to be turned so that the lead strips run perpendicular to the length of the patient to allow accurate centering (e.g., horizontal beam lateral lumbar spine).

Exception: decubitus—short dimension (SD)—type linear grids An exception to the more common lengthwise focused grid with the lead strips and center axis running lengthwise with the grid is the **decubitus-type** crosswise linear grid. This grid, in which the lead strips and center axis are running crosswise along the shorter dimension of the grid, is useful for horizontal beam decubitus-type projections. For these projections, the grid is placed lengthwise with the patient, but the CR is centered along the crosswise axis of the grid to prevent grid cutoff.

Off-level grid With angling, the CR must be angled along the long axis of the lead strips. Angling across the grid lines results in grid cutoff. Off-level grid cutoff also occurs if the grid is tilted; the CR hits the lead lines at an angle (Fig. 1.142).

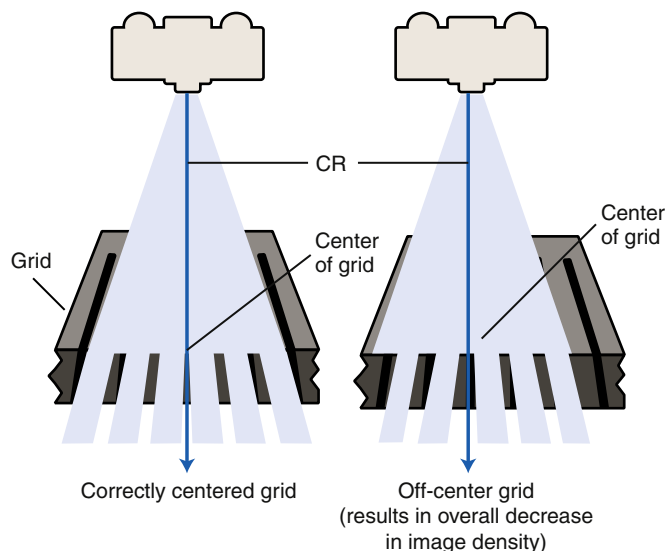


Fig. 1.141 Off-center grid cutoff.

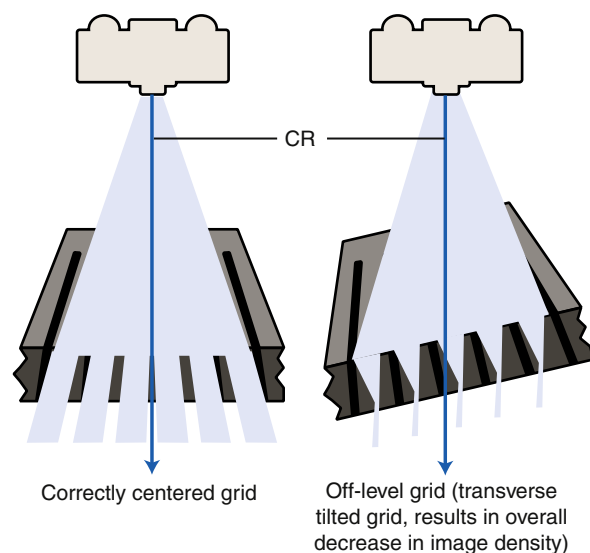


Fig. 1.142 Off-level grid cutoff.

Off-focus grid A focused grid must be used at a specified SID if grid cutoff is to be prevented. Grids typically have a minimum and a maximum usable SID; this is called the *focal range*. The focal range is determined by the **grid frequency** (number of grid strips per inch or centimeter) and the **grid ratio** (height of lead strips compared with the space between them). Portable grids generally have a lower grid frequency and a lower grid ratio than fixed grids or bucky-type grids. A common grid ratio for portable grids is **6:1** or **8:1** compared with **12:1** for bucky grids. This indicates a greater focal range for portable grids, but SID limitations still exist to prevent grid cutoff (Fig. 1.143). Each technologist should know which types of portable grids are available and should know the focal range of each.

Upside-down focused grid Each grid is labeled to indicate the side that must be positioned to face the x-ray tube. The lead strips are tilted or focused to allow the x-ray beam to pass through unimpeded (if the SID is within the focal range and the grid is correctly placed). If the grid is positioned upside-down, the image will show severe cutoff (Fig. 1.144).

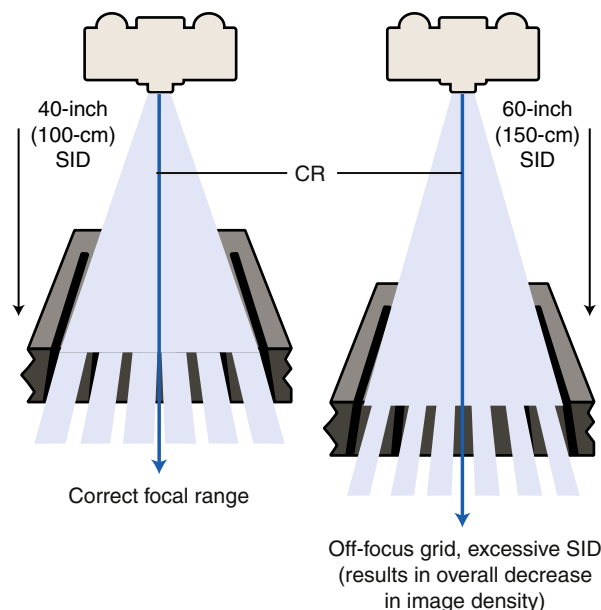


Fig. 1.143 Off-focus grid cutoff.

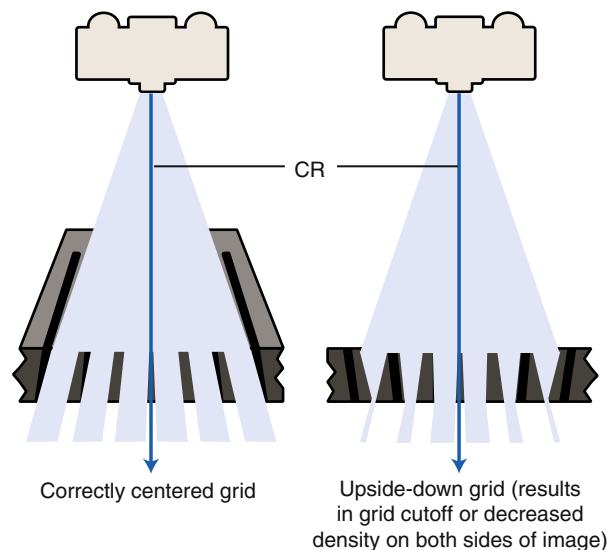


Fig. 1.144 Upside-down grid cutoff.

IMAGE QUALITY IN FILM-SCREEN (ANALOG) RADIOGRAPHY

Since the discovery of x-rays in 1895, methods of acquiring and storing x-ray images have evolved. Conventional film-screen technology with the associated chemical processing and film libraries is being replaced rapidly by digital technology. Digital technology uses computers and x-ray receptors to acquire and process images; specialized digital communication networks are used to transmit and store the x-ray images.

Analog Images

Analog (film) images provide a two-dimensional image of anatomic structures. The image acquisition device is a film-screen system that consists of a pair of intensifying screens with a film between them. The screens and film are housed in an x-ray cassette that protects the film from light and ensures that screens are in close contact with the film. When screens receive the remnant radiation from the patient, they fluoresce; this light exposes the film, which must be chemically processed so the image can be viewed. Chemical processing includes several steps (developing, fixing, washing, and drying) and typically takes 60 to 90 seconds.

The film image (radiograph), which actually is composed of a deposit of metallic silver on a polyester base, is permanent; it cannot be altered. The various shades of gray displayed on the image are representative of the densities and atomic numbers of the tissues being examined. The film image is often referred to as a *hard-copy image*.

Analog image receptors are best described as self-regulating systems with a limited dynamic range. Analog image receptors are also described using the term *exposure latitude*. Exposure latitude is the range of exposure over which a film produces an acceptable image. An image produced with a level of exposure outside of the exposure latitude is an unacceptable image. Figs. 1.145 and 1.146 illustrate the dynamic range and exposure latitude of an analog IR. Note the impact of doubling the mAs on the diagnostic quality of the images of the elbow. Analog images have relatively narrow exposure latitude.

Image Quality Factors

Film-based radiographic images are evaluated on the basis of **four quality factors**. These four primary image quality factors are:

- Density
- Contrast
- Spatial resolution (discussed earlier)
- Distortion (discussed earlier)

Each of these factors has specific parameters by which it is controlled.

DENSITY

Definition

Radiographic film density is defined as the **amount of “blackness” on the processed radiograph**. When a radiograph with high density is viewed, less light is transmitted through the image.

Controlling Factors

The **primary controlling factor** of film density is **mAs**. mAs controls density by controlling the quantity of x-rays emitted from the x-ray tube and the duration of the exposure. The relationship for our purpose can be described as linear; doubling the mAs doubles the quantity or duration of x-rays emitted, thus doubling the density on the film.

The distance of the x-ray source from the IR, or the **source–image receptor distance**, also has an effect on radiographic density according to the inverse square law. If the SID is doubled, at the IR, the intensity of the x-ray beam is reduced to one-fourth,

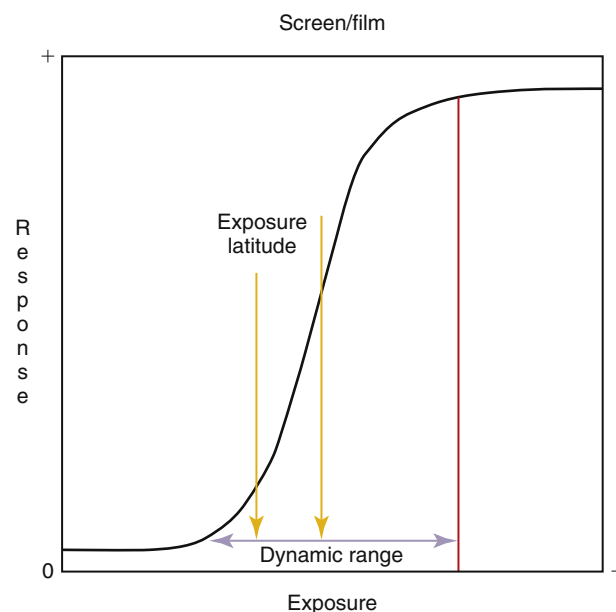


Fig. 1.145 Analog dynamic range.

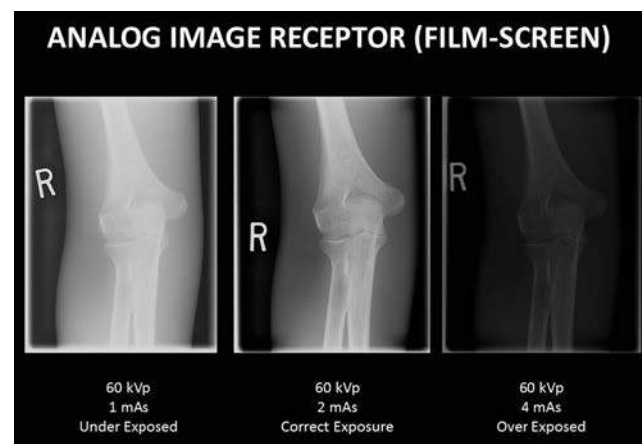


Fig. 1.146 Analog exposure latitude.

which then reduces radiographic density to one-fourth. A standard SID generally is used to reduce this variable.

Other factors that influence the density on a film image include kVp, part thickness, chemical development time/temperature, grid ratio, and film-screen speed.

Adjusting Analog Image Density

When film images (made with manual technique settings) are underexposed or overexposed, a general rule states that a minimum change in mAs of 25% to 30% is required to make a visible difference in radiographic density on the repeat radiograph. Some incorrectly exposed images may require a greater change, frequently 50% to 100%, or sometimes even greater. The radiograph of the elbow obtained with the use of 1 mAs shown in Fig. 1.147 was underexposed; the repeat radiograph was obtained with the use of 2 mAs (Fig. 1.148). Doubling the mAs in this example resulted in doubling of the density on the radiograph. kVp should not require an adjustment, provided that the optimal kVp for the part thickness was used. SID also should not require adjustment; it is a constant.



Fig. 1.147 1 mAs (60 kVp)—underexposed.



Fig. 1.148 2 mAs (60 kVp)—repeated, double mAs.

Density and Anode Heel Effect

The intensity of radiation emitted from the cathode end of the x-ray tube is greater than that emitted at the anode end; this phenomenon is known as the *anode heel effect*. Greater attenuation or absorption of x-rays occurs at the anode end because of the angle of the anode; x-rays emitted from deeper within the anode must travel through more anode material before exiting; thus, they are attenuated more.

Studies show that the difference in intensity from the cathode to the anode end of the x-ray field when a 17-inch (43-cm) IR is used at 40-inch (100-cm) SID can vary by 45%, depending on the anode angle¹³ (Fig. 1.149). The anode heel effect is more pronounced when a short SID and a large field size are used.

Applying the anode heel effect to clinical practice assists the technologist in obtaining quality images of body parts that exhibit significant variation in thickness along the longitudinal axis of the x-ray field. The patient should be positioned so that the **thicker portion of the part is at the cathode end** of the x-ray tube and the **thinner part is under the anode** (the cathode and anode ends of the x-ray tube usually are marked on the protective housing). The abdomen, thoracic spine, and long bones of the limbs (e.g., the femur and tibia/fibula) are examples of structures that vary enough in thickness to warrant correct use of the anode heel effect.

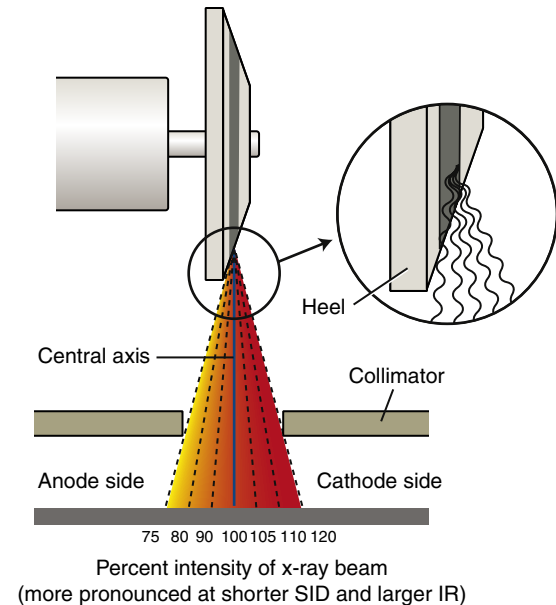


Fig. 1.149 Anode heel effect.

TABLE 1.7 SUMMARY OF ANODE HEEL EFFECT APPLICATIONS

PROJECTION	ANODE END	CATHODE END
Thoracic Spine		
AP	Head	Feet
Femur		
AP and lateral (see Fig. 1.123)	Feet	Head
Humerus		
AP and lateral	Elbow	Shoulder
Leg (Tibia/Fibula)		
AP and lateral	Ankle	Knee
Forearm		
AP and lateral	Wrist	Elbow

A summary chart of body parts and projections for which the anode heel effect can be applied is provided in Table 1.7; this information is also noted in the positioning pages for each of these projections throughout the text. In practice, the most common application of the anode heel effect is for AP projections of the thoracic spine.

It may not always be practical or even possible to take advantage of the anode heel effect; this depends on the patient's condition or the arrangement of specific x-ray equipment within a room.

Compensating Filters

As was discussed in the previous section, body parts of varying anatomic density may result in an image that is partially overexposed or underexposed because the anatomic parts attenuate the beam differently. This problem can be overcome through the use of *compensating filters*, which filter out a portion of the primary beam toward the thin or less dense part of the body that is being imaged. Several types of compensating filters are in use; most are made of aluminum; however, some include plastic as well. The type of compensating filter used by the technologist depends on the clinical application (Fig. 1.150).

Compensating filters in common use include the following:

- Wedge filter (Fig. 1.151A): Mounts on the collimator; the thicker portion of the wedge is placed toward the least dense part of the anatomy to even out the densities. This filter has numerous applications; the most common include AP foot, AP thoracic spine, and axiolateral projection of the hip.
- Trough filter: Mounts on the collimator and is used for chest imaging. The thicker peripheral portions of the filter are placed

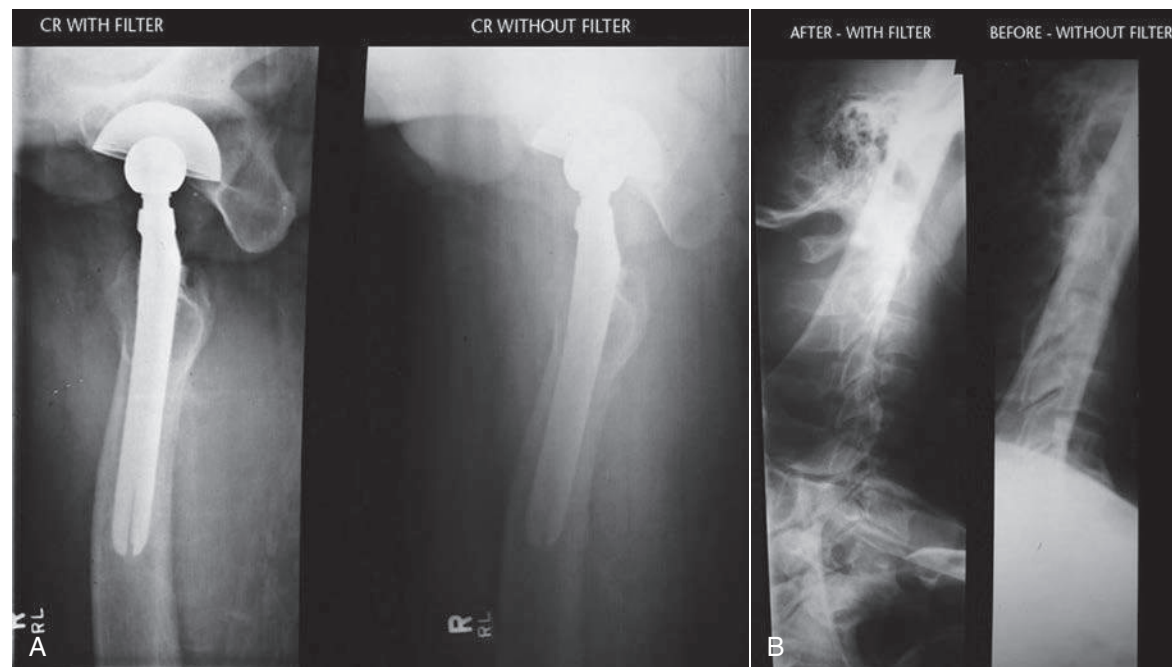


Fig. 1.150 Radiographic applications of compensating filters: hip (A) and upper thoracic spine (B). (Courtesy Ferlic Filters, Ferlic Filter Co, LLC.)

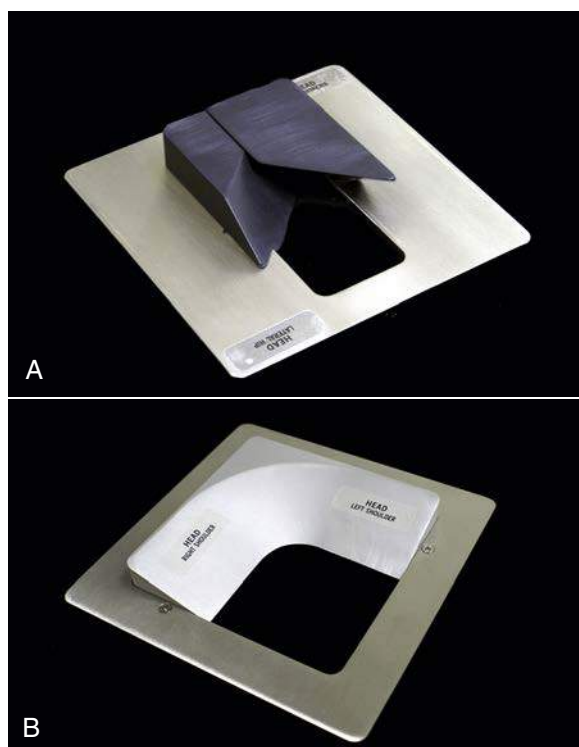


Fig. 1.151 Wedge (A) and boomerang (B) compensating filters (for use for upper thoracic spine and lateral hip projections). (Courtesy Ferlic Filters, Ferlic Filter Co, LLC.)

to correspond to the anatomically less dense lungs; the thinner portion of the filter corresponds to the mediastinum.

- Boomerang filter (Fig. 1.151B): Placed behind the patient and used primarily for shoulder and upper thoracic spine radiography; it provides improved visualization of soft tissues on the superior aspect of the shoulder and upper thoracic spine.

Summary of Density Factors

Adequate density, as **primarily controlled by mAs**, must be visible on processed film if the structures being radiographed are to be accurately represented. Too little density (underexposed) or too

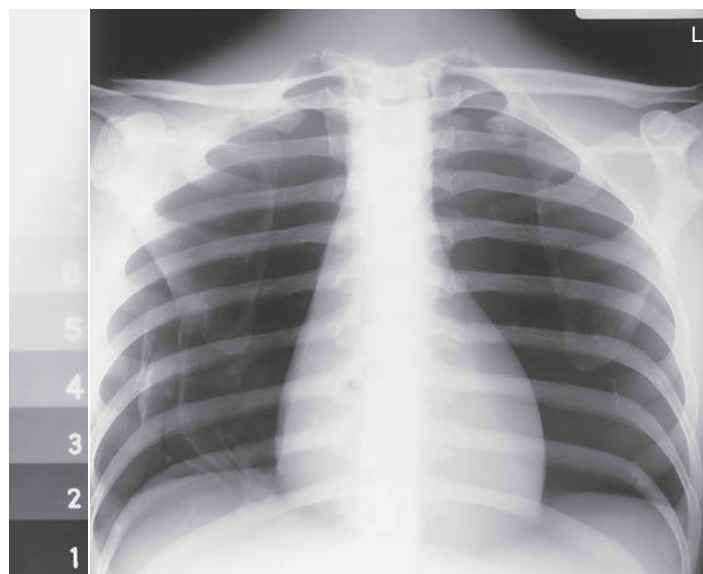


Fig. 1.152 High-contrast, short-scale 50 kVp, 800 mAs.

much density (overexposed) does not adequately demonstrate the required structures. Correct use of the anode heel effect and compensating filters helps to demonstrate optimal film density on anatomic parts that vary significantly in thickness.

CONTRAST

Definition

Radiographic contrast is defined as the **difference in density between adjacent areas of a radiographic image**. When the density difference is large, the contrast is high, and when the density difference is small, the contrast is low. This is demonstrated by the step wedge and by the chest radiograph in Fig. 1.152, which shows greater differences in density between adjacent areas; thus, this would be **high contrast**. Fig. 1.153 shows **low contrast** with less difference in density on adjacent areas of the step wedge and the associated radiograph.

Contrast can be described as **long-scale** or **short-scale contrast**, referring to the total range of optical densities from the lightest to the darkest part of the radiographic image. This is also demonstrated in Fig. 1.152, which shows short-scale/high-contrast (greater

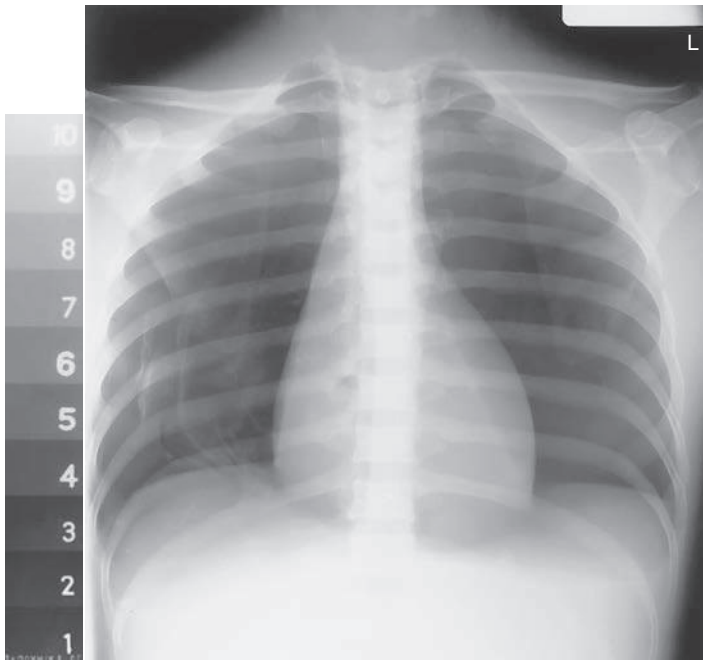


Fig. 1.153 Low-contrast, long-scale 110 kVp, 10 mAs.

differences in adjacent densities and fewer visible density steps), compared with Fig. 1.153, which illustrates long-scale/low-contrast.

Contrast allows the anatomic detail on a radiographic image to be visualized. Optimum radiographic contrast is important, and an understanding of contrast is essential for evaluating image quality.

Low or high contrast is not good or bad by itself. For example, low contrast (long-scale contrast) is desirable on radiographic images of the chest. Many shades of gray are required for visualization of fine lung markings, as is illustrated by the two chest radiographs in Figs. 1.152 and 1.153. The low-contrast (long-scale contrast) image in Fig. 1.153 reveals more shades of gray, as evident by the faint outlines of vertebrae that are visible through the heart and the mediastinal structures. The shades of gray that outline the vertebrae are less visible through the heart and the mediastinum on the high-contrast chest radiograph shown in Fig. 1.152.

Adjusting Analog Image Contrast

Contrast in film-based imaging may be adjusted in a variety of ways. The radiographer may choose to alter the kVp using the 15% rule; change the amount of beam restriction; or change grid ratio. In each of those circumstances, the technologist will need to make changes in the mAs settings in order to compensate for adjustment made for the change in contrast.

Controlling Factors

The **primary controlling factor** for contrast in film-based imaging is kilovolts peak (**kVp**). kVp controls the energy or penetrating power of the primary x-ray beam. The higher the kVp, the greater the energy and the more uniformly the x-ray beam penetrates the various mass densities of all tissues. Therefore, **higher kVp** produces less variation in attenuation (differential absorption), resulting in **lower contrast**.

kVp is also a **secondary controlling factor** of density. Higher kVp, resulting in both more numerous x-rays and greater energy x-rays, causes more x-ray energy to reach the IR, with a corresponding increase in overall density. A general rule of thumb states that a **15% increase in kVp will increase film density, similar to doubling the mAs**. In the lower kVp range, such as 50 to 70 kVp, an 8- to 10-kVp increase would double the density (equivalent to doubling the mAs). In the 80- to 100-kVp range, a 12- to 15-kVp increase is required to double the density. The importance of this relates to radiation protection, because as kVp is increased, mAs can be significantly reduced, resulting in absorption of less radiation by the patient.

Other factors may affect radiographic contrast. The amount of *scatter* radiation the film-screen receives influences the radiographic contrast. Scatter radiation is radiation that has been changed in direction and intensity as a result of interaction with patient tissue. The amount of scatter produced depends on the intensity of the x-ray beam, the amount of tissue irradiated, and the type and thickness of the tissue. Close collimation of the x-ray field reduces the amount of tissue irradiated, reducing the amount of scatter produced and increasing contrast. Close collimation also reduces the radiation dose to the patient and the technologist.

Irradiation of thick body parts produces a considerable amount of scatter radiation, which decreases image contrast. A device called a *grid* is used to absorb much of the scatter radiation before it hits the IR.

Summary of Contrast Factors

Selection of the appropriate kVp is a balance between optimal image contrast and lowest possible patient dose. A general rule states that the **highest kVp and the lowest mAs that yield sufficient diagnostic information should be used on each radiographic examination**.¹³ Close collimation and correct use of grids also ensure the processed radiographic image displays optimal contrast.

SPATIAL RESOLUTION

Controlling Factors Specific to Analog

Film-screen system With film-screen imaging systems, the *film-screen speed* used for an examination affects the detail shown on the resultant film. A faster film-screen system allows shorter exposure times, which are helpful in preventing patient motion and reducing dose; however, the image is less sharp than when a slower system is used.

IMAGE QUALITY IN DIGITAL RADIOGRAPHY

Digital imaging in radiologic technology involves application of the analog-to-digital conversion theory and computer software and hardware. Although digital imaging differs from film-screen imaging in terms of the method of image acquisition, factors that may affect x-ray production, attenuation, and geometry of the x-ray beam still apply. This section provides a brief practical introduction to a very complex topic.

Digital Images

Digital radiographic images also provide a two-dimensional image of anatomic structures; however, they are viewed on a computer monitor and are referred to as **soft-copy images**. These images are a **numeric representation of the x-ray intensities that are transmitted through the patient**. Each digital image is two-dimensional and is formed by a *matrix* of picture elements called *pixels* (Fig. 1.154). In diagnostic imaging, each pixel represents the smallest unit in the image; columns and rows of pixels make up the matrix. For illustrative purposes, consider a sheet of graph paper. The series of squares on the sheet can be compared with the matrix, and each individual square can be compared with a pixel.

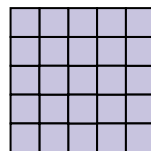
Digital imaging requires the use of computer hardware and software applications to view images (Fig. 1.155), whereas film-based images use chemical processing to visualize anatomic structures. Digital processing involves the **systematic application of highly complex mathematical formulas** called *algorithms*. Numerous mathematical manipulations are performed on image data to enhance image appearance and to optimize quality. Algorithms are applied by the computer to every data set obtained before the technologist sees the image.

Digital imaging systems are capable of producing a radiographic image across a large range of exposure values and are described as having a wide dynamic range (Fig. 1.156). Because of this wide dynamic range, it is essential that an institution define the exposure latitude for the digital imaging systems within its department. The exposure latitude for a digital imaging system is defined as the acceptable level of exposure that produces the desired image quality for the department. Fig. 1.157 demonstrates the dynamic range and exposure latitude of a digital imaging system. Note that the increase from 1 to 8 mAs still produces a diagnostic image of the elbow.

Exposure Factors for Digital Imaging

Although kVp and mA and time (mAs) must be selected if radiographic images are to be digitally acquired (see Fig. 1.157), they do not have the same direct effect on image quality as they do in film-screen imaging. It must be remembered, however, that the kVp and mAs used for the exposure affect patient dose.

Display matrix



Pixel

Fig. 1.154

Two-dimensional matrix display—pixel.



Fig. 1.155 Processing digital image.

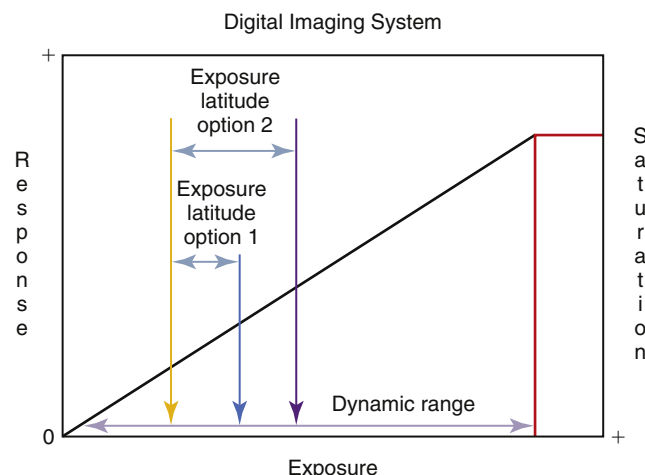


Fig. 1.156 Digital imaging systems.

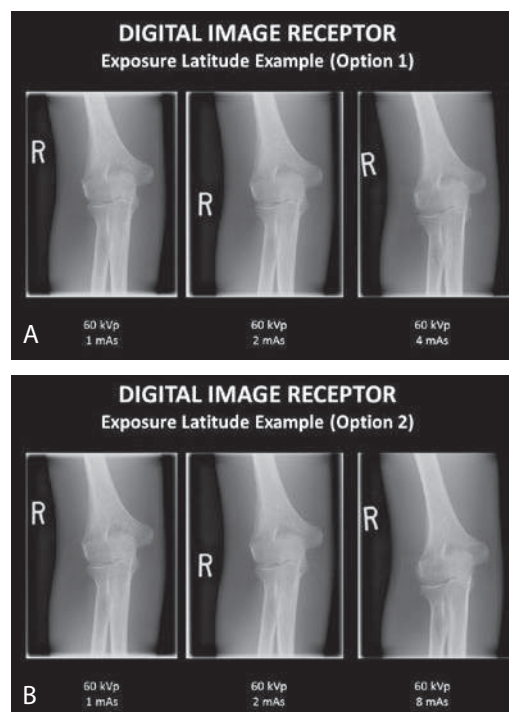


Fig. 1.157 Digital exposure latitudes. A, Option 1. B, Option 2.

Image Quality Factors

The factors used to evaluate digital image quality include the following:

- Brightness
- Contrast resolution
- Spatial resolution (discussed earlier)
- Distortion (discussed earlier)
- Exposure indicator
- Noise

BRIGHTNESS

Brightness is defined as the intensity of light that represents the individual pixels in the image on the monitor. In digital imaging, the term *brightness* replaces the film-based term *density* (Figs. 1.158 and 1.159).

Controlling Factors

Digital imaging systems are designed to display electronically the optimal image brightness under a wide range of exposure factors. Brightness is controlled by the processing software through the application of predetermined digital processing algorithms. In contrast to the linear relationship between mAs and density in film-screen imaging, changes in mAs do not have a controlling effect on digital image brightness. Although the density of a film image cannot be altered once it is exposed and chemically processed, the user can adjust the brightness of the digital image after exposure (see section on postprocessing later in this chapter).

CONTRAST RESOLUTION

In digital imaging, *contrast* is defined as the **difference in brightness between light and dark areas of an image**. This definition is similar to the definition used in film-based imaging, where contrast is the difference in density of adjacent areas on the film (Figs. 1.160 and 1.161 show examples of different contrast images). *Contrast resolution* refers to the ability of an imaging system to distinguish between similar tissues.

Controlling Factors

Digital imaging systems are designed to display electronically optimal image contrast under a wide range of exposure factors. Radiographic contrast is affected by the digital processing computer through the application of predetermined algorithms, in contrast to film-screen imaging, in which kVp is the controlling factor for image contrast. Although the contrast of a film image cannot be altered after exposure and processing, the user can manipulate the contrast of the digital image (see later section on postprocessing).

The ability of the image processing software to display a desired image contrast provides the radiographer with a potential opportunity to reduce entrance skin exposure to the patient through the use of higher kVp levels. Figs. 1.162 to 1.164 represent the ability to use the 15% rule and decrease patient entrance skin dose by approximately 22%. It is critical that the radiographer consult with the interpreting radiologist and medical physicist prior to implementing across-the-board kVp increases in order to ensure that acceptable image quality is maintained.

Pixels and Bit Depth

Each pixel in an image matrix demonstrates a single shade of gray when viewed on a monitor; this is representative of the physical properties of the anatomic structure. The range of possible shades of gray demonstrated is related to the *bit depth* of the pixel, which is determined by the manufacturer. Although a comprehensive description of bit depth is beyond the scope of this text, it is important to note that the **greater the bit depth of a system, the greater the contrast resolution (i.e., the greater the number of possible shades of gray that a pixel can have)**.

Because computer theory is based on the binary system, a 14-bit system, for example, is represented as 2^{14} ; the 14-bit-deep pixel could represent any one of 16,384 possible shades of gray, from black to white. Bit depth is determined by the manufacturer's system design and is closely related to the imaging procedures for which the equipment is designed. The most common bit depths available are 10, 12, and 16. For example, a digital system for chest imaging should have a bit depth greater than 10 bits (2^{10}) if it is to capture all required information; the x-ray beam that exits a patient who is having a chest x-ray can have a range of more than 1024 intensities.

Pixel size Two pixel sizes are used in medical imaging. These are *acquisition pixel size*, which is the minimum size that is inherent to the acquisition system, and *display pixel size*, which is the minimum pixel size that can be displayed by a monitor. A general radiography acquisition matrix may be 3000 × 3000 pixels—more than 9 million pixels (9 megapixels)—in a 17 × 17-inch (43 × 43-cm) image.

Scatter Radiation Control

Because digital receptors are more sensitive to low-energy radiation, controlling scatter radiation is an important factor in obtaining the appropriate image contrast. This is accomplished by the correct use of grids, by close collimation, and by selection of the optimal kVp. Grid cutoff occurring with digital image receptors will result in an image that has decreased contrast and has an exposure indicator reflecting a decrease in exposure. The change in exposure indicator is due to the decrease in amount of exit radiation striking the receptor.

Postprocessing software has been developed to minimize the presentation of scatter radiation on the final displayed image. It is important to recognize; however, that it does not prevent scatter radiation from reaching the image receptor, it simply minimizes to display of scatter on the final image.



Fig. 1.158 AP pelvis—high brightness (light).



Fig. 1.159 AP pelvis—less brightness (dark).



Fig. 1.160 AP pelvis—higher contrast.



Fig. 1.161 AP pelvis—lower contrast.

SPATIAL RESOLUTION

Spatial resolution in digital imaging is defined as the **recorded sharpness or detail of structures on the image**—the same as defined for film-screen imaging.

Resolution in a digital image represents a combination of the traditional factors explained previously for film-screen imaging (focal spot size, geometric factors, and motion) and, just as important, the **acquisition pixel size**. This pixel size is inherent to the digital imaging receptor. The smaller the acquisition pixel size, the greater the spatial resolution. Spatial resolution is measured in line pairs per millimeter. Current digital imaging systems employed for general radiography have spatial resolution capabilities ranging from approximately 2.5 lp/mm to 5 lp/mm.

Controlling Factors

In addition to acquisition **pixel size**, resolution is controlled by the **display matrix**. The perceived resolution of the image depends on the display capabilities of the monitor. Monitors with a larger display matrix can display images with higher resolution.

EXPOSURE INDICATOR

The *exposure indicator* in digital imaging is a **numeric value that is representative of the exposure that the IR has received**. Depending on the manufacturer of the system, the exposure indicator may also be called the *sensitivity (S) number*; *Exposure Index (EI, EXI)*; *Reached Exposure Value (REX)*; or possibly *Detector Exposure Index (DEI)*.

Deviation Index

In addition to the *Exposure Index*, there is also a *Deviation Index (DI)*. The DI provides feedback to the operator regarding receptor exposure. The DI ranges from -3 to $+3$. A DI value of 0 indicates that the level of exposure was appropriate. A positive DI value indicates overexposure and a negative DI value indicates underexposure. A DI value of $+1$ indicates a 26% overexposure whereas

a value of -1 indicates a 20% underexposure. The DI values of $+3$ and -3 represent exposure levels that are two times greater and less than the defined target exposure index.

Controlling Factors

The exposure indicator depends on the dose of radiation that strikes the receptor. It is a value that is calculated from the effect of mAs, the kVp, the total receptor area irradiated, and the objects exposed (e.g., air, metal implants, patient anatomy). Depending on the manufacturer and the technique used to calculate this value, the exposure indicator is displayed for each exposure.

An exposure indicator, as used by certain manufacturers, is **inversely related** to the radiation that strikes the receptor. For example, if the range for an acceptable image for certain examinations is 150 to 250, a value greater than 250 would indicate underexposure, and a value less than 150 would indicate overexposure.

An **exposure indicator** as used by other manufacturers is **directly related** to the radiation striking the IR, as determined by logarithmic calculations. For example, if an acceptable exposure indicator is typically 2.0 to 2.4, an indicator value less than 2.0 would indicate underexposure, whereas an indicator value greater than 2.4 would indicate overexposure.

This text uses the term **exposure indicator** when referring to this variable.

It has been stated previously that digital imaging systems are able to display images that have been obtained through the use of a wide range of exposure factors. Despite this wide dynamic range, there are limitations, and the technologist must ensure that the exposure factors used are acceptable and within the institution's defined exposure latitude (similar to reviewing an analog [film] image to confirm that adequate contrast and density are present) (see Figs. 1.162 to 1.164). Checking the exposure indicator is key in verifying that acceptable **quality digital radiographic images have been obtained with the least possible dose to the patient**.



Fig. 1.162 Low exposure indicator indicates underexposure with “noisy” undesirable image.



Fig. 1.163 Example of desirable exposure with acceptable exposure indicator.



Fig. 1.164 High exposure indicator indicates overexposure.

If the exposure indicator is outside the recommended range for the digital system, the image may still appear acceptable when viewed on the monitor of the technologist's workstation. The monitor the technologist uses to view the image typically provides lower resolution than is provided by the radiologist's reporting workstation. The technologist's workstation is intended to allow verification of positioning and general image quality; however, this image is typically not of diagnostic quality. The monitor of a radiologist's reporting workstation typically provides superior spatial and contrast resolution caused by an increased display matrix with smaller pixels and superior brightness characteristics.

NOISE

Noise is defined as a **random disturbance that obscures or reduces clarity**. In a radiographic image, this translates into a grainy or mottled appearance of the image.

Signal-to-Noise Ratio

One way to describe noise in digital image acquisition is the concept of *signal-to-noise ratio* (SNR). The number of x-ray photons that strike the receptor (mAs) can be considered the "*signal*." Other factors that negatively affect the final image are classified as "*noise*." A **high SNR is desirable** in imaging, in which the signal (mAs) is greater than the noise, so that low-contrast soft tissue structures can be demonstrated. A **low SNR is undesirable**; a low signal (low mAs) with accompanying high noise obscures soft tissue detail and produces a grainy or mottled image.

High SNR Although a high SNR is favorable (Fig. 1.165), technologists must ensure exposure factors used are not beyond what is required for the projection so as not to overexpose the patient needlessly. Overexposed images are not readily evident with digital processing and display, so checking the exposure indicator as described on the previous page is the best way to determine this.

Low SNR When insufficient mAs is selected for a projection, the receptor does not receive the appropriate number of x-ray photons, resulting in a low SNR and a *noisy* image (Fig. 1.166). This insufficient signal may not be readily visible on the lower resolution monitor of the technologist's workstation, but the exposure indicator, as checked for each projection, can aid in determining this. The technologist may check for noise at the workstation by using the magnify feature and magnifying the image to determine the level of noise present within the image. In the event that noise is clearly visible in the image without any magnification, the image should be reviewed by the radiologist to determine if the image needs to be repeated.

Scatter radiation leads to a degradation of image contrast that can be controlled by the use of grids and correct collimation, as was described previously.

A secondary factor related to noise in a radiographic image is *electronic noise*. Although a comprehensive discussion of electronic noise is beyond the scope of this text, electronic noise typically results from inherent noise in the electronic system, nonuniformity of the image receptor, or power fluctuations.

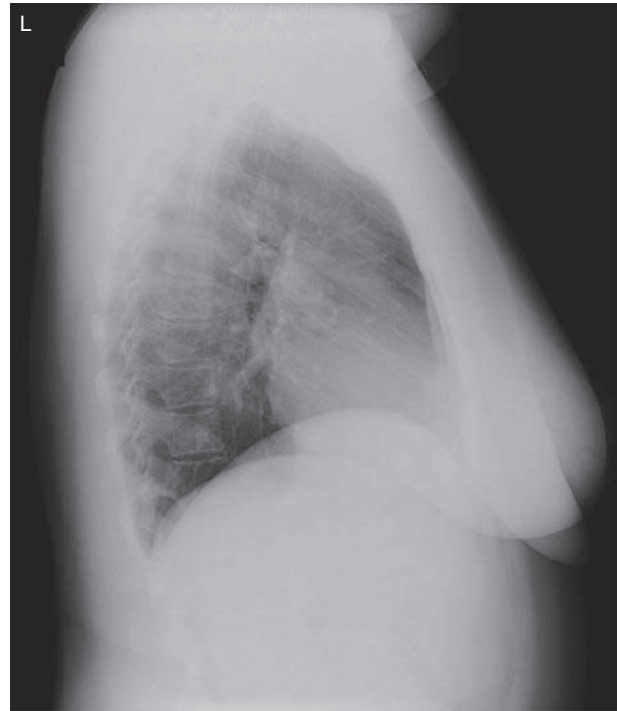


Fig. 1.165 Good-quality image—acceptable SNR.

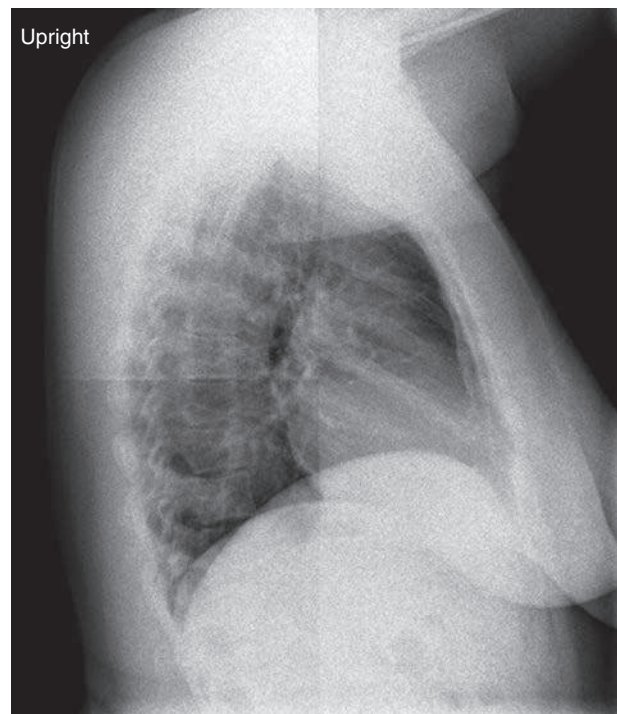


Fig. 1.166 Poor-quality image, "noisy" (grainy)—low SNR.

POSTPROCESSING

One of the advantages of digital imaging technology over film-screen technology is the ability to *postprocess* the image at the technologist's workstation. Postprocessing refers to **changing or enhancing the electronic image for the purpose of improving its diagnostic quality**. In postprocessing, algorithms are applied to the image to modify pixel values. Once viewed, the changes made may be saved, or the image default settings may be reapplied to enhance the diagnostic quality of the image. It is important to note that an image modified at the technologist's workstation and sent to the picture archiving and communication system (PACS) cannot be unmodified on the PACS. As a result of this inability to undo changes made at the technologist's workstation once the image is sent to the PACS, postprocessing of images at the technologist's workstation should be avoided.

Postprocessing and Exposure Indicator Range

After an acceptable exposure indicator range for the system has been established, it is important to determine whether the image is inside or outside this range. If the exposure indicator is below this range (indicating low SNR), postprocessing would not be effective in minimizing noise; more "signal" cannot be created through postprocessing. Theoretically, if the algorithms are correct, the image should display with the optimal contrast and brightness. However, even if the algorithms used are correct and exposure factors are within an acceptable range, as indicated by the exposure indicator, certain postprocessing options may still be applied for specific image effects.

POSTPROCESSING OPTIONS

Various postprocessing options are available in medical imaging (Figs. 1.167 through 1.170). The most common of these options include the following:

Windowing and Leveling: The user can adjust image contrast and brightness on the monitor. Two types of adjustment are possible: *window width (windowing)* which controls the **contrast** of the image (within a certain range), and *window level (leveling)* which controls the **brightness** of the image, also within a certain range. It is important to note that when adjusting the display window for a digital radiograph, the manner in which the values assigned for each characteristic vary is dependent on the viewing system software. In some PACS, increasing the window level results in a darker image, and in others, it results in a brighter image.

Smoothing: Specific image processing is applied to reduce the display of noise in an image. The process of smoothing the image data does not eliminate the noise present in the image at the time of acquisition.

Magnification: All or part of an image can be magnified.

Edge enhancement: Specific image processing that alters pixel values in the image is applied to make the edges of structures appear more prominent compared with images with less or no edge enhancement. The spatial resolution of the image does not change when edge enhancement is applied.

Equalization: Specific image processing that alters the pixel values across the image is applied to present a more uniform image appearance. The pixel values representing low brightness are made brighter, and pixel values with high brightness are made to appear less bright.

Subtraction: Background anatomy can be removed to allow visualization of contrast media-filled vessels (used in angiography).

Image reversal: The dark and light pixel values of an image are reversed—the x-ray image reverses from a negative to a positive.

Annotation: Text may be added to images.



Fig. 1.167 AP hip image applied to create angiographic mask.



Fig. 1.168 AP hip image with subtraction.

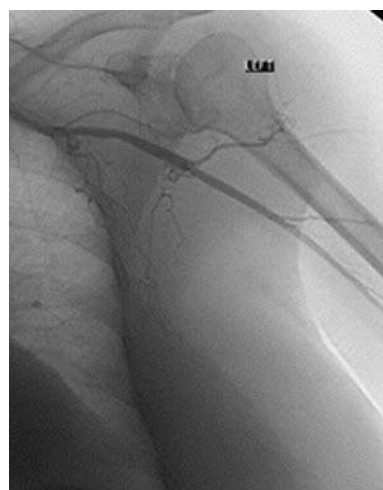


Fig. 1.169 Subtracted AP shoulder angiogram image.

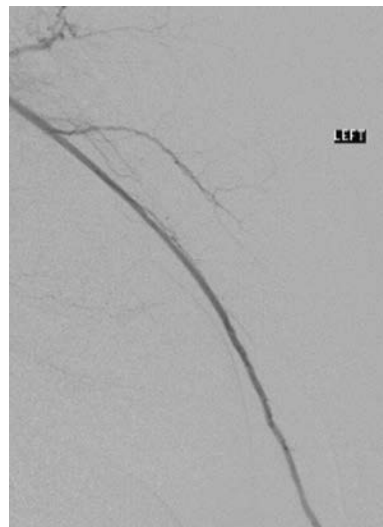


Fig. 1.170 Subtracted and magnified option of shoulder angiogram.

APPLICATIONS OF DIGITAL TECHNOLOGY

Although digital technology has been used for years in digital fluoroscopy and CT (further information on these modalities is available in [Chapters 12 and 18](#)), its widespread application to general imaging is more recent. This section introduces and briefly describes the digital imaging technology used in general radiography. Each of the systems described starts the imaging process using an x-ray beam that is captured and converted into a digital signal.

DIGITAL IMAGING SYSTEMS

The many acronyms associated with digital imaging have created a plethora of misconceptions regarding digital imaging systems, and these misconceptions have resulted in technologists not having a thorough understanding of how various digital imaging systems work. The following sections describe the current digital imaging systems, based first on how the image data are captured and extracted, and second on their appearance. Regardless of appearance and how the image data are captured and extracted, each of the digital imaging systems described has a wide dynamic range that requires a defined set of exposure indices to enable the technologist to adhere to the ALARA principle—as low as reasonably achievable.

Computed Radiography (CR) (Photostimulable Storage Phosphor Plate [PSP])

PSP technology was the first widely implemented digital imaging system for general radiography. It is most commonly called *computed radiography (CR)*. A CR digital imaging system relies on the use of a storage phosphor plate that serves the purpose of capturing and storing the x-ray beam exiting the patient. When the plate is exposed to radiation, electrons migrate to electron traps within the phosphor material. The greater the exposure to the plate, the more electrons move to the electron traps. The exposed plate containing the latent image undergoes a reading process following the exposure ([Fig. 1.171](#)). The reading of the plate involves scanning the entire plate from side to side using a laser beam. As the laser moves across the plate, the trapped electrons in the phosphor are released from the electron traps and migrate back to their resting location. This migration results in the emission of light from the phosphor. The greater the exposure to the plate, the greater the intensity of the light emitted from the plate during the reading process. The light released is collected by an optical system that sends the light to a device responsible for converting the light into an analog electrical signal. The device may be a photomultiplier tube or charge-coupled device (CCD). The analog electrical signal is sent to an analog-to-digital converter (ADC) so that the image data may be processed by the computer to create the desired digital image. Depending on the manufacturer, the image may be viewed on the technologist's workstation as quickly as 5 seconds after plate reading. After the reading process, the PSP plate is exposed to bright white light so that any remaining latent image is erased from the plate and the plate may be used for the next exposure.



Fig. 1.171 PSP cassette and reader.

A CR digital imaging system may be cassette based or cassette-less. A cassette-based system allows the technologist to place the IR physically in a variety of locations. The cassette-less system ([Figs. 1.172 and 1.173](#)) provides the technologist with a larger device that encloses the IR. The IR in a cassette-less system has a limited amount of movement to align with the x-ray beam and anatomic structure owing to its design. The appearance of the device is not an indication of what is happening inside of the device after exposure to the x-ray beam. Therefore, it is critical that technologists recognize and understand what is inside the equipment with which they work.

Technologist Workstation

The workstation includes a bar-code reader (optional), a monitor for image display, and a keyboard with a mouse or trackball for entering commands for postprocessing. The technologist verifies the patient position and checks the exposure indicator at this workstation.

IMAGE ARCHIVING

After the image quality has been verified and any needed adjustments have been made, the image can be transmitted to the digital archive for viewing and reading by the referring physician or radiologist. Images also may be printed onto film by a laser printer.



Fig. 1.172 Cassette-less imaging system.



Fig. 1.173 Cassette-less chest imaging system.

APPLICATION OF CR DIGITAL SYSTEMS

Regardless of the technology used to acquire radiographic images, accurate positioning and attention to technical details are important. However, when digital technology is used, attention to these details becomes more important because of the following factors.

COLLIMATION

In addition to the benefit of reducing radiation dose to the patient, collimation that is closely restricted to the part that is being examined is key to ensuring optimal image quality. The software processes the entire x-ray field as a data set; any unexpected attenuation of the beam may be included in the calculations for brightness, contrast, and exposure indicator. If the collimation is not closely restricted, the exposure indicator may be misrepresented, and the image may exhibit lower contrast or possibly incorrect brightness.

ACCURATE CENTERING OF PART AND IR

Because of the way the extracted image data are analyzed, the body part and collimated exposure field should be centered to the IR to ensure proper image display. Failure to align the part to the receptor accurately and collimate the exposure field properly may result in poor image quality on initial image display.

USE OF LEAD MASKS

Use of lead masks or a blocker for multiple images on a single IR is recommended when a cassette-based CR or analog system is used (Fig. 1.174). This recommendation is due to the hypersensitivity of the PSP plate to lower energy scatter radiation; even small amounts may affect the image.

NOTE: Some manufacturers recommend that only one image be centered and placed per IR. Check with your department to find out whether multiple images can be placed on a single IR.

USE OF GRIDS

Use of grids (as explained in an earlier section of this chapter) for body parts larger (thicker) than 10 cm is especially important when images are acquired with the use of PSP image receptors because of the hypersensitivity of the image plate phosphors to scatter radiation.

EXPOSURE FACTORS

Because of their wide dynamic range, CR systems are able to display an acceptable image from a broad range of exposure factors (kVp, mAs). It is important to remember, however, that the ALARA principle (exposure to patient as low as reasonably achievable) must be followed, and the lowest exposure factors required to obtain a diagnostic image must be used. When the image is available for viewing, the technologist must check the exposure indicator to verify that the exposure factors used are consistent with the ALARA principle and diagnostic image quality. In some circumstances it is possible to increase kVp by 5 to 10 while decreasing mAs by the equivalent ratio with digital imaging equipment to maintain image quality while significantly reducing entrance skin exposure dose to the patient (refer to Figs. 1.162 to 1.164).

EVALUATION OF EXPOSURE INDICATOR

As soon as the image is available for viewing at the workstation, it is critiqued for positioning and exposure accuracy. The technologist must check the exposure indicator to verify the exposure factors used were in the correct range for optimum quality with the lowest radiation dose to the patient.

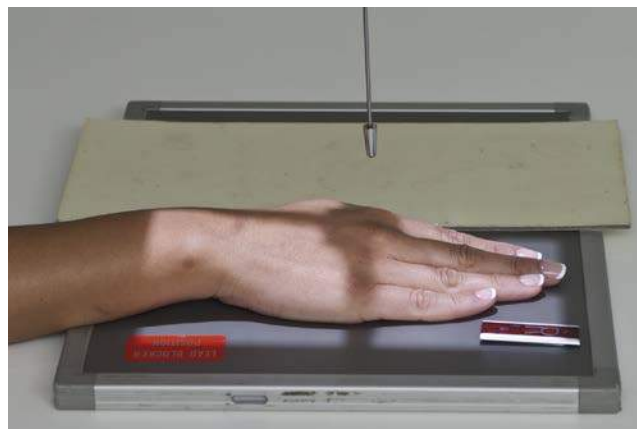


Fig. 1.174 Lead blockers on cassette and close collimation are important with the use of cassette-based PSP systems.

Direct Radiography (DR) (Flat-Panel Detector with Thin-Film Transistor)

The **direct radiography** imaging system for general radiography is a second type of digital imaging system. The flat-panel detector with thin-film transistor (FPD-TFT) system is commonly referred to as **digital radiography (DR)** or **direct digital radiography (DDR)**.

The DR unit may be constructed using amorphous selenium or amorphous silicon. The purpose of those two materials is to provide a source of electrons to the thin-film transistor that collects the electrons during the exposure. The creation of the electrons for the TFT is different with the two materials. The exposure of amorphous selenium to x-ray photons causes electrons to move through the material and into the electron collection portion of the TFT. Amorphous silicon requires the use of a scintillator, which produces light when struck by x-ray photons. The light exiting the scintillator causes the movement of electrons through the amorphous silicon and into the electron collection centers of the TFT. The TFT collects the electrons in an ordered manner and then sends the analog electrical signal to an ADC. The signal from the ADC is sent to the computer to create the digital image. The display of the radiographic image on the technologist's workstation with the DR system occurs within seconds after the exposure ends.

A DR imaging system may be cassette-less (Fig. 1.175) or cassette based (Fig. 1.176). The appearance of the IR does not indicate how the device captures and produces the image. Therefore, it is important for the technologist to know what type of IR is being used.

ADVANTAGES OF DR SYSTEMS

One advantage of DR-based systems compared with CR systems is that the DR system is capable of displaying the image faster. The faster image display applies to both the cassette-less and the cassette-based DR systems. One other advantage is the potential to produce diagnostic radiographs with lower levels of exposure. However, the ability to produce these images using a lower level of exposure depends on the manufacturer's choice of materials used to construct the system.

DR and CR systems both give the technologist the advantage of being able to view a *preview image* to evaluate for positioning errors and confirm the exposure indicator. The projection may be repeated immediately if necessary. Also, the operator is able to postprocess and manipulate the image.

As with CR systems and film-screen acquisition, DR-based systems can be used for both grid and nongrid examinations. In reality, however, when cassette-less DR-based systems are used for traditional nongrid examinations, the grid often is not removed for practical reasons: It is expensive and fragile and may be damaged easily. Because of the high efficiency of the receptor, the increase in exposure that is required when a grid is used is less of an issue; the exception to this would be pediatric examinations (because of the greater sensitivity of pediatric patients to radiation exposure).

APPLICATION OF DR-BASED SYSTEMS

Regardless of the digital technology used to acquire radiographic images, accurate positioning and attention to certain technical details are important, as described previously for CR systems. For



Fig. 1.175 FPD-TFT cassette-less imaging system.



Fig. 1.176 FPD-TFT cassette. (Courtesy Konica Minolta Medical Imaging, Inc.)

DR-based systems, these details include **careful collimation**, correct use of **grids**, and careful attention to **exposure factors and evaluation of exposure indicator values**, combined with adherence to the ALARA principle. When either CR or DR technology is used, attention to these details is essential.

The most current FPD-TFT-based receptors are cassette based with wireless capability. The wireless connectivity allows the radiographer to easily move the DR detector from the conventional table bucky to the upright bucky. Furthermore, the wireless DR detectors are routinely adapted to mobile radiographic units.

It is highly recommended that a department that mixes vendors and CR and DR technology develop standardized protocols to ensure image quality regardless of the method of image capture. Specifically, the department needs to establish exposure field sizes; projections per receptor, if permitted; and uniform processing parameters on all technologist workstations.

Image Receptor Sizes and Orientation

As noted previously, the term *image receptor*, or *IR*, applies to the device that captures the radiation that exits the patient; IR refers to the film cassette and to the digital acquisition device. Use of metric *Système Internationale* (SI) units to describe the size of analog cassettes and image receptors in CR has primarily replaced use of the English units. See Tables 1.8, 1.9, and 1.10 for a list of available IR size options for analog, CR, and DR.

TABLE 1.8 AVAILABLE IR SIZES IN ANALOG (FILM-SCREEN) IMAGING

METRIC (SI) SIZE (CM)	ENGLISH UNIT REFERENCE (INCHES)	CLINICAL APPLICATION
18 × 24	8 × 10	General imaging, mammography
24 × 24	9 × 9	Fluoroscopic spot imaging
18 × 43	7 × 17	General imaging
24 × 30	10 × 12	General imaging, mammography
30 × 35; 35 × 35; 30 × 40	11 × 14	General imaging
NA	14 × 36; 14 × 51	Full spine/lower limbs
35 × 43	14 × 17	General imaging
NA	5 × 12; 6 × 12	Mandible/orthopantomography

TABLE 1.9 AVAILABLE IR SIZES IN CR SYSTEMS

METRIC (SI) SIZE (CM)	ENGLISH UNIT REFERENCE (INCHES)	CLINICAL APPLICATION
18 × 24	8 × 10	General imaging, mammography
24 × 30	10 × 12	General imaging, mammography
35 × 35	14 × 14	General imaging
35 × 43	14 × 17	General imaging

TABLE 1.10 AVAILABLE IR SIZES IN DR SYSTEMS

METRIC (SI) SIZE (CM)	ENGLISH UNIT REFERENCE (INCHES)	CLINICAL APPLICATION
18 × 24	8 × 10	General imaging, mammography
24 × 30	10 × 12	General imaging, mammography
35 × 43	14 × 17	General imaging
43 × 43	17 × 17	General imaging

The selection of IR size depends primarily on the body part that is to be examined. The size and shape of the body part being examined also determine the orientation of the IR. If the IR is positioned with the longer dimension of the IR parallel to the long axis of the body part, the orientation is **portrait**; if the IR is positioned with the shorter dimension of the IR parallel to the long axis of the body part, the orientation is **landscape**. A common example applied to clinical practice relates to chest radiography. Patients who are hypersthenic are imaged with the IR in landscape orientation, so the lateral aspects of the chest may be included in the image (Fig. 1.177).

Students also may hear the terms *lengthwise* and *crosswise* used to describe IR orientation. These correspond to *portrait* and *landscape*, respectively.

The size of the image displayed will be affected by the size of the CR imaging plate chosen or the size of the collimated exposure field for DR systems. The change in image size is based on the number of pixels in the image matrix that in turn must be displayed on the technologist workstation or radiologist's reading station.

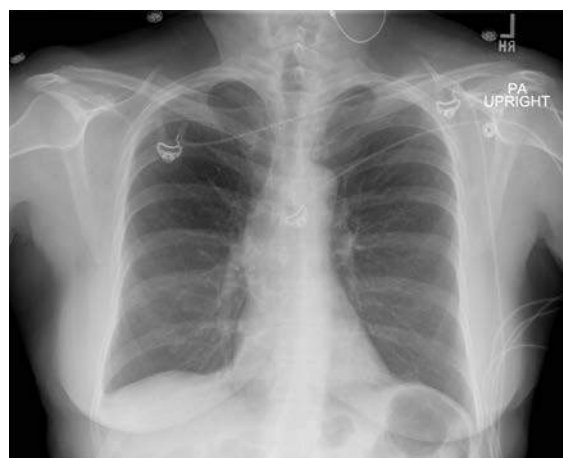


Fig. 1.177 PA chest landscape (crosswise) IR alignment.

Picture Archiving and Communication System

As imaging departments move from film-based acquisition and archiving (hard-copy film and document storage) to digital acquisition and archiving (soft-copy storage), a complex computer network has been created to manage images. This network, called a *picture archiving and communication system*, can be likened to a virtual film library. Images stored on digital media are housed in PACS archives.

PACS is a sophisticated array of hardware and software that can connect all modalities with digital output (nuclear medicine, ultrasound, CT, MRI, angiography, mammography, and radiography), as illustrated in Fig. 1.178. The acronym PACS can best be defined as follows:

P	Picture	Digital medical image(s)
A	Archiving	"Electronic" storage of images
C	Communication	Routing (retrieval/sending) and displaying of images
S	System	Specialized computer network that manages the complete system

The connection of various equipment types and modalities to a PACS is complex. Standards have been developed to ensure that all manufacturers and types of equipment are able to communicate and transmit images and information effectively. Current standards include **DICOM** (Digital Imaging and Communications in Medicine) and **HL7** (health level 7). Although standards may not always provide for an instantaneous functionality between devices, they do allow for resolution of connectivity problems.

For optimum efficiency, PACS should be integrated with the radiology information system (**RIS**) or the hospital information system (**HIS**). Because these information systems support the operations of an imaging department through examination scheduling, patient registration, report archiving, and film tracking, integration with PACS maintains integrity of patient data and records and promotes overall efficiency.

When PACS is used, instead of hard-copy radiographs that must be processed, handled, viewed, transported, and stored, the soft-copy digital images are processed with the use of a computer, viewed on a monitor, and stored electronically. Most PACS use web browsers to enable easy access to the images by users from any location. Physicians may view these radiologic images from a personal computer at virtually any location, including their homes.

ADVANTAGES OF PACS

Advantages of PACS include the following:

- Elimination of less efficient traditional film libraries and their inherent problem of physical space requirements for hard-copy images.
- Convenient search for and retrieval of images.
- Rapid (electronic) transfer of images within the hospital (e.g., clinics, operating rooms, treatment units).
- Ease in consulting outside specialists—teleradiology. Teleradiography is the electronic transmission of diagnostic images from one location to another for purposes of interpretation or consultation.
- Simultaneous viewing of images at multiple locations.
- Elimination of misplaced, damaged, or missing films.
- Increase in efficiency of reporting examinations with soft-copy images (compared with hard-copy images).
- Reduction of the health and environmental impact associated with chemical processing as a result of decreased use.

The growth of computer applications in radiologic technology has led to new career paths for radiologic technologists. PACS Administrator and the Diagnostic Imaging Information Technologist are new positions that many radiologic technologists are pursuing.

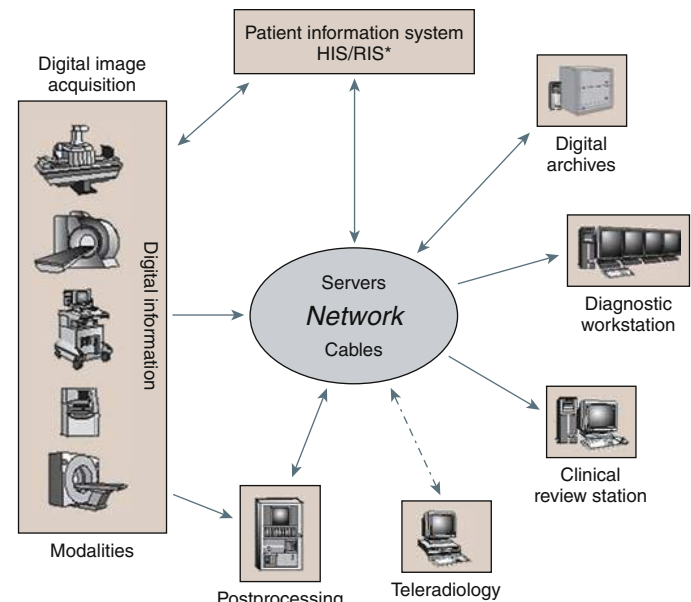


Fig. 1.178 A full PACS network that includes digital acquisition, communication, reporting, and archiving. *HIS/RIS*, Hospital information system/radiology information system. (Modification of diagram from Philips Medical Systems.)

PART THREE ■ RADIATION PROTECTION

Contributor Frank Goerner, PhD, DABR

As professionals, radiologic technologists have the important responsibility to protect their patients, themselves, and fellow workers from unnecessary radiation. A complete understanding of radiation protection is essential for every technologist, but a comprehensive review¹³ is beyond the scope of this anatomy and positioning text. The basic principles and applied aspects of radiation protection, as described in this part, should be an essential component of a course in radiographic positioning. Every technologist has the obligation **to ensure that the radiation dose to both the patient and other health care professionals is kept as low as reasonably achievable (ALARA).**

Radiologic technologists are the last line of defense between patients and unnecessary radiation exposure, making radiation protection awareness of primary importance.

It is important to keep in mind that the three most effective ways to protect patients and staff from ionizing radiation are:

1. **Time**—Minimize radiation beam-on time
2. **Distance**—Maximize distance from the radiation source of both patients and staff
3. **Shielding**—Use shielding on staff and patients

Radiation Units

Determining the amount of radiation is a critical element in protecting patients and staff. The most important quantities for a radiographic technologist to be familiar with are absorbed dose (Gy/rad) and equivalent dose (Sv/rem). Personnel badge reports typically use equivalent dose to quantify radiation, whereas radiography and fluoroscopy equipment tends to display absorbed dose to help estimate patient dose during procedures. Additional terms used to quantify radiation are defined next and include exposure, air kerma, absorbed dose, equivalent dose, and effective dose.

Exposure (C/Kg or R)—measures the amount of ionization created in air by x-rays. Roentgen (R) or coulomb per kilogram (C/Kg) are the units of exposure. X-ray tube output, patient entrance exposure, and scattered radiation levels are usually indicated by measurements of exposure.

Air kerma (Gy or rad)—indicates the amount of energy transferred to a mass of air by the photons; this has replaced exposure as the preferred quantity for the applications listed above. The unit of measurement for air kerma is the gray (Gy) or rad.

Absorbed dose (Gy or rad) is the amount of energy deposited per unit mass by the interactions of ionizing radiation with tissue and has units of **gray (Gy)** or **rad**.

Equivalent dose (Sv or rem) quantifies the risk for different types of radiation using the same relative scale; the units are **sieverts (Sv)** or **rem**. The product of the absorbed dose and the radiation-weighting factor yields the equivalent dose. The radiation-weighting factor depends on the type and energy of the radiation. In radiography, the radiation-weighting factor is always one. When a patient is x-rayed, air kerma, absorbed dose, and equivalent dose all are considered numerically equal but have different conceptual meanings.

Effective dose (Sv or rem) indicates the risk from a partial body exposure by multiplying the equivalent dose by the tissue-weighting factor that corresponds to the area of the body that is exposed. Effective dose allows comparisons of the relative risk from various imaging procedures and exposure of different areas of the body.

Traditional and SI Units

The SI system has been the international standard for units of radiation measurement since 1958. However, just as the United States has been slow to convert to the metric system for other applications, traditional units of radiation measurement such as the roentgen, rad, and rem are still in common use in the United States. Dose limits and patient doses in this section are designated in both SI and traditional units (1 gray = 100 rads and 1 rad = 10 mGy) (Table 1.11). The gray is an extremely large unit for most dose considerations in medicine. A smaller unit of milligray is often used (1000 mGy = 1 Gy).

DOSE LIMITS

High doses of radiation are harmful. Scientists have established dose limits to help eliminate the risk of adverse effects (Table 1.12). The rationale for the dose limits is to make risk from occupational exposure comparable to the risks for workers in other safe industries (excluding mining and agriculture). The annual dose limit for occupationally exposed workers is **50 mSv** (5000 mrem) whole-body effective dose equivalent. Higher annual dose limits are applied for partial body exposure: 150 mSv (15,000 mrem) for the lens of the eye and 500 mSv (50,000 mrem) for the skin, hands, and feet. Medical radiation received as a patient and background radiation are not included in these occupational dose limits.

The annual dose limit for the general public is 1 mSv (100 mrem) for frequent exposure and 5 mSv (500 mrem) for infrequent exposure. For practical purposes, the shielding design for x-ray facilities is based on the lower limit. In essence, the facility must demonstrate that x-ray operation is unlikely to deliver a dose greater than 1 mSv to any member of the public over a period of 1 year.

The recommended cumulative lifetime dose for the occupationally exposed worker is 10 mSv (1 rem) times the age in years. For example,

TABLE 1.11 CONVERSION TABLE—TRADITIONAL TO SI UNITS

TO CONVERT FROM (TRADITIONAL UNITS)	TO (SI UNITS)	MULTIPLY BY
Roentgen (R)	C/kg	2.58×10^{-4} (0.000258)
Rad	Gray (Gy)	10^{-2} (1 rad = 0.01 Gy)
Rem	Sievert (Sv)	10^{-2} (1 rem = 0.01 Sv)

TABLE 1.12 SUMMARY OF DOSE-LIMITING RECOMMENDATIONS

OCCUPATIONAL WORKERS ^a		GENERAL PUBLIC		INDIVIDUALS <18 YEARS OLD		PREGNANT WORKERS	
Annual	50 mSv (5 rem)	Annual	1 mSv (100 mrem)	Annual	1 mSv (100 mrem)	Month	0.5 mSv (50 mrem)
Lifetime accumulation	10 mSv (1 rem) × years of age					Gestation period	5 mSv (500 mrem)

^aWhole-body effective dose equivalent.

a 50-year-old technologist has a recommended accumulated dose of no more than 500 mSv (50 rem). However, the principle of ALARA should be practiced so that the occupational dose is accrued at a rate that is very much less than the dose limit of 50 mSv (5 rem) per year.

Individuals younger than 18 years of age should not be employed in situations in which they are occupationally exposed. The dose limit for minors is the same as that for the general public—1 mSv (0.1 rem) per year.

PERSONNEL MONITORING

Personnel monitoring refers to the measurement of the amount of radiation dose received by occupationally exposed individuals. The monitor offers no protection but simply provides an indication of radiation dose received by the wearer of the monitoring device. On a periodic basis (monthly or quarterly), the personnel monitor (film badge, thermoluminescent dosimeter [TLD], or optically stimulated luminescence dosimeter [OSL]) is exchanged for a new one. A commercial personnel dosimetry company processes the dosimeter, and the radiation dose for the period is determined. Measurement of occupational dose is an essential aspect of radiation safety as a means to ensure that workers do not exceed the dose limit and to assess that the dose received is reasonable for the work activities.

Each worker who is likely to receive 10% of the dose limit must be issued a personnel monitor. Generally, health care professionals, including emergency department and operating room nursing personnel, who are occasionally present when mobile x-ray equipment is in operation, do not require personnel monitoring devices. The radiation dose received by nursing personnel (with the exception of some angiography and cath lab nursing staff) is typically less than 10% of the annual dose limit. Clerical and support staff working near the x-ray room do not need and should not be monitored with a personnel dosimeter.

The personnel dosimeter is worn at the level of the chest or waist during radiography (Fig. 1.179). If an individual is involved in fluoroscopic procedures, the dose under the apron is known to be a small fraction of the dose to the head and neck.^{14,15} The dosimeter should be positioned on the collar outside the protective apron (or outside the thyroid collar) during fluoroscopy. The personnel dosimeter should not be worn on the sleeve. The collar reading greatly overestimates the dose to the total body. To account for the protective effect of the apron and determine an effective whole-body dose (called the *effective dose equivalent*), the collar reading is multiplied by a factor of 0.3. A measured value of 3 mSv (300 mrem) for the collar dosimeter is equivalent to a whole-body dose of 0.9 mSv (90 mrem). The annual dose limit of 50 mSv (5000 mrem) applies to the effective dose equivalent.

When not in use, personnel monitoring devices should remain at the place of employment in a low-background area, such as a locker or office. Personnel monitoring devices should not be stored in areas of x-ray use.

ALARA

In recent years, radiation protection measures have been devised according to the principle of ALARA. Radiation exposure should be maintained at the lowest practicable level and very much below the dose limits. All technologists should practice the ALARA principle so that patients and other health care professionals do not receive unnecessary radiation. Following is a summary of four important ways that ALARA can be achieved:

1. **Always wear a personnel monitoring device.** Although the device does not reduce the dose to the wearer, exposure history has an important impact on protection practices. Radiologic technologists should ensure that individuals present during x-ray operation wear personnel monitors as appropriate.
2. **Mechanical holding devices** (e.g., compression bands, sponges, sandbags, and 2-inch-wide tape) are effective tools for the



Fig. 1.179 Technologist wearing a personnel dosimeter.

immobilization of patients and should be used if the procedure permits. Only as a last resort should someone hold the patient. The following criteria are applicable for the selection of someone to hold a patient during a radiographic procedure:

- No individual shall be regularly used to hold patients.
 - An individual who is pregnant shall not hold patients.
 - An individual younger than 18 years of age shall not hold patients.
 - Whenever possible, an individual occupationally exposed to radiation shall not hold patients during exposures.
 - A parent or family member should be used to hold the patient if necessary.
 - A hospital employee who is not occupationally exposed may be used to hold the patient if necessary.
 - If an individual holds the patient, he or she is provided with a protective apron and gloves. The individual is positioned so that the primary beam exposes no part of his or her body except hands and arms. Only individuals required for the radiographic procedure should be in the room during exposure. All persons in the room except the patient are provided protective devices.
3. Close collimation, filtration of the primary beam, optimum kVp technique, high-speed IRs, and avoidance of repeat projections reduce the dose to the patient.
 4. Practice the three cardinal principles of radiation protection: time, distance, and shielding. The technologist should minimize the time in the radiation field, stand as far away from the source as possible, and use shielding (protective devices or control booth barrier). For individuals not shielded by a protective barrier during x-ray operation, the radiologic technologist should ensure that these persons wear lead aprons and gloves as appropriate.

Exposure to persons outside a shielded barrier is due primarily to scattered radiation from the patient. Therefore, a reduction in patient exposure results in decreased dose to workers in unshielded locations. Protection from scatter radiation is an important consideration during mobile C-arm fluoroscopy, as described in detail in [Chapter 15](#) in the discussion of trauma and mobile radiography.

In the absence of a radiologist during an x-ray examination, the radiologic technologist generally has the highest level of training in radiation protection in a patient care setting. An essential component of a radiation safety program is that individuals present during x-ray operation wear protective lead aprons and personnel monitors as appropriate. Management must have a clearly defined and enforced radiation safety policy, which includes the use of protective apparel and designates who can hold patients. Individuals who do not follow radiation safety policy of the institution should be subject to disciplinary action.

PREGNANT TECHNOLOGISTS

Studies have shown that the fetus is sensitive to high doses of ionizing radiation, especially during the first 3 months of gestation. A small risk of harmful effects from low doses of radiation is assumed, but not proven, to exist. That is, any radiation dose, however small, is considered to increase probability of harm to the fetus.

Effective, fair management of pregnant employees exposed to radiation requires the balancing of three factors: (1) the rights of the expectant mother to pursue her career without discrimination based on gender, (2) the protection of the fetus, and (3) the needs of the employer. Each health care organization should establish a realistic policy that addresses these three concerns by clearly articulating the expectations of the employer and the options available to the employee. A sample pregnancy policy for radiation workers has been published in the literature.¹⁶ The pregnant technologist should review the institutional policy and other professional references and consult with the radiation safety officer (if available) to determine expectations and the best practices to protect her unborn child.

The recommended equivalent dose limits to the embryo/fetus is 0.5 mSv (50 mrem) during any 1 month and 5 mSv (500 mrem) for the gestation period. To recognize the increased radiosensitivity of the fetus, the total fetal dose is restricted to a level that is much less than that allowed for the occupationally exposed mother. However, the expectant mother's exposure from other sources, such as medical procedures, is excluded from the fetus dose limit. The fetal dose limit can be applied only if the employer is informed of the pregnancy. The regulations define the declared pregnant woman as one who voluntarily informed her employer, in writing, of her pregnancy and the estimated date of conception.

In recent years, radiation protection measures have been devised according to the principle of ALARA. Radiation exposure should be maintained at the lowest practicable level. **Radiation protection practices do not change because the worker becomes pregnant.** The measures that reduce the dose to the worker also reduce the dose to the fetus. The major ways to decrease the dose further are to restrict the type of tasks performed or to limit the number of times a particular task is performed.

When an employee first discovers she is pregnant, an individualized review of her exposure history and work assignments should be conducted. If she averaged 0.3 mSv (30 mrem) per month for the last several months, a reasonable projection is that this individual, and her unborn child, will not receive more than 5 mSv (500 mrem) during the period of gestation. This radiologic technologist could continue to work in her current capacity during her pregnancy. However, she should monitor her dosimeter readings and report any unusual reading to the radiation safety officer. Contrary to general belief, fluoroscopy procedures do not cause high exposures to the fetus. In fluoroscopy, attenuation by the lead apron and by overlying maternal tissues reduces the dose to the fetus. Personnel dosimeter readings totaling

5 mSv (500 mrem) at the collar correspond to a fetal dose of 0.075 mSv (7.5 mrem). Therefore, radiologic technologists can continue their work assignments in fluoroscopy throughout pregnancy.

A declared pregnant radiation worker may monitor the dose to the fetus by placing a second dosimeter at waist level under the protective apron. This monitoring method generally produces readings below the detectable limit of the dosimeter and is useful only in demonstrating that the fetus received no measurable radiation exposure. The fetal badge must be clearly marked to distinguish the device worn under the apron from that worn at the collar.

RADIOGRAPHIC PATIENT DOSE

For a particular radiographic examination, several different "doses" may be used to characterize patient dose. The most common descriptor is the exposure to the skin in the region where the x-rays enter the body, called the *entrance skin exposure*. Air kerma is rapidly replacing exposure because it is easily converted to skin dose with the application of the backscatter factor. The backscatter factor takes into account the additional dose at the surface caused by scattering from tissue within the irradiated volume. As the x-rays directed toward the IR pass through the body, attenuation causes a dramatic reduction in dose (Fig. 1.180). Exit dose is often a small percentage of the entrance dose. Specific organ dose varies depending on depth and radiation quality. If the organ is located outside the primary beam, dose is from scattered radiation only and is a small fraction of the in-beam dose. Entrance air kerma and organ doses from common radiographic examinations are shown in Table 1.13. These values represent multiple facilities but vary according to technique factors, type of IR, field size, and patient size.

The **effective dose (ED)** takes into account the respective dose to each organ and the cumulated relative risk from all organs that received dose. This dose metric essentially specifies a whole-body dose that yields the same overall risk as incurred by the nonuniform dose distribution in the patient. Effective dose becomes a means to compare different imaging procedures with respect to potential for harm (Table 1.14).

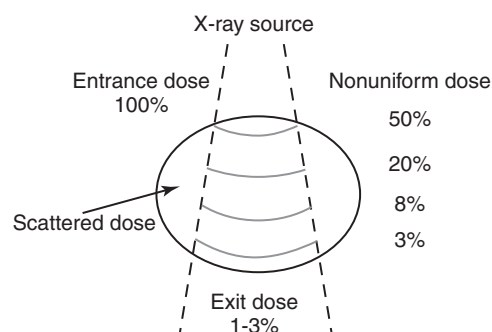


Fig. 1.180 Radiation dose from an AP abdomen decreases markedly from the entrance side to the exit side of the patient.

TABLE 1.13 PATIENT DOSE CHART

PROJECTION	ENTRANCE AIR KERMA (MGY)	ORGAN DOSE (MGY)				
		TESTES	OVARIES	THYROID	MARROW	UTERUS
Chest PA	0.2	<0.001	<0.001	0.008	0.02	<0.001
Skull (lateral)	1.7	<0.001	<0.001	0.05	0.06	<0.001
Abdomen AP	4.0	0.09	1.0	<0.001	0.19	1.3
Retrogram pyelogram	6.0	0.13	1.5	<0.001	0.29	2.0
Cervical spine AP	1.1	<0.001	<0.001	0.9	0.02	<0.001
Thoracic spine AP	4.0	<0.001	0.003	0.5	0.16	0.002
Lumbar spine AP	3.4	0.02	0.52	0.002	0.16	1.0

TABLE 1.14 EFFECTIVE DOSE (ED)

EXAMINATION	EFFECTIVE DOSE (MSV)	EXAMINATION	EFFECTIVE DOSE (MSV)
Skull	0.07	Cerebral angiography	2.0
Chest	0.14	Cardiac angiography	7.3
Abdomen	0.53	PTCA	22
Lumbar spine	1.8	Barium enema	20
Thoracic spine	1.4	CT head	2.3
Cervical spine	0.27	CT abdomen	13
Extremities	0.06	CT coronary angiography	20
Mammography	0.22		
Upper GI	3.6		
Small bowel	6.4		

PTCA, Percutaneous transluminal coronary angiography.

Patient Protection in Radiography

Radiologic technologists subscribe to a code of ethics that includes responsibility to control the radiation dose to all patients under their care. This is a serious responsibility, and each of the following seven ways of reducing patient exposure must be understood and put into practice as described in the next sections:

1. Minimum repeat radiographs
2. Correct filtration
3. Accurate collimation
4. Specific area shielding (gonadal and female breast shielding)
5. Protection of the fetus
6. Optimum imaging system speed
7. Select projections and technique factors appropriate for the examination
 - Use high-kVp and low-mAs techniques
 - Use PA rather than AP projections to reduce dose to anterior upper thoracic region (thyroid and female breasts) (see Chapter 8)
 - Use techniques consistent with system speed for digital radiography as confirmed by exposure indicator values

MINIMUM REPEAT RADIOGRAPHS

The first basic and most important method to prevent unnecessary exposure is to **avoid repeat radiographs**. A primary cause of repeat radiographs is poor communication between the technologist and the patient. Unclear and misunderstood breathing instructions are a common cause of motion, which requires a repeat radiograph.

When the procedures are not explained clearly, the patient can have added anxiety and nervousness from fear of the unknown. Stress often increases the patient's mental confusion and hinders his or her ability to cooperate fully. To engage the patient, the technologist must take the time, even with heavy workloads, to explain carefully and fully the breathing instructions and the procedure in general in simple terms that the patient can understand (Fig. 1.181). Patients must be forewarned of any movements or strange noises by equipment that may occur during the examination. In addition, any burning sensation or other possible effects of injections should be explained to the patient.

Avoid carelessness in positioning and selection of erroneous technique factors, as these are common causes of repeats. Correct and accurate positioning requires a thorough knowledge and understanding of anatomy, which enables the technologist to visualize the size, shape, and location of structures to be radiographed. This is the reason that every chapter in this text combines anatomy with positioning.

CORRECT FILTRATION

Filtration of the primary x-ray beam reduces exposure to the patient by preferentially absorbing low-energy "unusable" x-rays, which mainly expose the patient's skin and superficial tissue without contributing to image formation. The effect of filtration is a "hardening" of the x-ray beam, which shifts the beam to a higher effective energy, resulting in increased penetrability (Fig. 1.182).

Filtration is described in two ways. First is inherent or built-in filtration from components of the x-ray tube itself. For most radiographic tubes, this is approximately 0.5 mm aluminum equivalent. Second, and more important to technologists, is **added filtration**, which is accomplished by placing a metal filter (aluminum or copper or combination of these) in the beam within the collimator housing. The amount of **minimum total filtration** as established by federal regulations depends on the operating kVp range. The manufacturers of x-ray equipment are

required to meet these standards. Minimum total filtration (inherent plus added) for diagnostic radiology (excluding mammography) is 2.5 mm aluminum for equipment operating at 70 kVp or higher.

Often, radiographic equipment has variable added filtration, which can be selected by the technologist. Added filtration becomes another component to adapt the acquisition parameters to the patient. Generally, as patient size increases, more added filtration provides skin dose sparing. The technique chart should specify the use of added filtration. The technologist has the responsibility to ensure that proper filtration is in place.

The filtration of each x-ray tube **should be checked annually and after major repair** (x-ray tube or collimator replacement). Qualified personnel, such as the medical physicist, should test the filtration of each tube. The test typically used to ensure adequate filtration is measurement of the half-value layer.



Fig. 1.181 Clear, precise instructions help relieve patient anxieties and prevent unnecessary repeats.

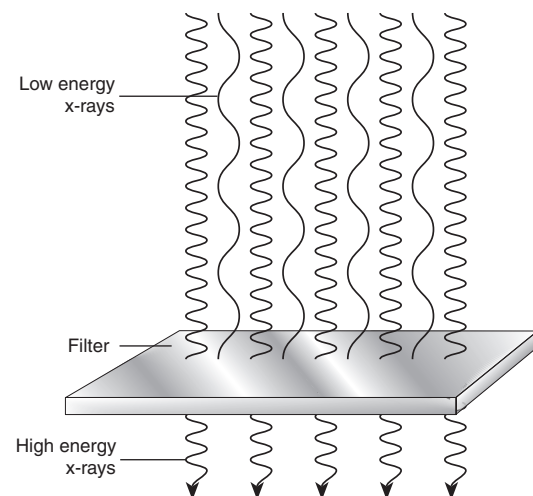


Fig. 1.182 A metal filter preferentially removes low-energy x-rays, shifting the x-ray beam to higher effective energy.

ACCURATE COLLIMATION

Accurate collimation reduces patient exposure by **limiting the size and shape of the x-ray field to the area of clinical interest**. Careful and accurate collimation is emphasized and demonstrated throughout this textbook. The adjustable collimator is used routinely for general diagnostic radiography. The illuminated light field defines the x-ray field on accurately calibrated equipment and can be used effectively to determine the tissue area to be irradiated. Safety standards require light field and x-ray field concurrence within **2% of the selected SID**.

The concept of divergence of the x-ray beam must be considered for accurate collimation. Therefore, the illuminated field size as it appears on the skin surface appears smaller than the actual size of the anatomic area, which would be visualized on the IR. This is most evident on a projection such as lateral thoracic or lumbar spine (Fig. 1.183), in which the distance from the skin surface to the IR is considerable. In such cases, the light field, when collimated correctly to the area of interest, appears too small unless one considers the divergence of the x-ray beam.

Collimation and Tissue Dose

Accurate and close collimation to the area of interest results in a dramatic drop-off in tissue dose as distance from the border of the collimated x-ray field is increased. For example, the dose **3 cm** from the edge of the x-ray field is about **10%** of that within the x-ray field. At a distance of **12 cm** the reduction in dose is about **1%** of that within the x-ray field.

Positive Beam Limitation

All general-purpose x-ray equipment built between 1974 and 1993 in the United States required collimators with positive beam limitation (PBL) that automatically adjusts the useful x-ray beam to the film size (this requirement became optional after May 3, 1993, as a result of a change in U.S. Food and Drug Administration [FDA] regulations). The PBL feature consists of sensors in the cassette holder that, when activated by placing a cassette in the bucky tray, automatically signal the collimator to adjust the x-ray field to that film size. PBL can be deactivated or overridden with a key, but this should be done only under special circumstances, in which collimation by manual control is needed. A red warning light is illuminated to indicate that PBL has been deactivated. The key cannot be removed while PBL is overridden (Fig. 1.184).

Manual Collimation

Even with automatic collimation (PBL), the operator can manually reduce the collimation field size. This adjustment should be made for every projection in which the IR is larger than the critical area to be radiographed. Accurate manual collimation also is required for upper and lower limbs that are radiographed on the tabletop, in which PBL is not engaged. Throughout the positioning pages in this textbook, collimation guidelines are provided to maximize patient protection through careful and accurate collimation.

The practice of close collimation to only the area of interest reduces patient dose in **two ways**. First, the **volume of tissue directly irradiated is diminished**, and second, the **amount of accompanying scattered radiation is decreased**. Scatter radiation produced by additional tissue in the x-ray field from improper collimation or lack of shielding not only adds unnecessary patient dose but also degrades image quality through the “fogging” effect of scatter radiation. (This is especially true in high-volume tissue imaging such as abdomen and chest.)

The practice of visible collimation on all four sides of a radiograph reduces patient exposure, improves image quality, and acts as a method to ensure that appropriate collimation did occur. If no collimation border is visible on the radiograph, evidence does not exist that the primary beam was restricted to the area of clinical interest. An added benefit of showing the extent of collimation on all four sides is the ability to check the final radiograph for correct central ray location. As described previously, this is done by imagining a large X extending from the four corners of the collimation field, the center of which is the CR location.

Collimation Rule

A general rule followed throughout this text indicates that collimation should **limit the x-ray field to only the area of interest, and collimation borders should be visible on the IR on all four sides** if the IR size is large enough to allow four-sided collimation without cutting off essential anatomy.

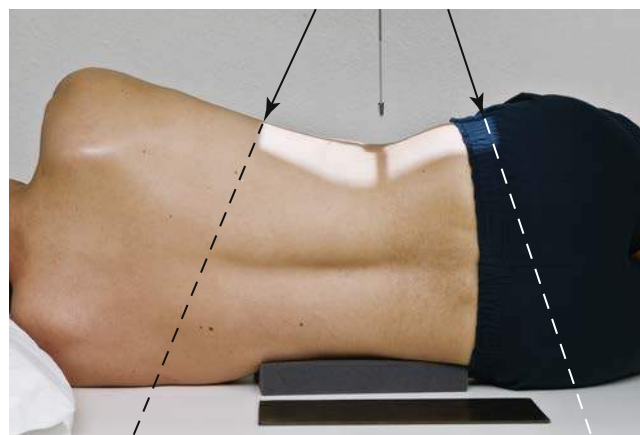


Fig. 1.183 Close four-sided collimation. The collimated light field may appear too small because of divergence of x-rays.



Fig. 1.184 Automatic collimation (PBL).

SPECIFIC AREA SHIELDING

Specific area shielding can be used to protect radiosensitive organs, such as the thyroid gland, breasts, and gonads, when they are in the useful beam and the use of such shielding does not interfere with the objectives of the examination. Historically the most commonly shielded area has been the gonads, which significantly lowers the dose to the reproductive organs. Gonadal shields, if placed correctly, reduce the gonadal dose by 50% to 90% if the gonads are in the primary x-ray field. Recent studies have shown that the breasts are actually more radiosensitive than the gonads and thus the efficacy of gonadal shielding is being brought into question. Additionally, improper placement of gonadal shields is a common and well-documented problem. Furthermore, the use of the same contact shield on different patients can lead to infection control issues, particularly in sensitive patients such as those in the neonatal intensive care unit (NICU). These factors have led some institutions to eliminate or more closely monitor gonadal shielding, because often the benefit does not outweigh the risk. Although there is evidence that gonadal shielding may not offer as much protection as originally anticipated, it is still recommended in many countries and in most states within the U.S.

The two general types of specific area shielding are shadow shields and contact shields.

Shadow Shields

As the name implies, **shadow shields**, which are attached to the collimator, are placed between the x-ray tube and the patient and cast a shadow on the patient when the collimator light is turned on. The position of the shadow shield is adjusted to define the shielded area. One such type of breast shadow shields, as shown in Fig. 1.185, is affixed to the collimator exit surface with Velcro. Another type of shadow shield, as shown in Fig. 1.186, is mounted with magnets directly to the bottom of the collimator. These shields may be combined with clear lead compensating filters to provide more uniform exposure for body parts that vary in thickness or density, such as for a thoracic and lumbar spine scoliosis radiograph (Fig. 1.187).

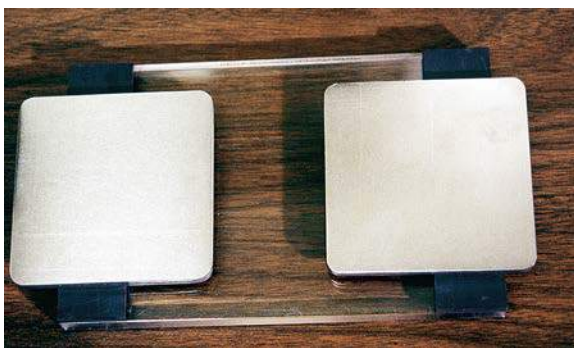


Fig. 1.185 Breast shadow shields designed to be attached to collimator exit surface with Velcro.

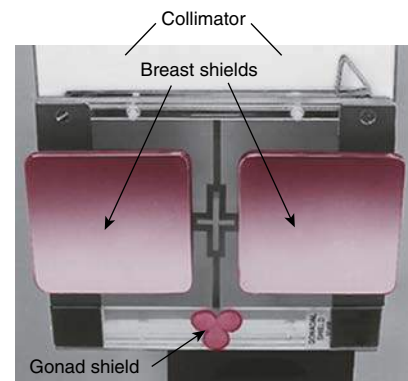


Fig. 1.186 Shadow shields in place under collimator (attached with magnets). (Courtesy Nuclear Associates, Carle, NY.)



Fig. 1.187 AP spine for scoliosis with compensating filter and breast and gonadal shields in place. (Courtesy Nuclear Associates, Carle, NY.)

Contact Shields

Flat gonadal contact shields are used most commonly for patients in recumbent positions. Vinyl-covered lead shields are placed over the gonadal area to attenuate scatter or leakage radiation or both (Fig. 1.188). These shields usually are made from the same lead-impregnated vinyl materials that compose lead aprons. Gonadal contact shields, 1 mm lead equivalent, absorb 95% to 99% of primary rays in the 50- to 100-kVp range. Examples of these include small vinyl-covered lead material cut into various shapes to be placed directly over the reproductive organs, as shown in Figs. 1.189 and 1.190.

Male

Gonadal shields should be placed distally to the symphysis pubis, covering the area of the testes and scrotum (see Fig. 1.189A). The upper margin of the shield should be at the symphysis pubis. These shields are tapered slightly at the top and are wider at the bottom to cover the testes and scrotum without obscuring pelvic and hip structures. Smaller sizes should be used for smaller males or children.

Female

Gonadal shielding is placed to cover the area of the ovaries, uterine (fallopian) tubes, and uterus but may be more difficult to achieve. A general guideline for women is to shield an area 4½ to 5 inches (11 to 13 cm) proximal or superior to the symphysis pubis extending 3 to 3½ inches (8 to 9 cm) each way from the pelvic midline. The lower border of the shield should be at or slightly above the symphysis pubis, with the upper border extending just above the level of the anterior superior iliac spines (ASIS) (see Fig. 1.190A).

Various-shaped female ovarian shields may be used, but they should be wider in the upper region to cover the area of the ovaries and narrower toward the bottom to offer less obstruction of pelvic or hip structures. The shielded area should be proportionally smaller on children. For example, a 1-year-old girl would require a shield that is only about 2 to 3 inches (6 to 7 cm) wide and 2 inches (5 cm) high placed directly superior to the symphysis pubis.¹⁸

Summary of Rules for Specific Area Shielding

Proper specific area shielding is a challenge for each technologist because its use requires additional time and equipment. Additionally, improper shielding can lead to repeat examinations, thus defeating their intention:

1. **Gonadal shielding should be considered for all patients.** A common policy of many imaging facilities directs the use of specific area shielding for all children and adults of reproductive age.
2. Placement of gonadal shielding is necessary **when the organ of concern lies within or near (2 inches [5 cm]) the primary beam**, unless such shielding obscures essential diagnostic information or may introduce pathogens to a sensitive patient population.
3. **Accurate beam collimation** and careful positioning are essential. Specific area shielding is an additional protective measure but no substitute for accurate collimation.

PREGNANT PATIENT

All women of childbearing age should be screened for the possibility of pregnancy before an x-ray examination of the abdomen or pelvis area. This concern is particularly critical during the first 2 months of pregnancy, when the fetus is most sensitive to radiation and the mother may not yet be aware of the pregnancy. Posters or signs (Figs. 1.191 and 1.192) should be prominently displayed in examination rooms and waiting room areas, reminding the



Fig. 1.188 Vinyl-covered lead shield in place over pelvis for lateral mid and distal femur.

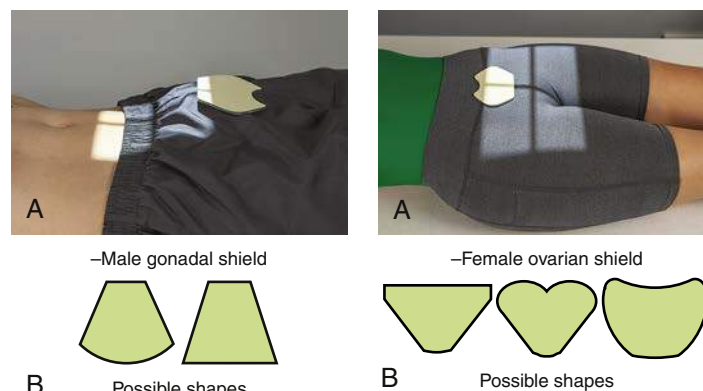


Fig. 1.189 A, AP pelvis with flat contact shield (1 mm lead equivalent). B, Male gonadal shield shapes.

Fig. 1.190 A, AP right hip with flat contact shield (1 mm lead equivalent). B, Female ovarian shield shapes.



Fig. 1.191 Warning sign. (From Ehrlich RA, Coakes D: *Patient care in radiography*, ed 9, St. Louis, 2017, Elsevier.)

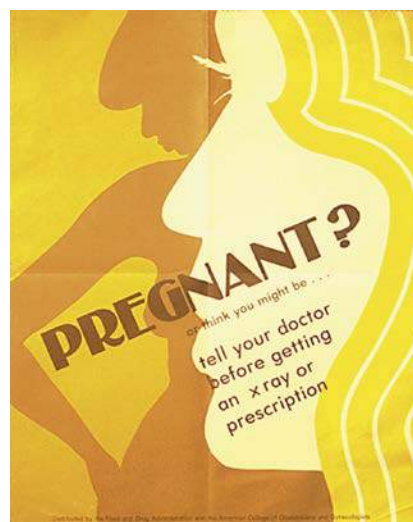


Fig. 1.192 Warning poster.

patient to inform the technologist of any known or potential pregnancy.

If the patient **indicates that she is pregnant or may be pregnant**, the technologist should consult the radiologist before proceeding with the examination. If the mother's health is at risk and clear indications for an imaging study exist, the examination should not be denied or delayed because of the pregnancy. Radiation protection practices already described, especially **careful collimation**, should be used.

For examinations of body parts above the diaphragm or below the hips, the scattered dose to the fetus is very low, and the examination may proceed normally. For examinations in which the fetus is in the direct beam and the estimated fetal dose is less than 10 mGy (1 rad), the radiation dose should be kept as low as possible consistent with obtaining the desired diagnostic information. Shielding of the abdomen and pelvis with a lead apron should be considered. Limiting the number of views should be considered. For examinations in which the fetus is in the direct beam and the estimated fetal dose is greater than 10 mGy (1 rad), the radiologist and referring physician should discuss other options such as sonography and MRI that can provide the needed information.

If the x-ray imaging procedure is deemed appropriate, the patient should be informed of the risks and benefits of the procedure. The clinician responsible for the care of the patient should document in the medical record that the test is indicated for the management of the patient.

In the past, the **10-day or LMP rule** (last menstrual period) was applied to prevent exposure to the embryo/fetus early in pregnancy, when the pregnancy is not known. This rule stated that all radiologic examinations involving the pelvis and lower abdomen should be scheduled during the first 10 days following the onset of menstruation because conception will not have occurred during this period. Currently, this rule is considered obsolete because the potential harm associated with canceling essential x-ray procedures may greatly exceed the risk of the fetal radiation dose.

The following examinations deliver a dose of less than 10 mGy (1 rad) to the embryo/fetus:

- Extremities
- Chest
- Skull
- Thoracic spine
- Head CT
- Chest CT

The following examinations have the potential to deliver a dose of more than 10 mGy (1 rad) to the fetus and embryo:

- Lumbar spine series
- Fluoroscopic procedures (abdomen)
- Abdomen or pelvis with three or more views
- Scoliosis: full series
- CT abdomen
- CT pelvis

OPTIMUM SPEED

Analog imaging systems. As a general guideline, the highest-speed film-screen combination that results in diagnostically acceptable radiographs is desirable to manage patient dose. The presence of the screen does result in some loss of spatial resolution and becomes more pronounced as the speed is increased. The radiologist must balance the reduction in patient exposure with the potential loss of detail in the resultant image. A common practice is to select a slow 100-speed (detail) screen with tabletop procedures, such as upper and lower limbs, when a grid is not used and spatial detail is important. A 400-speed screen is commonly preferred for larger body parts when grids and higher-exposure techniques are required. For other applications, a 200-speed screen may be preferred. Departmental protocol generally indicates the film-screen combination for each procedure. This is not a decision that is usually made by individual technologists.

Digital imaging systems have essentially replaced film-screen for most radiographic applications. These digital receptors are more sensitive than film-screen and have the potential to reduce patient dose greatly. In addition, their wide dynamic range can result in fewer repeated "films." Automatic exposure control (AEC) for digital systems is usually set at an exposure indicator level that produces images with an acceptable level of noise. However, the technologist may adjust the AEC density control to change the effective system speed. The wide dynamic range of digital receptors enables this variation in dose while still producing a quality image (although noise becomes more pronounced as the dose is reduced). Because the FPD-TFT is often integrated into the radiographic unit, the variable-speed option is readily available to customize the speed for each imaging protocol.

SELECTION OF PROJECTIONS AND EXPOSURE FACTORS WITH THE LEAST PATIENT DOSE

The seventh and final method to reduce patient dose requires an understanding of the factors that affect patient dose. For example, technologists should know that patient dose is decreased with AEC use when the kVp is increased. For manual technique, an increase in kVp with no change in mAs results in a higher dose to the patient. The goal is to use the combination of technique factors that will provide acceptable image quality and minimize patient dose.

There is a substantial difference in dose to the thyroid and female breasts for the AP projection compared with the PA projection for the head, neck, and upper thorax region. The ovarian dose can be reduced for certain projections, such as a female hip, if a specific area shield is correctly placed. An axiolateral or inferosuperior lateral hip projection compared with a lateral hip projection delivers a higher dose to the testes.

RADIATION SAFETY PRACTICES

Technologists must adhere to ethical and safe practice when using digital technology. The wide dynamic range of digital imaging enables an acceptable image to be obtained with a broad range of exposure factors. During the evaluation of the quality of an image, the technologist must ensure the exposure indicator is within the recommended range. Any attempt to process an image with a different algorithm to correct overexposure is unacceptable; it is vital that patient dose be minimized at the outset and the ALARA principle be upheld.

To maintain dose at a reasonable, consistent dose level, the following practices are recommended:

1. Use protocol-specific kVp and mAs values for all procedures. If no exposure protocol exists, consult with the lead technologist, physicist, or manufacturer to establish one. Increasing kVp by 5 to 10 and decreasing mAs by the equivalent ratio can produce a quality image with digital imaging systems while reducing patient dose.
2. Monitor dose by reviewing all images to ensure radiographs were obtained with the established exposure indicator.
3. If the exposure indicator for a given procedure is outside of the acceptable range, review all factors, including kVp and mAs, to determine the cause of this disparity. Processing of digital images can be adversely affected if the exposure indicator deviates from the manufacturer's acceptable values.

Fluoroscopic Patient Dose

Because fluoroscopy can potentially deliver a **high patient dose**, federal standards have set a limit of 10 R/min for the tabletop exposure rate, which corresponds to an air kerma rate of 88 mGy/min. In **high-level fluoroscopy (HLF) mode**, the exposure rate at tabletop cannot exceed 20 R/min or an air kerma rate of 176 mGy/min. For C-arm fluoroscopic units, the point of measurement is specified as 30 cm from the image receptor. HLF mode should be reserved for instances in which the lack of penetration creates a poor image (large patients). There is no exposure rate limit when the image is recorded, as in digital cine and serial digital spot filming. With most modern equipment, the average tabletop fluoroscopy exposure rate is 1 to 3 R/min (air kerma rate of 8.8 to 26 mGy/min). Use of magnification mode increases the instantaneous exposure rate but decreases the volume of tissue irradiated.

Typical patient doses during gastrointestinal fluoroscopy procedures are shown in [Table 1.15](#), which includes approximate entrance air kerma during fluoroscopy and spot filming. Fluoroscopic procedures generally involve a much higher patient dose than conventional overhead-tube radiographic examinations because of the need to penetrate the contrast medium and the time required to conduct the study. The volume of tissue exposed during fluoroscopy and spot filming is fairly small.

Dose Area Product

The FDA requires fluoroscopic units manufactured after 2006 to provide a means for the operator to monitor radiation output. Two types of readout, **dose area product (DAP)** and **cumulative total dose**, have been developed for this purpose. The total dose in mGy represents the dose to a point at a specific distance from the focal spot. DAP is a quantity that indicates a combination of dose and the amount of tissue irradiated. It is calculated as the product of the air kerma and the cross-sectional area of the beam, expressed in units of $\mu\text{Gy}\cdot\text{m}^2$ or $\text{cGy}\cdot\text{cm}^2$ or $\text{rad}\cdot\text{cm}^2$. Some manufacturers also provide DAP readings for general diagnostic radiography as well.

SKIN INJURY

The FDA has issued a **Public Health Advisory** regarding radiation-induced skin injuries from fluoroscopic procedures. These injuries are usually delayed, so the physician cannot discern damage by observing the patient immediately after the procedure. The radiation dose required to cause skin injury is typically 3 Gy (300 rad) for temporary epilation (onset 2 to 3 weeks after exposure), 6 Gy (600 rad) for main erythema (onset 10 to 14 days after exposure), and 15 to 20 Gy (1500 to 2000 rad) for moist desquamation (onset several weeks after exposure).

The procedures of concern are primarily interventional procedures during which fluoroscopy is used to guide instruments. The risk of skin injury is associated with prolonged fluoroscopy time and multiple digital cine acquisitions **to a single skin site**. At the maximum rate of 10 R/min, the fluoroscopy time must exceed 30 minutes to cause skin injury (see [Table 1.15](#)). However, during angiography, the patient may be positioned close to the x-ray tube, where the fluoroscopy exposure rate can exceed 10 R/min. Digital recording may use very high exposure rates. If digital recording is performed, fluoroscopic skin injuries occur much more rapidly. Monitoring of the total dose or DAP during interventional procedures is essential for the prevention of skin injury.

DOSE REDUCTION TECHNIQUES DURING FLUOROSCOPY

Most operators are trained to activate the x-ray beam for a few seconds at a time, long enough to view the current catheter position or the bolus of contrast agent. Total fluoroscopic times can be reduced dramatically with **intermittent fluoroscopy**. This technique is particularly effective when combined with last image hold. Many modern fluoroscopy systems have the capability to retain the last fluoroscopic image on the monitor after x-ray exposure is terminated. This allows the physician to study the most recent acquisition and plan the next task without radiation exposure to the patient.

During pulsed fluoroscopy, the x-ray beam is emitted as a series of short pulses rather than continuously. For conventional fluoroscopy, the image is acquired and displayed at a constant 30 frames per second. Pulsed fluoroscopy at 15 frames per second compared with the usual 30 frames per second demonstrates substantial dose reduction (factor of 2). However, manufacturers may increase the

TABLE 1.15 TYPICAL PATIENT DOSE DURING FLUOROSCOPY

UPPER GI	
DIVISION OF USE	MAXIMUM IN ONE LOCATION
17 spot films	5 spot films at 1.75 mGy each
5 minutes of fluoroscopy	1.5 minutes at 26 mGy/min
Total maximum entrance air kerma: 48 mGy	
Total maximum entrance exposure: 5.5 R	
DOUBLE-CONTRAST BARIUM ENEMA	
DIVISION OF USE	MAXIMUM IN ONE LOCATION
11 spot films	3 spot films at 1 mGy each
7 minutes of fluoroscopy	1.5 minutes at 35 mGy/min
Total maximum entrance air kerma: 55 mGy	
Total maximum entrance exposure: 6.3 R	

radiation level per frame to achieve a more pleasing visual appearance (less noise), and the dose reduction may be only 25%. Mobile C-arm fluoroscopic units make pulsed fluoroscopy available at low frame rates (e.g., 8 frames per second). **Low frame rates** adversely affect the ability to display rapidly moving structures.

Large field size increases the amount of scatter radiation produced. Additional scatter radiation enters the receptor and degrades the resulting video image. Collimation to the area of interest improves image quality but also reduces the total volume of tissue irradiated by excluding tissue with little diagnostic value.

The design of the fluoroscopy system may incorporate **variable or operator-selectable filtration**. Substantial reductions in skin dose can be achieved by inserting appropriate metal filters (aluminum or copper) into the x-ray beam at the collimator. Filtration reduces skin dose by preferentially removing low-energy x-rays, which generally do not penetrate the patient to contribute to the image.

The presence of a grid improves contrast by absorbing scattered x-rays. However, the dose to the patient is increased by a factor of 2 or more. For pediatric cases, the removal of the grid reduces the dose with little degradation of image quality. Grids should be used with discretion when fluoroscopic studies are performed on children. These systems should have the capability for easy removal and reintroduction of the grid.

In most interventional fluoroscopic procedures, most of the fluoroscopic time the x-ray beam is directed toward a particular anatomic region. Some reduction in maximum skin dose can be achieved by **periodically rotating the fluoroscopic x-ray tube** to image the anatomy of interest from a different direction. This method tends to spread the entry dose over a broader area, reducing the maximum skin dose.

SCATTERED RADIATION

During routine fluoroscopy of the gastrointestinal tract, personnel are exposed to radiation scattered by the patient and other structures in the x-ray beam. Scattered radiation levels depend on entrance exposure rate, field size, beam quality, and patient thickness but decrease rapidly with distance from the patient. The pattern of scattered radiation is shown in Fig. 1.193, in which the tower drape shielding is not in place.

The IR, tower lead drapes, bucky slot shield, x-ray table, foot rest (if present), and radiologist all provide a source of shielding for the technologist. The bucky slot shield covers the gap under the tabletop that allows the bucky to move along the length of the table for radiography. The area behind the radiologist and away from the table has the lowest scattered radiation level (<10 mR/hr) (Fig. 1.194).

When the receptor is lowered as close as possible to the patient, much of the scatter to the worker's eyes and neck is eliminated. The vertical and lateral extents of the scattered radiation field contract dramatically as the distance between patient and receptor is reduced.

Radiation Protection Practices During Fluoroscopy

Even with correct shielding in place and the IR as close to the patient as possible, scatter radiation is still present during fluoroscopy. Radiation levels are highest in the region close to the table on each side of the radiologist (Table 1.16). The presence of tower drapes greatly reduces the dose to the radiologist. Technologists and others in the room can decrease their dose by not standing close to the table on either side of the radiologist.

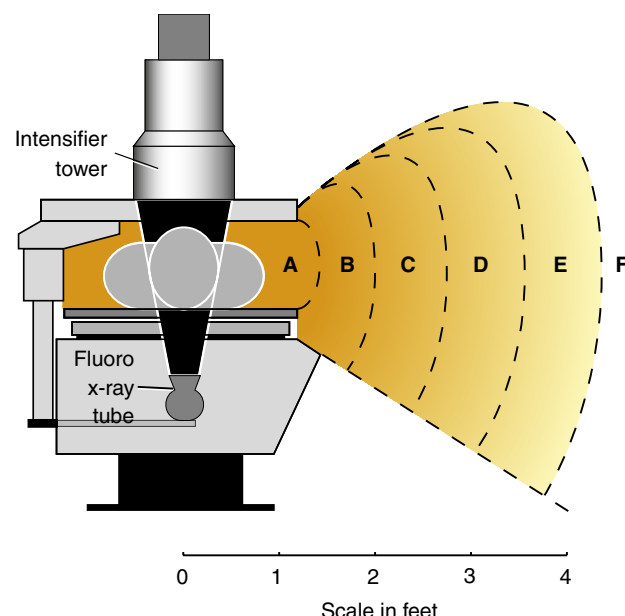


Fig. 1.193 Fluoroscopy scattered radiation pattern without tower drape shields in place.

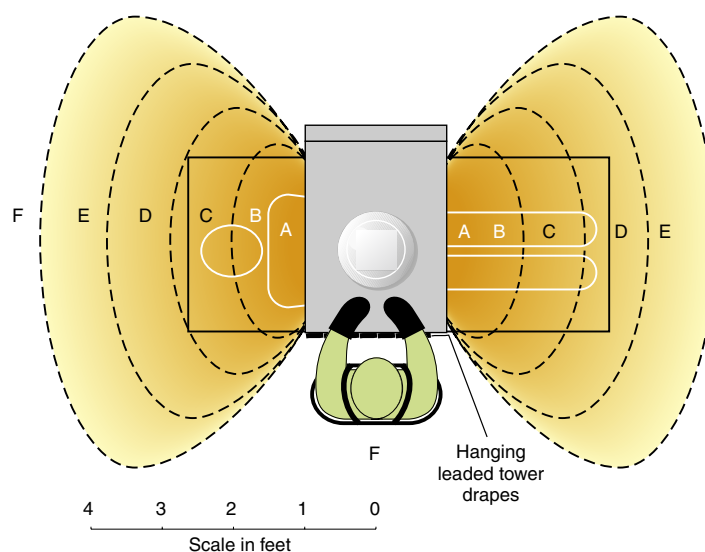


Fig. 1.194 Fluoroscopy scattered radiation pattern with tower drape shields in place and image receptor close to the patient.

TABLE 1.16 EXPOSURE LEVELS

ZONE	EXPOSURE RATE (MR/HR)	AIR KERMA RATE (MGY/HR)
A	>400	>3.5
B	400	3.5
	200	1.75
C	200	1.75
	100	0.88
D	100	0.88
	50	0.44
E	50	0.44
	10	0.088
F	<10	0.088



Fig. 1.195 Thyroid shields with regular neck cutout apron.

All individuals participating in fluoroscopic procedures must wear a protective apron. A 0.5 mm lead equivalent apron, which reduces the exposure by a factor 50 over the diagnostic x-ray energy range, is recommended.¹ Typical doses under the apron are below the threshold of detectability for personnel monitors. Dosimeters placed under the apron show readings only for individuals approaching the dose limit, which are typically less than 20 mrem. Aprons of multiple element composition with a 0.5 mm lead equivalence between 80 and 110 kVp offer the advantage of reduced weight. However, some manufacturers of “light” aprons achieve a weight reduction by the removal of lead vinyl layers, sacrificing some protection. Technologists should be cautious about using aprons with large cutouts around the arms and low necklines. These allow greater exposure to the thyroid and breasts. Although some protective aprons have a thyroid shield built into them, most do not. A separate **thyroid shield** can be worn with the apron to protect the neck region (Fig. 1.195).

Although a thyroid shield is not required for an individual participating in fluoroscopic procedures, a thyroid shield should be available (provided by the health care facility) for use at the option of the radiation worker. Wearing the thyroid shield is consistent with the ALARA principle, but the overall reduction in effective dose provided by this device is small. For an individual approaching a significant fraction of the dose limit, the thyroid shield is recommended.

In 2012, the International Commission on Radiological Protection (ICRP) published the modified threshold for a vision-impairing cataract from an acute dose of 5 Gy to an acute dose of 0.5 Gy.¹⁸ This was due to new evidence indicating that the eye is more radiosensitive than originally thought. Most occupationally exposed personnel will not exceed this threshold under normal working

conditions if recommended practices are followed. However, very busy **interventionalists** may be at risk to exceed this threshold. There are various ways to protect the eye from unnecessary radiation exposure, including leaded face shields. The use of leaded eyewear is not usually necessary or recommended unless the occupationally exposed worker is consistently participating in long fluoroscopy cases that require very close proximity to the patient. Most interventional radiologists and interventional cardiologists should wear leaded glasses.

Radiation-attenuating surgical gloves offer minimal protection of the operator’s hands, provide a false sense of protection, and are not recommended. The instantaneous dose from scatter radiation is reduced when hands covered with **one layer of glove material** are located near the radiation field. However, the total time near the radiation field depends on the speed with which the procedure is performed, as well as the distance from the imaged anatomy when the x-ray beam is activated. The increased thickness of these gloves reduces dexterity and can increase procedure time. The automatic exposure control system in fluoroscopy increases the radiation output to penetrate the glove when the hand is present in the beam. This can be confirmed by noting that anatomy is seen even though the glove is present. The dose to the hand is comparable to the dose when the radiation-attenuating glove is not present. The cost of radiation-attenuating surgical gloves and the minimal dose reduction do not justify the use of these devices according to the ALARA principle.

Image Wisely and Image Gently

Image Wisely¹⁹ is an awareness program, developed jointly by the American College of Radiology, Radiological Society of North America, American Association of Physicists in Medicine, and American Society of Radiologic Technologists, to promote radiation safety in adult medical imaging. The goal is to eliminate unnecessary radiation associated with adult imaging by avoiding non—medically indicated imaging procedures, by conducting the most appropriate imaging procedure, and by using the lowest optimal dose in all imaging practices.

Printed and electronic educational resources have been developed for radiologists, medical physicists, radiologic technologists, referring physicians, patients, and the general public. Topics include dose, dose reduction techniques, appropriateness of imaging procedures, and risks. The information is directed at each respective target audience. A similar campaign, called **Image Gently**²⁰, is designed to minimize the radiation exposure in children, whose long life expectancy and increased radiosensitivity contribute to higher lifetime cancer risk. More information on the Image Wisely and Image Gently campaigns can be found at www.imagewisely.org and www.imagegently.org.